Endpoint crossings for random walks with memory

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Abstract

Drop a ball through a Galton board on which every bounce after the first repeats the previous direction with probability p. Kagey asked when the middle bin is as likely as each end bin. For this Markov walk with N steps the crossing comes when the expected number of direction changes is about $\log N$. We ask what changes when the ball remembers more than its last step. In the elephant random walk each step copies a uniformly chosen earlier step and reverses it with probability ε. We prove that the crossing is unique and that $N\varepsilon$ at the crossing is $2\log N - \log\log N - \log 16 + o(1)$, about twice the Markov value, because a walk that ends in the middle needs a reversal among its first few steps. At the crossing the law has a hump near each end, and the middle is the lowest bin outside a window of width $O(N^{1/2}\log N)$. Next we take alternating runs whose lengths have tail $k^{-\alpha}$ with $1 < \alpha < 2$, started in equilibrium. Wang, Li and Hänggi observed, as a heuristic for heat conduction, that the ends and the middle decay at the same rate when α is the golden ratio. We prove the lattice form of their observation, with its constant. Namely, the ends beat the middle for large N exactly when α is below the golden ratio. Finally, an aging walk switches at step k with probability c/k. With a fixed first step each of the bins $1, \dots, N-1$ ties with the end exactly at $c = 1$, for every N. We prove the same for an aging walk with d states, whose vector of counts is exactly uniform at the tie.

Keywords: persistent random walk; elephant random walk; random recursive tree; Pólya urn; alternating renewal process; heavy tails; golden ratio; aging random walk; Galton board.
MSC2020: 60G50; 60K05; 60C05; 60F05; 60J10.

1 Introduction

Suppose you drop a ball through a Galton board with N rows of pegs. At the first peg the ball goes left or right with probability $\frac{1}{2}$ each. At every later peg it repeats its previous direction with probability p and switches with probability $q = 1 - p$. This is the board of Problem 131 in Kagey's Open Problem Collection [1], and one of his questions is when the middle bin is exactly as likely as each of the two end bins. If p is close to 1 the ball rarely switches, so the ends win. If $p = \frac{1}{2}$ the ball does a simple random walk, so the middle wins. Paper A [2] finds the crossing in between. The middle catches up with the ends when the expected number of switches $T = Nq$ is about $\log N$.

Kagey's ball remembers one step, since its next direction depends only on its last one. In this paper we ask what happens to the middle-versus-ends question if the ball remembers more. We study three kinds of memory.

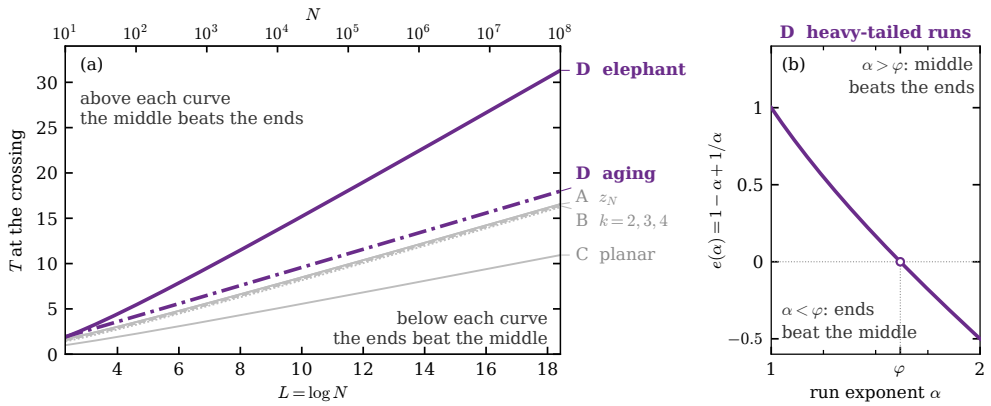


Figure 1: The crossing on each model's switching scale against $L = \log N$ (a), and the heavy-tailed exponent $e(\alpha) = 1 - \alpha + 1/\alpha$ of this paper for a stationary start (b). The scale is T for Papers A to C, $x = N\varepsilon$ for the elephant walk (drawn through the root g_N of Theorem 2.7), and the expected number $H_N - 1$ of switches at the fixed-start crossing $c = 1$ for the aging walk. The curves of this paper are in bold, and a version of this figure also appears in Paper C.

- In the elephant random walk of Schütz and Trimper [5] the ball remembers its whole past. At each peg it picks one of its earlier directions uniformly at random and repeats it, except that with probability ε it reverses it. We write $x = N\varepsilon$, which is the expected number of reversals up to ε.
- In a walk with heavy-tailed runs the ball remembers how long it has gone in its current direction. Its path is a sequence of runs of alternating direction, and the run lengths ξ are independent with $\mathbb{P}(\xi \geq k) = k^{-\alpha}$ for $k \geq 1$.
- In an aging walk the ball remembers how old it is. It switches at peg $k \geq 2$ with probability c/k. We also give the ball d directions. Then at peg $k \geq 2$ it turns to each of the other $d - 1$ directions with probability $\theta/(k + d - 2)$, which is c/k with $c = \theta$ when $d = 2$.

The answer depends on the kind of memory. For the elephant walk the middle needs about twice as many reversals as Kagey's board needs switches. Namely, the crossing value of x is about $2 \log N$, not $\log N$. For heavy-tailed runs started in equilibrium the ends win for large N if α is below the golden ratio $\varphi = (1 + \sqrt{5})/2$, and the middle wins if α is above φ. For the aging walk with a fixed first step the crossing is exactly $c = 1$, for every N and every bin. With d directions the same happens at $\theta = 1$, where the vector of counts is exactly uniform (Section 5).

Why the elephant needs twice as many reversals. The ends are easy. The ball lands in the right end bin exactly when its first step goes right and it never changes direction. This has probability $\frac{1}{2}(1 - \varepsilon)^{N-1} \approx \frac{1}{2}e^{-x}$ for the elephant walk, and the same with q in place of ε for Kagey's board. So only the middle differs. On Kagey's board the T switches are spread over the whole walk, and each moves the final position by about a run length N/T. So the number of right steps spreads over about $N/\sqrt{T}$ bins, the middle bin has probability of order $\sqrt{T}/N$, and the ends lose when e^{-T} falls to about $1/N$, that is, when $T \approx \log N$.

The elephant walk spreads a reversal in a different way. A reversal at step $t + 1$ is copied by later steps, those copies are copied, and so on. The steps that descend from it form a family, whose share of the walk behaves like the share of one ball in Pólya's urn started with one ball against t. So a late reversal has a small family and hardly moves the ball. To land exactly in the middle,

the walk needs a family that fills half the walk, so it needs a reversal among its first few steps. This has probability of order ε, and hitting the exact middle brings another factor of order $1/N$. So the middle bin has probability of order $\varepsilon/N = x/N^2$ (in fact about $4x/N^2$), and the ends lose only when e^{-x} falls to about $1/N^2$, that is, when $x \approx 2\log N$. More precisely, at Kagey's crossing $\frac{1}{2}e^{-T} \approx \sqrt{2T/\pi}/N$, and squaring this relation turns it into the elephant relation $\frac{1}{2}e^{-x} \approx 4x/N^2$ with $x \approx 2T - \log(2\pi)$.

The small families also give the shape of the law. About x/m^2 reversals have a family of m steps, so the bin at distance m from an end has probability about $x/(2m^2)$ over a wide range of m. So the law has a hump near each end, and the middle is where the two tails meet. Indeed, twice $x/(2m^2)$ at $m = N/2$ is the $4x/N^2$ above.

Why the golden ratio. For heavy-tailed runs take $1 < \alpha < 2$, so a run has a finite mean μ but an infinite variance. Start the walk in equilibrium, at a typical time of a long sequence of runs. Then the first run is already in progress, and long runs are more likely to be in progress. The chance that this run covers all N steps is of order $N^{1-\alpha}$, and this is the end. For the middle, the runs to the right and the runs to the left form two independent renewal processes, each making about $N/(2\mu)$ runs, and the ball ends in the middle when they meet at level $N/2$. Each process fluctuates on the scale $N^{1/\alpha}$ of a stable law, so they meet exactly with probability of order $N^{-1/\alpha}$. Hence the ratio of end to middle is of order $N^{1-\alpha}/N^{-1/\alpha} = N^{1-\alpha+1/\alpha}$. The exponent vanishes when $\alpha^2 - \alpha - 1 = 0$, that is, at $\alpha = \varphi$. Below φ the ends win, and above φ the middle wins. If the first run is a fresh run instead, the end is of order $N^{-\alpha}$, a factor N smaller, and the middle wins for every $1 < \alpha < 2$.

1.1 Relation to Papers A, B and C

This is the fourth of four papers on Problem 131, which we call Papers A, B, C and D. Paper A [2] solves Kagey's board. Paper B [3] treats d directions, where the ball turns to each other direction with the same probability. Paper C [4] gives the general theory for slowly switching chains. These three papers study Markov walks, whose next step depends only on the current one, and their answer is that the middle (or a face) ties with the vertices when T is of order $\log N$. This paper, Paper D, keeps Kagey's question and changes the walk. It can be read on its own. From the earlier papers we use only the crossing equation of Paper A, namely (1) below, and we use it only to compare the elephant crossing with the Markov crossing in Section 2. The four papers share the following notation.

- Kagey's board has N steps (Kagey's row n has $N = n - 1$). The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p. We put $q = 1 - p$ and $u = q/p$ (the odds of a switch).
- $P_N(k)$ is the probability of k right steps, that is, the probability that the ball lands in bin k. The endpoints have $P_N(0) = P_N(N) = p^{N-1}/2$.
- $L = \log N$, and $T = Nq$ is the expected number of switches (up to the first step).
- The crossing is the value of p (or of T) at which a bin is exactly as likely as an endpoint. For the middle bin it is p_N, and $z_N = N(1 - p_N)$.
- Papers B and C have d states. A face F is a nonempty set of states, and the vertices are the faces of size 1. For a chain with transition matrix $I + (T/N)Q$, λ_F is the principal Dirichlet eigenvalue of $-Q$ on F, and K_F is a second-order constant.

For the elephant walk $x = N\varepsilon$ corresponds to T, and its crossing x_N corresponds to z_N. We compare the two crossings through the crossing equation of Paper A [2, Proposition 3.8], which

says that for every $N \geq 2$

$$z_N + \tfrac{1}{2}\log z_N = L + c_0 + \frac{1}{8z_N} + O(L^{-2}), \qquad c_0 = \tfrac{1}{2}\log\tfrac{\pi}{8}, \tag{1}$$

where the error term is at most $\frac{1}{4L^2} + \frac{L^2+L}{N}$ in absolute value, and that $L/2 \leq z_N \leq L$. As in Paper B the start law matters. For heavy-tailed runs it decides whether the threshold is φ or 1, and for the aging walk it decides whether the crossing is exactly $c = 1$. Figure 1 shows the crossings of all four papers.

1.2 Setting and notation

Every walk in this paper takes N steps $X_1, \ldots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \leq N : X_i = +1\}$ be the number of right steps, and keep writing $P_N(k) = \mathbb{P}(A_N = k)$ for the probability that the ball lands in bin k, now for whichever walk we study. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the two directions alike, so $P_N(k) = P_N(N-k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. The crossing is the value of the parameter at which $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \leq k} 1/j$ is the k-th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \to \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The elephant walk (Definition 2.1) has the reversal probability $\varepsilon \in (0, 1)$ and $x = N\varepsilon$. Its crossing is ε_N, and $x_N = N\varepsilon_N$. Heavy-tailed runs (Section 4) have the exponent α, the mean run length $\mu = \mathbb{E}\xi = \zeta(\alpha)$ for $\alpha > 1$, and the length D of the first run. A fresh start has D distributed as ξ, and the stationary start has $\mathbb{P}(D = d) = \mathbb{P}(\xi \geq d)/\mu$. We put $R_N = E_N/C_N$, so the ends win exactly when $R_N > 1$. The aging walk (Section 5) has the constant $c \in (0, 2)$, and its version with d states has the constant θ and the count vector $(n_1, \ldots, n_d)$, where n_i is the number of steps in state i. The symbols that only one section uses are defined at the start of that section. The letters p, q and u have their meaning for Kagey's board only in this introduction, and later sections reuse them locally.

1.3 Results

The elephant walk. For every even $N \geq 2$ the crossing ε_N exists, is unique and is below $\frac{1}{2}$ (Proposition 2.3). The center has derivative exactly $4/(N+2)$ at $\varepsilon = 0$ (Proposition 2.5), and $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$ for even $N \geq 64$ and $\varepsilon(1 + \log N) \leq \frac{1}{100}$, with explicit constants (Theorem 2.6). Let g_N be the root of $g + \log g = 2L - \log 8$. Then $x_N = g_N + O(L^2/N)$, and so $x_N = 2L - \log L - \log 16 + O(\log L/L)$ (Theorem 2.7). Compared with Kagey's board, $x_N = 2z_N - \log(2\pi) + c_*/L + O(\log L/L^2)$ with $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$ (Theorem 2.8).

We also go one order further. In the same range of ε the center is $\frac{4\varepsilon}{N+2}N^{2\varepsilon}(1 + O(\varepsilon))$, with explicit constants (Proposition 2.9). This gives $x_N = g_N - (2g_N L + g_N^2/2)/(N(1+1/g_N)) + O(L/N)$, so the root g_N is above the crossing for large N (Proposition 2.10). At fixed N the center is a polynomial $c_1\varepsilon + c_2\varepsilon^2 + \cdots$ in ε, and $c_2/c_1 - 2L \to 2\gamma - 2 - \log 2$ (Proposition 2.11).

The remaining results describe the whole law near the crossing. At the crossing the end is below its neighbor for every even $N \geq 8$ (Corollary 3.6 and Proposition 3.8). For $N < 1688$ this rests on a floating-point computation whose rounding error we bound (Lemma 3.7). Let M_N be the number of steps that disagree with the first step. For $N \geq 2$ and $\varepsilon \leq \frac{1}{4}$ the law of M_N is within

an explicit δ_B of a compound Poisson law, bin by bin, and $\delta_B = O(x^2 \log^2 N/N)$ when $x^2 \le N$ (Theorem 3.9). If $x \to \infty$ and $x^3 \log^2 N/N \to 0$, then $(M_N - x \log x)/x$ converges in law to the Landau law, with a bound for each bin (Theorem 3.10). So at the crossing every mode lies at distance $x_N \log x_N + \lambda_0 x_N + O(\sqrt{x_N \log x_N})$ from the nearer end, where $\lambda_0 \approx -0.2228$ is the mode of the Landau density (Corollary 3.12, and Corollary 3.17 gives the weaker bound $O(x_N \log^2 x_N)$). Farther from the ends, bin $N - m$ has probability $\frac{x}{2m^2}(1 + o(1))$ for $x \log^2(x+1) \ll m \ll N$, when $x \ge 1$ and $x \log^2 N/N \to 0$ (Theorem 3.14). Finally, for large even N the center at the crossing is below every interior bin outside a window of width of order $N^{1/2} \log N$ around it (Theorem 3.15). Inside the window we use second differences. For every $N \ge 3$ and every $0 < \varepsilon < \frac{1}{2}$ the second difference of the law changes sign at most four times (Theorem 3.21), so the law has at most two concave stretches, one for each hump. For large even N the law at the crossing is strictly convex between the humps (from $2m^*$ to $N - 2m^*$, with m^* the distance of a mode from the nearer end), and the center is its unique interior minimum (Theorem 3.24 and Corollary 3.25). The convexity also holds for large odd N, because the two central bins of an odd N are the center of $N + 1$ up to an explicit factor (Proposition 3.27). A certified computation gives the convexity for $10 \le N \le 2000$ (Proposition 3.26), and the N in between are open (Conjecture 3.16).

Heavy-tailed runs. For $1 < \alpha < 2$ and every first run that is finite almost surely, C_N is asymptotic to an explicit constant times $N^{-1/\alpha}$ (Theorem 4.4). With the stationary start $R_N \sim C(\alpha) N^{1-\alpha+1/\alpha}$, and $C(\varphi) = 0.7761\ldots$, so at $\alpha = \varphi$ the center still wins. With a fresh start $R_N \to 0$ (Corollary 4.5). For $\alpha \ge 2$ the center is of order $N^{-1/2}$, with an extra factor $(\log N)^{-1/2}$ at $\alpha = 2$, and $R_N \to 0$ with either start (Theorem 4.7). For $0 < \alpha < 1$ the runs have infinite mean and only the fresh start makes sense. Then $NC_N \to \frac{2}{\pi} \tan \frac{\pi\alpha}{2}$ and $R_N \to \infty$ (Theorem 4.8). So with a fresh start the ends win for $\alpha < 1$ and the center wins for $\alpha > 1$. We do not prove what happens at $\alpha = 1$ (Remark 4.9).

Aging. With a fixed first step and $c = 1$ the number of right steps is exactly uniform on $\{1, \ldots, N\}$. The ratio of each interior bin to the end increases strictly in c, so every interior bin ties with the end exactly at $c = 1$, for every $N \ge 2$ (Proposition 5.1). With d states, a start in state 1 and $\theta = 1$, the count vector is exactly uniform over all compositions $(n_1, \ldots, n_d)$ of N with $n_1 \ge 1$, and every other such bin ties with the corner $(N, 0, \ldots, 0)$ exactly at $\theta = 1$, for every $N \ge 2$ (Theorem 5.2). A small change of the aging rate gives a chain that redraws or repeats its state, and its counts have exactly the law of Pólya's urn for every weight (Proposition 5.4). With a fair first step each interior bin is twice as likely as an end at $c = 1$, and its crossing is unique and below 1 (Proposition 5.6). For the center the crossing is $c_N = 1 - \log 2/(\log N + \gamma + 2 - 2\log 2) + O((\log N)^{-3})$ (Proposition 5.7).

1.4 Prior work

The steps of the elephant walk split into clusters, since a step either copies an earlier step exactly, with probability $1 - 2\varepsilon$, or starts a new cluster with a fair sign. This is Kürsten's coupling with bond percolation on random recursive trees [6], and Baur and Bertoin [7] relate the walk to Pólya-type urns. Our regime $N\varepsilon \asymp \log N$ lies in the strongly supercritical regime of Baur [8], who extends the results of Bertoin [9] for retention probability $1 - t/\log n$. Baur's Corollary 1 gives the sizes of the largest clusters other than the root's (a Poisson limit with intensity $a^{-2}\, da$), which is the picture of about x/m^2 families of size m behind Theorem 3.14. These results describe cluster sizes, not the center probability, which is a large deviation of the root cluster. The law

that appears in the bulk (Theorems 3.9 and 3.10) is also known. Without its truncation, our compound Poisson law is the Lea–Coulson form of the Luria–Delbrück distribution, and Kessler and Levine [15, Eqs. (14), (18) and (19)] describe its profile $x/(m(m+1))$ for small x and its Landau limit for large x (see also [16]). The same stable law governs the fluctuations of the root cluster when the retention probability is $1 - c/\log n$ [14]. What we add is the comparison of the elephant walk with this law in our regime (with an explicit error in Theorem 3.9) and a certified enclosure of the mode. Guérin, Laulin, Raschel and Simon [10, proof of Theorem 1.1] prove that the law with a fixed first step is unimodal, and their other results fix the memory parameter as N grows. So do the local limit theorem of Trinas and Valle [11, Theorem 3.1] for the superdiffusive walk and their bounds on where log-concavity fails. We did not find an asymptotic formula for the center probability when $\varepsilon \to 0$ as N grows.

For heavy-tailed runs, Godrèche and Luck [12] give the transform of the occupation time of an alternating renewal process and the scalings of the center and of both edges, and [18, 19] describe the profile of a Lévy walk. Wang, Li and Hänggi [20] already observe, as a heuristic for heat conduction, that equal decay of the central peak and the side peaks gives the golden ratio. The two decay rates of the end, $N^{1-\alpha}$ and $N^{-\alpha}$, are the scalings of the ballistic peaks of a Lévy walk with and without an equilibrated start [17, Table I]. We prove the lattice statement with a local limit at the center, treat both starts, and compute $C(\alpha)$. For $\alpha < 1$ the limit law of A_N/N is Lamperti's [13]. For aging walks, Dietz and Sethuraman [30] proved the limit law of A_N/N, Engländer and Volkov [31] gave a second proof, and Engländer, Volkov and Wang [37] found the scaling limit of the walk. For d states the limit is a Dirichlet law [30, Theorem 1.4]. The exact uniform law for two states is an exercise in [37, footnote 1 on p. 6], which also announces a general connection with Pólya urns, and the chain that redraws or repeats is a conservative random walk in the sense of [34]. What is new in Section 5 is the explicit joint law of the counts and the last state for d states and the ties that follow from it.

1.5 Organization

Section 2 reduces the elephant walk to a counting chain, proves that its crossing is unique, finds the crossing, and then gives the center and the crossing to second order. Section 3 describes the whole law near the crossing, namely the dip at each end, the compound Poisson and Landau approximations of the bulk with the place of the modes, the tail $x/(2m^2)$, and the window around the center. Section 4 treats heavy-tailed runs, first the golden ratio for $1 < \alpha < 2$ and then the exponents $\alpha \geq 2$ and $\alpha < 1$. Section 5 treats the aging walk with a fixed first step, with d states, and with a fair first step. Section 6 describes the checks. The longer proofs are in the appendices. Appendix A holds the proofs for Section 2. Appendix B holds the proofs for the tail and the modes in Section 3, and Appendix C holds the proofs for the bulk. Appendix D holds the proofs for Section 4, and Appendix E proves Proposition 5.7.

2 The elephant walk and its crossing

We reduce the elephant walk to a counting chain, prove that the crossing is unique, compute the center to first and second order in ε, and locate the crossing. The last subsection gives one more term of the center and of the crossing.

Definition 2.1 (elephant walk)**.** Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε. All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in

Section 1.2, $A_N = \#\{i \le N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$, and $x_N = N\varepsilon_N$.

A flip is what the introduction called a reversal. In the notation of [5] the memory parameter is $1-\varepsilon$, and [10] write $a = 1-2\varepsilon$. Given the first step, all that matters is how many steps disagree with it. So let M_t be the number of $i \le t$ with $X_i \ne X_1$, and put

$$r_t(b) = \varepsilon + (1-2\varepsilon)\frac{b}{t}, \qquad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t\ge1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b+1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1. For $0 \le k \le N$, $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$, and $P_N(k) = P_N(N-k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t. So $X_{t+1} \ne X_1$ with conditional probability $(1-\varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1, which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N-k$. Finally $M_N \le N-1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1}(1 - r_t(0)) = (1-\varepsilon)^{N-1}$. □

So the end is Kagey's end $p^{N-1}/2$ with $p = 1-\varepsilon$. More flips always help the middle against the end.

Proposition 2.3. *Let $N \ge 2$ and $1 \le k \le N-1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0,1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $C_N \ge 2E_N$ for $\varepsilon \ge \frac{1}{2}$.*

Proof. We compare each path of M with the path that never flips, one step at a time. The probability of a path of M is a product of $N-1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s. The path that never flips has probability $f_N(0) = (1-\varepsilon)^{N-1}$, which is one factor $1-\varepsilon$ for each step. So we divide each factor by $1-\varepsilon$. With $y = b_s/s \in [0,1)$ the two kinds of factors become $\frac{\varepsilon+(1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \ge 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1-y) + (1-\varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε, and it is strictly increasing if the path has an up step. Summing over the paths with $k \ge 1$ up steps proves the claim for $f_N(k)/f_N(0)$. By Lemma 2.2, $P_N(k)/E_N = [f_N(k) + f_N(N-k)]/f_N(0)$, and both k and $N-k$ are at least 1. So the claim for $P_N(k)/E_N$ follows.

It remains to locate the crossing. The ratio C_N/E_N is continuous and strictly increasing in ε. As $\varepsilon \to 0$ it tends to 0, since $C_N \to 0$ (at $\varepsilon = 0$ the chain never leaves 0) and $E_N \to \frac{1}{2}$. For $\varepsilon \ge \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1-\varepsilon, \varepsilon]$, so each path of M that ends at h has probability at least $(1-\varepsilon)^{N-1} = 2E_N$. Hence $C_N \ge 2E_N$ there. So the ratio equals 1 at exactly one ε, and this ε lies in $(0, \frac{1}{2})$. □

2.1 First order, one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε, it is Pólya's urn.

Lemma 2.4 (Pólya's urn)**.** *Let $2 \le v \le N$. At time v an urn holds one ball of one color and $v-1$ of the other. At each time $k = v, \dots, N-1$ we draw a ball uniformly and return it with one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first*

color. Then $\beta_v(d) = \binom{N-d-1}{v-2}/\binom{N-1}{v-1}$ *for* $1 \le d \le N-v+1$*, and* $\beta_v(d) = 0$ *for other* d*. Also* β_v *is nonincreasing in* d *and* $\sum_d d\,\beta_v(d) = N/v$*. For* $1 \le d \le N-1$*,*

$$\sum_{v=2}^{N} \beta_v(d) = \frac{N}{d(d+1)}, \qquad \sum_{v=2}^{N} (v-2)\,\beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. We compute the probability of one order of the draws and count the orders. At time k the urn holds k balls. So if s of them are of the first color, the next draw is of that color with probability s/k. Take one order of the $N-v$ draws in which i draws are of the first color. The draws of the first color contribute the numerators $1, 2, \dots, i$, the other draws contribute $v-1, v, \dots, N-i-2$, and the denominators are $v, v+1, \dots, N-1$. So this order has probability $\frac{i!\,(v-1)v\cdots(N-i-2)}{v(v+1)\cdots(N-1)} = \frac{(v-1)\,i!\,(N-i-2)!}{(N-1)!}$, whatever the order is. There are $\binom{N-v}{i}$ orders. With $d = i+1$ this gives $\beta_v(d) = a_d(v-1)\binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, and writing out the factorials shows that this is $\binom{N-d-1}{v-2}/\binom{N-1}{v-1}$. From the last form, $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d} \le 1$ for $2 \le d \le N-v+1$, so β_v is nonincreasing. The fraction of balls of the first color is a martingale that starts at $1/v$. So the mean number of such balls at time N is N/v.

For the two sums we use the first form of $\beta_v(d)$ and put $i = N-v$, so that $v-1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. The Chu–Vandermonde identity says that $\sum_i \binom{i}{a}\binom{N-1-i}{b} = \binom{N}{a+b+1}$ (sort the subsets of $\{1, \dots, N\}$ of size $a+b+1$ by their element number $a+1$, which we call $i+1$). With $a = d-1$ and $b = 1, 2$ it shows that the two sums are $a_d\binom{N}{d+1}$ and $2a_d\binom{N}{d+2}$, and these are the values in the statement. □

We need some more notation. Let $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ be the chance of ending at the center from state b at time s. For $1 \le t \le h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}, \qquad c_1 = \sum_{t=1}^{h}\pi_t, \qquad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1\le t\le h}$ is a probability vector.

Proposition 2.5. *Let* $N \ge 2$ *be even.*

(a) $C_N = \sum_{t=1}^{h} \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1)$*.*

(b) C_N *is a polynomial in* ε *with* $C_N(0) = 0$*. At* $\varepsilon = 0$ *we have* $V_{t+1}(1) = \pi_t$*, and* $C_N'(0) = c_1 = \frac{4}{N+2}$ *exactly.*

(c) $\sum_t w_t t = 3 - \frac{6}{h+2}$*, and* $w_t \le 2t2^{-t}$ *for* $1 \le t \le h$*.*

Proof. (a) We condition on the time of the first flip. Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \le t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. A path with no flip ends at $0 \ne h$. So C_N is the sum of $\varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$ over $1 \le t \le N-1$. Only $t \le h$ contribute. Indeed at time $t+1$ there are t steps equal to X_1, this number never decreases, and at the center it is h. So $V_{t+1}(1) = 0$ for $t > h$.

(b) Every transition probability is affine in ε, so C_N and $V_{t+1}(1)$ are polynomials in ε. By (a), $C_N(0) = 0$ and $C_N'(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M counts the balls of the first color in the urn of Lemma 2.4 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h+1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^{N}\beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4, with $d = h$ and $v = t+1$, gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$. Dividing by $c_1 = \frac{N}{h(h+1)}$ gives $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2} = 2 - \frac{6}{h+2}$, and adding $\sum_t w_t = 1$ gives the first claim. For the second claim, each factor $\frac{h-i}{N-i}$ in π_t is at most $\frac{1}{2}$, so $\pi_t \le \frac{t}{N-t} 2^{1-t}$. Hence $w_t = \frac{N+2}{4}\pi_t \le \frac{N+2}{2(N-t)}\, t 2^{-t}$. For $t \le h$ we have $N - t \ge h$ and $\frac{N+2}{2h} = 1 + \frac{2}{N} \le 2$, so $w_t \le 2t2^{-t}$. □

In words, to first order the walk flips once, at some step $t+1$, and then runs as Pólya's urn from t balls of one color and one of the other. The minority fraction is then close to a Beta$(1,t)$ variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$, and $\sum_t t2^{1-t} = 4$ explains the constant $4/(N+2)$. Since $w_t \le 2t2^{-t}$, only an early flip counts.

2.2 Second order

Theorem 2.6. *Let $N \ge 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$, $C_N \ge \frac{4\varepsilon}{N+2}(1-2\varepsilon)$.

(b) $C_N \le \frac{4\varepsilon}{N+2}\, N^{2\varepsilon} e^{5200\varepsilon}$ whenever $N \ge 64$ and $0 < \varepsilon(1+\log N) \le \frac{1}{100}$.

In particular $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon \log N)$ in the range of (b), with an absolute implied constant.

In words, the first-order term is correct up to a factor $1 + O(\varepsilon \log N)$, which is $1 + O(L^2/N)$ at the crossing.

Idea of the proof. The proof is in Appendix A. After the first flip, M at time s is Pólya's urn with βs extra balls of each color, where $\beta = \varepsilon/(1-2\varepsilon)$ (Lemma A.1). They pull the minority fraction toward $\frac{1}{2}$. For (a) we drop them. For (b) we compare M with an urn with $\kappa(s+10)$ extra balls, where $\kappa = 2\varepsilon/(1-4\varepsilon)$. The step at time s then loses a factor of about $e^{\kappa/s}$, and the product of these factors is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1-x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2\log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \qquad G(x) = x + \log x,$$

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0,\infty)$.

Theorem 2.7. *As $N \to \infty$ over even N, the following hold.*

(a) $x_N + \log x_N = \Lambda_N + O(L^2/N)$, so $x_N = g_N + O(L^2/N)$.

(b) $x_N = 2\log N - \log\log N - \log 16 + O(\log\log N/\log N)$.

In words, the root g_N locates the crossing to within $O(L^2/N)$, and the expected number of flips at the crossing is about $2\log N$.

N	x_N	g_N	$x_N - x_{\mathrm{2nd}}$	$x_N/\log N$	$D(N)\log N$
50	3.940487	4.288636	+0.3452	1.007	−0.083
400	7.629217	7.843768	+0.0621	1.273	−0.199
1600	10.240565	10.340051	+0.0179	1.388	+0.215
3200	11.546834	11.610464	+0.0097	1.431	+0.386
12800	14.158845	14.182921	+0.0028	1.497	+0.607
51200	16.778468	16.786946	+0.0008	1.547	+0.707

Table 1: Exact elephant crossings, the root g_N of $G(g) = \Lambda_N$, and the difference $D(N)$ of Theorem 2.8 times $\log N$. Here x_{2nd} is the formula of Proposition 2.10 without its error term.

Idea of the proof. Let $F(x)$ be the logarithm of C_N/E_N at $\varepsilon = x/N$. It increases in x, and its only zero is x_N. Theorem 2.6 and the formula for E_N show that $F(x)$ is within $O(L^2/N)$ of $G(x) - \Lambda_N$ when x is of order L. Since $G' \geq 1$, the zero of F is within $O(L^2/N)$ of the zero g_N of $G - \Lambda_N$.

Proof. (a) *Step 1: both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε. By Proposition 2.3, F is strictly increasing and x_N is its only zero. We show that

$$|F(x) - G(x) + \Lambda_N| \leq \rho_N \quad (1 \leq x \leq 2\Lambda_N), \qquad \text{with } \rho_N = O(L^2/N). \tag{$*$}$$

Let $1 \leq x \leq 2\Lambda_N$ and $\varepsilon = x/N$. Then $\varepsilon \leq 4L/N$, so for large N both parts of Theorem 2.6 apply, and together they say that $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon L)$. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + 2/N)$, and $\log(1 + 2/N) \leq 2/N \leq 2xL/N$ because $x \geq 1$. For the end we use $\log(1 - y) = -y + O(y^2)$ for $0 \leq y \leq \frac{1}{2}$ in $\log E_N = -\log 2 + (N-1)\log(1 - \frac{x}{N})$. So

$$\log C_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big), \qquad \log E_N = -\log 2 - x + O\Big(\frac{x^2}{N}\Big).$$

We subtract. Since $\log 4 + \log 2 = \log 8$, the main terms give $x + \log x - 2L + \log 8 = G(x) - \Lambda_N$, and both errors are $O(L^2/N)$ because $x \leq 2\Lambda_N \leq 4L$. This is ($*$).

Step 2: trap the zero. We have $G' = 1 + 1/x \geq 1$ and $G(g_N) = \Lambda_N$. So $G(x) - \Lambda_N < -\rho_N$ for $x < g_N - \rho_N$, and $G(x) - \Lambda_N > \rho_N$ for $x > g_N + \rho_N$. By ($*$), $F(x) < 0$ for $1 \leq x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \leq 2\Lambda_N$. Both ranges are nonempty for large N, because $\Lambda_N - \log\Lambda_N \leq g_N \leq \Lambda_N$ (note that $G(\Lambda_N - \log\Lambda_N) \leq \Lambda_N \leq G(\Lambda_N)$). Since F is increasing, its zero x_N lies between the two ranges, so $|x_N - g_N| \leq \rho_N$. Then x_N lies in $[1, 2\Lambda_N]$, and $F(x_N) = 0$ in ($*$) gives $|G(x_N) - \Lambda_N| \leq \rho_N$.

(b) We solve $g_N = \Lambda_N - \log g_N$ by one substitution. Since $\Lambda_N - \log\Lambda_N \leq g_N \leq \Lambda_N$, the ratio g_N/Λ_N is $1 + O(\log\Lambda_N/\Lambda_N)$, so $\log g_N = \log\Lambda_N + O(\log\Lambda_N/\Lambda_N)$. Also $\Lambda_N = 2\log N - \log 8$ gives $\log\Lambda_N = \log 2 + \log\log N + O(1/\log N)$. So $g_N = 2\log N - \log 8 - \log 2 - \log\log N + O(\log\log N/\log N)$, which is (b) for g_N, and (a) gives it for x_N. □

Table 1 gives exact crossings on the grid $N = 50 \cdot 2^k$, computed from the recursion for f_N in Lemma 2.2. The limit 2 of $x_N/\log N$ is approached slowly, because of the term $-\log\log N$.

Now we compare with Kagey's board. Doubling (1) without its last two terms gives $2z_N + \log(2z_N) = \Lambda_N + \log(2\pi)$, so $2z_N - \log(2\pi)$ nearly solves $G(x) = \Lambda_N$. What is left is of order $1/L$.

Theorem 2.8. *Let z_N be the crossing value of T for Kagey's board with N steps, as in Section 1.1. As $N \to \infty$ over even N,*

$$D(N) := x_N - \big(2z_N - \log(2\pi)\big) = \frac{c_*}{\log N} + O\Big(\frac{\log\log N}{(\log N)^2}\Big), \qquad c_* = \tfrac{1}{2}\log(2\pi) - \tfrac{1}{4} = 0.668939\ldots$$

In words, the elephant needs twice as many flips as Kagey's board needs switches, less $\log(2\pi) = 1.837\ldots$, and the difference from this tends to 0 like $c_*/\log N$.

Idea of the proof. Both crossings solve an equation of the form $x + \log x = 2L + \text{constant}$ up to small errors. We put the Markov crossing into the equation $G(x) = \Lambda_N$ of the elephant, find that $2z_N - \log(2\pi)$ misses it by c_*/L, and divide by $G' \approx 1$.

Proof. Write $\ell = \log(2\pi)$ and $Y = 2z_N$. We compute $G(Y - \ell)$ in three steps and then compare it with $G(x_N)$.

Step 1: the size of z_N. By Section 1.1, the error term in (1) is $O(L^{-2})$ and $L/2 \le z_N \le L$. So $\log z_N = \log L + O(1)$ and $\frac{1}{8z_N} = O(1)$, and (1) gives $z_N = L - \frac{1}{2}\log L + O(1)$. Hence $Y = 2L - \log L + O(1)$, and

$$\frac{1}{8z_N} = \frac{1}{8L} + \frac{L - z_N}{8Lz_N} = \frac{1}{8L} + O\Big(\frac{\log L}{L^2}\Big), \qquad \frac{1}{Y} = \frac{1}{2L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 2: the equation for Y. We double (1) and add $\log 2$. The left side becomes $2z_N + \log z_N + \log 2 = Y + \log Y$. On the right, $2c_0 + \log 2 = \log\frac{\pi}{4}$, and $\frac{1}{4z_N} = \frac{1}{4L} + O(\log L/L^2)$ by Step 1. So $Y + \log Y = 2L + \log\frac{\pi}{4} + \frac{1}{4L} + O(\log L/L^2)$.

Step 3: the value $G(Y - \ell)$. We have $\log(Y - \ell) = \log Y + \log(1 - \ell/Y)$, and $\log(1 - \ell/Y) = -\ell/(2L) + O(\log L/L^2)$ by Step 1. With Step 2 and $\log\frac{\pi}{4} - \ell = -\log 8$ this gives

$$G(Y - \ell) = Y + \log Y - \ell + \log\Big(1 - \frac{\ell}{Y}\Big) = \Lambda_N + \frac{1}{4L} - \frac{\ell}{2L} + O\Big(\frac{\log L}{L^2}\Big) = \Lambda_N - \frac{c_*}{L} + O\Big(\frac{\log L}{L^2}\Big).$$

Step 4: compare with x_N. By Theorem 2.7(a), $G(x_N) = \Lambda_N + O(L^2/N)$. So $G(x_N) - G(Y - \ell) = c_*/L + O(\log L/L^2)$. Both x_N and $Y - \ell = 2z_N - \log(2\pi)$ are $2L + O(\log L)$, so $G'(x) = 1 + 1/x = 1 + O(1/L)$ between them. By the mean value theorem we may divide by G', and we get $x_N - (Y - \ell) = c_*/L + O(\log L/L^2)$. □

We computed z_N from the exact law of Kagey's board [2]. The difference $D(N)$ changes sign between $N = 400$ and $N = 1600$, and $D(N)\log N$ is 0.707 at $N = 51200$ against the limit $c_* = 0.669$ (Table 1). So the asymptotic regime is reached slowly.

2.4 The crossing to second order

Theorem 2.6 gives $\log C_N$ up to an error $O(\varepsilon \log N)$. A lower bound that matches part (b) of that theorem brings the error down to $O(\varepsilon)$.

Proposition 2.9 (the center to second order)**.** *Let $N \ge 64$ be even and $0 < \varepsilon(1 + \log N) \le \frac{1}{100}$. Then*

$$-11\varepsilon - \varepsilon^2(84\log N + 1.05\cdot 10^5) \le \log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \le 5200\,\varepsilon.$$

In particular $\big|\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N\big| \le 5200\,\varepsilon$ there.

In words, C_N is $\frac{4\varepsilon}{N+2}N^{2\varepsilon}$ times a factor $1 + O(\varepsilon)$.

Idea of the proof. The proof is in Appendix A. The upper bound is Theorem 2.6(b). For the lower bound we compare M after the first flip with Pólya's urn conditioned to end at h, which we call the bridge. Then $V_{t+1}(1)/\pi_t$ is the mean over the bridge of a product of factors $1+O(\varepsilon)$, one for each step (Proposition A.5). By Jensen's inequality the mean of the product is at least the exponential of the mean of the sum of the logarithms. The step at time u adds about $2\varepsilon/u$ to this mean, and the sum over $u \le N$ is about $2\varepsilon \log N$.

With this bound the proof of Theorem 2.7 gives one more term of the crossing.

Proposition 2.10 (the crossing to second order)**.** *As $N \to \infty$ over even N,*

$$x_N = g_N - \frac{2g_N \log N + g_N^2/2}{N(1+1/g_N)} + O\Big(\frac{\log N}{N}\Big).$$

So $x_N < g_N$ for large N.

In words, the root g_N overshoots the crossing by about $6\log^2 N/N$.

Proof. We repeat the proof of Theorem 2.7(a) with one more term on each side. The center gains the term $2\varepsilon \log N$ and the end gains the term $-x^2/(2N)$, and both push F up.

Step 1: both sides. Let F be as in the proof of Theorem 2.7, and let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. For large N Proposition 2.9 applies, and its error $5200\,\varepsilon$ is $O(x/N)$, as is the term $\log(1+2/N)$ in $\log \frac{4\varepsilon}{N+2}$. For the end we use $\log(1-y) = -y - \frac{y^2}{2} + O(y^3)$ for $0 \le y \le \frac{1}{2}$ with $y = x/N$. So

$$\log C_N = \log x + \log 4 - 2L + \frac{2xL}{N} + O\Big(\frac{x}{N}\Big), \qquad \log E_N = -\log 2 - x - \frac{x^2}{2N} + O\Big(\frac{x}{N}\Big).$$

In the second formula we multiplied the expansion by $N-1$, and the terms left over are $O(x/N)$ because x^2/N is bounded for $x \le 4L$. Subtracting gives

$$F(x) = G(x) - \Lambda_N + \frac{2xL + x^2/2}{N} + O\Big(\frac{x}{N}\Big) \qquad (1 \le x \le 2\Lambda_N).$$

Step 2: solve for x_N. By Theorem 2.7, x_N lies in this range for large N. So $F(x_N) = 0$ gives $G(x_N) = \Lambda_N - \eta_N$ with $\eta_N = (2x_N L + x_N^2/2)/N + O(L/N)$. Since $G(g_N) = \Lambda_N$, the mean value theorem gives $x_N - g_N = -\eta_N/G'(\xi)$ for some ξ between x_N and g_N. Here $G'(\xi) = 1 + 1/\xi = (1+1/g_N)(1+O(1/N))$, since $|\xi - g_N| = O(L^2/N)$ and $g_N \ge L$ for large N. As $\eta_N = O(L^2/N)$, we get $x_N - g_N = -\eta_N/(1+1/g_N) + O(L^2/N^2)$. Replacing x_N by g_N in the main term of η_N changes it by $O(L^3/N^2)$, again because $|x_N - g_N| = O(L^2/N)$. All these errors are $O(L/N)$, which gives the formula. Its main term is of order L^2/N and is larger than the error for large N, so $x_N < g_N$. □

Proposition 2.9 does not identify the term of order ε. For fixed N and small ε we can find it. By Proposition 2.5(b), C_N is a polynomial in ε, and we write $C_N = c_1\varepsilon + c_2\varepsilon^2 + O(\varepsilon^3)$ as $\varepsilon \to 0$ and $k_2(N) = c_2/c_1$.

Proposition 2.11 (the coefficient of ε^2)**.** *As $N \to \infty$ over even N,*

$$k_2(N) - 2\log N \to 2\gamma - 2 - \log 2 = -1.5387\ldots$$

In words, for fixed large N and small ε the factor $1+O(\varepsilon)$ in Proposition 2.9 is about $1 - 1.54\,\varepsilon$.

Idea of the proof. The proof is in Appendix A. Differentiating the bridge formula at $\varepsilon = 0$ gives $k_2(N) = \sum_t w_t[A_t - (t-1)]$, where A_t is the mean over the bridge of an explicit sum (Lemma A.8). For fixed t and large N the bridge is close to a walk with fair steps, and the sum can be computed in closed form. It is $2\log N + 2(\gamma - 1 - \log 2) + (t-2)H_{t-1} + o(1)$. The weights w_t tend to $t2^{-t-1}$, and the series over t sums to $2\gamma - 2 - \log 2$.

Remark 2.12 (numerical values and open points). From the Taylor coefficients of C_N, computed in double precision (`verify_elephant_crossing.py`), we get $k_2(N) - 2\log N = -1.585878$, -1.546572 and -1.541005 at $N = 1600$, 12800 and 51200. The formula of Proposition 2.10 without its error term is within 0.07 of x_N already at $N = 400$ (Table 1). We do not know an upper bound $\log C_N \le \log(c_1\varepsilon) + \varepsilon k_2(N) + O(\varepsilon^2(\log N)^k)$, or an explicit N beyond which $g_N > x_N$.

3 The elephant walk near its crossing

We now look at the whole law P_N near the crossing (Figure 2). The picture is that of the introduction. Each flip turns a whole family of steps, most families are small, and the law falls like $x/(2m^2)$ from each end. Lemmas 3.1 to 3.4, the results of Section 3.2 and Theorem 3.9 hold as stated. Elsewhere in this section and in Appendices B and C we assume, unless we say otherwise, that

$$N \ge 64, \qquad 0 < \varepsilon \le \tfrac{1}{4}, \qquad x = N\varepsilon \ge 1.$$

Recall that $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$. Some numerical statements also use odd N, and there x_N is the value of x at which the two central bins are as likely as the ends (it is unique by Proposition 2.3). For a mode $a^* \ge N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N.

3.1 Families and the first flip

For $k_0 \ge 1$ let $(M_k^{[k_0]})_{k \ge k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M. It is the count of disagreeing steps for a walk whose first k_0 steps are forced to agree. So $M^{[1]}$ has the law of M. We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \ge 2$ and $t \ge j-1$. In a random recursive tree on $[N] = \{1, \dots, N\}$, vertex $k+1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \dots, N-1$. We write $\mathcal{T}_v$ for the subtree of v, that is, v and its descendants, and we call it the family of v.

Lemma 3.1 (tree and marks)**.** *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k+1$ is U_k. For $2 \le v \le N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \ge k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k, so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k, $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \cdots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \ge k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. □

So a single flip at v turns every step of its family. Kürsten's coupling [6] uses bond percolation with retention probability $1 - 2\varepsilon$ instead, which gives the same law of the walk. For $k \ge v$, if

$\mathcal{T}_v$ contains s of the vertices in $[k]$, then $k+1$ joins $\mathcal{T}_v$ with probability s/k. This is the urn of Lemma 2.4, so $|\mathcal{T}_v|$ has the law β_v and mean N/v, and on average there are $N/(d(d+1))$ families of size d. If v is the only marked vertex, the odd vertices are those of $\mathcal{T}_v$. Two or more marks have probability $O(\varepsilon^2)$ at fixed N, and no mark gives $M_N = 0$. So at fixed N the part of $f_N(m)$ of first order in ε is exactly $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$, for $1 \le m \le N-1$ (at $m = h$ this is $c_1\varepsilon$ of Proposition 2.5). This is also proved in Lean (Section 6).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting)**.** *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \le v \le N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S, the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v, which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \ge 2$ has its parent outside S. So, given this event, each element of S other than v picks its parent freely among the smaller elements of S, v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. □

Lemma 3.3 (the first flip)**.** *Let $V = \min\{v \ge 2 : \mu_v = 1\}$, with $V = \infty$ if there is no mark, and let $D = |\mathcal{T}_V|$. Then $\mathbb{P}(V = j,\, D = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and for $m \ge 1$*

$$f_N(m) = \sum_{j=2}^{N} \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d)\, \mathbb{P}\big(d - Y_d + Z_{N-d}^{(j)} = m\big),$$

where Y_d and $Z_{N-d}^{(j)}$ are independent.

Proof. Marks and tree are independent, which gives the law of (V, D). Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \dots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \dots, j-1$ carry no mark, so j is odd, and $u \in S$ is odd exactly when the path from u up to j, with j excluded, has an even number of marks. So by Lemma 3.1 for the tree on S rooted at j, the number of odd vertices in S has the law of $d - Y_d$. Paths from outside S stay outside S, and the tree on $[N] \setminus S$ has the unmarked vertices $1, \dots, j-1$ first. So by Lemma 3.1 its odd count has the law of $Z_{N-d}^{(j)}$. The two counts are independent, and their law depends on S only through d. Sum over j and d, and note that $M_N = 0$ if there is no mark. □

In words, the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j-1$ steps agree with X_1. Only the law of each pair $(Y_d, Z_{N-d}^{(j)})$ enters the lemma. So from now on $(Y_d)_{d \ge 1}$ is one path of $M^{[1]}$ and $(Z_t^{(j)})_{t \ge j-1}$ one path of $M^{[j-1]}$, independent of each other and of (V, D).

Lemma 3.4 (unimodality given the first step)**.** *The function f_N is unimodal.*

This is shown in the proof of Theorem 1.1 of Guérin, Laulin, Raschel and Simon [10], for every N and every memory parameter $1-\varepsilon$, and it is also proved in Lean (Section 6). So f_N is nondecreasing on $\{0, \dots, m_h\}$ and nonincreasing on $\{m_h, \dots, N\}$. With a fair first step, P_N mixes f_N and its mirror image. When the mode of f_N is far from the center, the mixture can have two humps, one near each end.

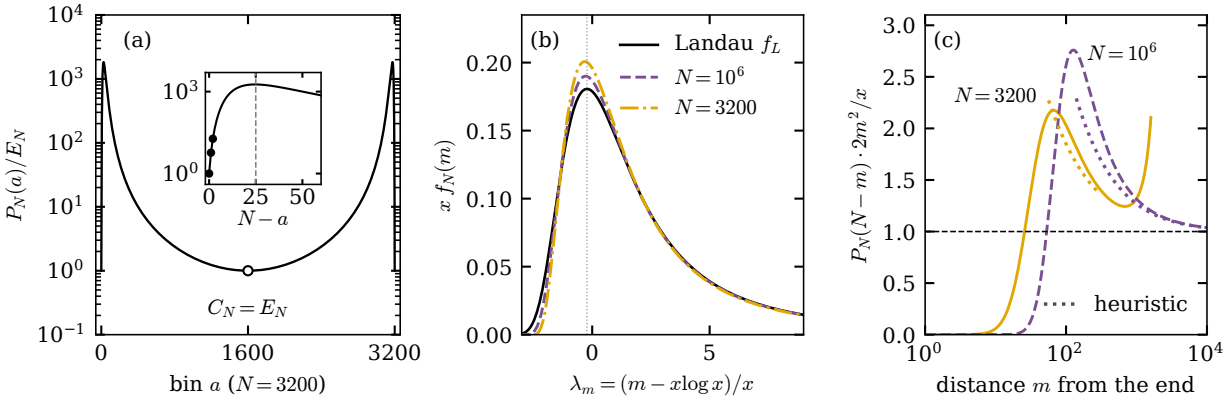


Figure 2: The elephant walk at its crossing. (a) The law P_N at $N = 3200$ and its crossing $x = 11.5468$, divided by the end E_N. The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins N, $N-1$ and $N-2$. The end is below its neighbor, and the modes are 25 steps from each end (dashed). (b) The law of M_N, scaled as $x f_N(m)$, against $\lambda_m = (m - x\log x)/x$, with the Landau density f_L. The dotted line is its mode λ_0. (c) The tail $P_N(N-m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of Theorem 3.14, and the dotted curves are the heuristic correction $1 + x(2\log m + 2\gamma - 3)/m$ for $2x\log x \le m \le N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N-m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

3.2 The end and its neighbor

We first say why the end should fall below its neighbor once $x > 2$. Given $X_1 = +1$, the end needs no flip at all. Its neighbor, bin $N-1$, needs one flip that is never copied. A flip at step j is never copied with probability about j/N. Summing $\varepsilon j/N$ over j gives a factor of about $x/2$ in favor of the neighbor.

Theorem 3.5 (endpoint dip)**.** *Let $N \ge 3$ and $0 < \varepsilon \le \frac{1}{2}$, and put $\rho_1 = P_N(N-1)/E_N$. Then*

$$\frac{x}{2(1-\varepsilon)} < \rho_1 \le \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \ge 2(1-\varepsilon)$, and strictly above it if the right side is below 1.

In words, when ε is small the neighbor of the end is about $x/2$ times as likely as the end.

Proof. By Lemma 2.2, $\rho_1 = \big(f_N(1) + f_N(N-1)\big)/f_N(0)$ with $f_N(0) = (1-\varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up exactly once, say from time $j-1$ to time j with $2 \le j \le N$. Before that it stays at 0 for $j-2$ steps, each with probability $1 - r_t(0) = 1 - \varepsilon$. The move up has probability ε. After it the chain stays at 1 at the times $k = j, \dots, N-1$, each with probability $1 - r_k(1) = (1-\varepsilon)(1-c/k)$, where $c = (1-2\varepsilon)/(1-\varepsilon) \in [0,1]$. Multiplying and summing over j gives

$$f_N(1) = \varepsilon(1-\varepsilon)^{N-2} \sum_{j=2}^{N} P_j, \qquad P_j = \prod_{k=j}^{N-1} \Big(1 - \frac{c}{k}\Big).$$

For the lower bound we use $c \le 1$. Then $P_j \ge \prod_{k=j}^{N-1}(1-1/k) = (j-1)/(N-1)$, and these lower bounds sum to $N/2$ over $2 \le j \le N$. So $f_N(1)/f_N(0) \ge x/(2(1-\varepsilon))$. For the upper bound, $1 - c/k \le e^{-c/k}$ and $\sum_{k=j}^{N-1} 1/k \ge \log(N/j)$ give $P_j \le (j/N)^c$. As $(t/N)^c$ is nondecreasing in t, we can compare the sum with an integral, namely

$$\sum_{j=2}^{N} P_j \le \int_2^{N+1} \Big(\frac{t}{N}\Big)^c dt \le \frac{N(1+1/N)^{c+1}}{c+1} \le \frac{N(1+1/N)^2(1-\varepsilon)}{2-3\varepsilon},$$

using $c \le 1$ and $1/(c+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \le x(1+1/N)^2/(2-3\varepsilon)$.

The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \ge 2$ has probability $r_k(k-1) = 1 - \varepsilon - (1-2\varepsilon)/k \in [\frac{1}{2}, 1-\varepsilon]$. So $0 < f_N(N-1) \le \varepsilon(1-\varepsilon)^{N-2}$. This makes the lower bound strict and gives the term $\varepsilon/(1-\varepsilon)$ in the upper bound. □

Corollary 3.6 (the dip at the crossing)**.** *Let $N \ge 4$ be even and suppose that $x_N \ge 2(1-\varepsilon_N)$. Then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. By Proposition 2.3, $\varepsilon_N < \frac{1}{2}$, so Theorem 3.5 gives $P_N(N-1) > E_N = C_N$. Suppose that P_N is nondecreasing on $\{0, \dots, a_0\}$ and nonincreasing on $\{a_0, \dots, N\}$. It is symmetric about h, so we may replace a_0 by $N - a_0$ and assume $a_0 \le h$. Then $P_N(a) \le C_N$ for every $a \ge h$, including $a = N-1$, which is a contradiction. □

The hypothesis $x_N \ge 2(1-\varepsilon_N)$ says that $\varepsilon_N \ge 2/(N+2)$. By Proposition 2.3 it holds exactly when $C_N \le E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \ge 8$. For $N \ge 1688$ we use Theorem 2.6(b). For smaller N that theorem does not apply, and we compute C_N/E_N in floating-point arithmetic and bound the rounding error. At $\varepsilon = 2/(N+2)$ the transition probabilities of M are ratios of integers, namely

$$1 - r_t(b) = \frac{N(t-b)+2b}{(N+2)t}, \qquad r_t(b) = \frac{2(t-b)+Nb}{(N+2)t}. \tag{2}$$

By Lemma 2.2, $C_N/E_N = 2f_N(h)/f_N(0)$. Write $f_t(b) = \mathbb{P}_+(M_t = b)$ for the law of M_t at this ε, for $1 \le t \le N$. The law of M_{t+1} follows from the law of M_t by

$$f_{t+1}(b) = f_t(b)\big(1 - r_t(b)\big) + f_t(b-1)\, r_t(b-1), \qquad f_1(0) = 1. \tag{3}$$

Here $f_t(b) > 0$ exactly for $0 \le b \le t-1$. So the first term is absent for $b = t$ and the second for $b = 0$. The values $f_{t+1}(b)$ with $b \le h$ only need the values $f_t(b)$ with $b \le h$. Every number in (3) is positive, and nothing is subtracted. This is what makes the rounding error easy to bound.

Lemma 3.7 (rounding error)**.** *Let $N \ge 2$ be even with $N(N+2) < 2^{53}$, let $\varepsilon = 2/(N+2)$, and put $u = 2^{-53}$. Compute numbers $\hat f_t(b)$ for $1 \le t \le N$ and $0 \le b \le \min(t-1, h)$ in IEEE 754 double precision with rounding to nearest, as follows. Put $\hat f_1(0) = 1$. In (3) each weight is one division of the two integers in (2), each term on the right is one multiplication of a computed value $\hat f_t$ by a computed weight, and if both terms are present they are added by one addition. Finally let $\hat R$ be twice the computed quotient of $\hat f_N(h)$ by $\hat f_N(0)$. Suppose that no computed weight, product or sum in the recursion is below 2^{-1000}. Then*

$$(1-u)^{6N-5} \le \frac{C_N/E_N}{\hat R} \le (1-u)^{-(6N-5)}.$$

For $N < 1688$ the right side is at most $1 + 3 \cdot 10^{-12}$.

In words, for $N < 1688$ the computed ratio is correct to about 11 digits, because each of the $N-1$ steps of the recursion rounds each term three times and all terms are positive.

Proof. We use the standard model of floating-point arithmetic [27, Section 2.2]. If a and b are floating-point numbers and $\circ$ is addition, multiplication or division, then the computed value of $a \circ b$ is $(a \circ b)(1+\delta)$ with $|\delta| \le u$, as long as the result does not overflow or underflow. Underflow means that the exact result is nonzero and below the smallest normal number 2^{-1022}, and then the rounded result is at most 2^{-1022}. So a computed result that is at least 2^{-1000} did not underflow, and by hypothesis this covers every operation in the recursion. The integers in (2) are positive and at most $N(N+2) < 2^{53}$, so they are floating-point numbers, and each computed weight is the true weight times some $1+\delta$.

We claim that $\hat{f}_t(b) = f_t(b)(1+\vartheta)$ with $(1-u)^{3(t-1)} \le 1+\vartheta \le (1+u)^{3(t-1)}$. We use induction on t, and the case $t=1$ is clear. In the step from t to $t+1$, each term on the right of (3) is computed as $\hat{f}_t(b')$ times the true weight times $(1+\delta_1)(1+\delta_2)$, and the sum of two terms gets one more factor $1+\delta_3$. All numbers are positive. So $\hat{f}_{t+1}(b)$ lies between $(1-u)^3$ and $(1+u)^3$ times the right side of (3) with $\hat{f}_t$ in place of f_t, and the claim follows. Nothing overflows on the way, because the true weights and the $f_t(b)$ are at most 1 and $(1+u)^{3N} < 2$.

In the last division both numbers lie between 2^{-1000} and 2. So the quotient lies between 2^{-1001} and 2^{1001}, it neither underflows nor overflows, and the division adds one more factor $1+\delta$. Doubling is exact. Hence $(C_N/E_N)/\hat{R}$ lies between $(1-u)^{3(N-1)}(1+u)^{-3(N-1)-1}$ and $(1+u)^{3(N-1)}(1-u)^{-3(N-1)-1}$. Since $1+u \le (1-u)^{-1}$, this gives both bounds with the exponent $6(N-1)+1 = 6N-5$. For $N \le 1686$ put $k = 6N-5 \le 10111$. Then $(1-u)^{-k} \le (1-ku)^{-1} \le 1+2ku \le 1+3\cdot 10^{-12}$. □

Proposition 3.8 (the hypothesis holds from $N=8$ on)**.** *For every even $N \ge 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \ge 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2$, 4 and 6 the hypothesis of Corollary 3.6 fails.*

In words, from $N=8$ on the end is below its neighbor at the crossing, so the law has a hump on each side of the center.

Proof. The range $N \ge 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1+\log N) \le \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \le \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4\cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \ge -y-y^2$ for $0 \le y \le \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} \ge \frac{1}{2}e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \le N < 1688$. The script `certify_elephant.py` runs the computation of Lemma 3.7 for each of the 840 even N in this range, with exactly the operations listed there (elementwise operations on arrays of doubles, with no fused operation, and for a sample of N the same bits come out of plain scalar arithmetic). The smallest weight, product or sum that occurs is above 10^{-9}, so the lemma applies. The largest computed ratio is $\hat{R} = 0.947307\ldots$, at $N = 8$, and the next largest is $0.729947\ldots$, at $N = 10$. By Lemma 3.7, $C_N/E_N \le 0.9474$ for every even $8 \le N < 1688$.

The three smallest N. At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2$, 4 and 6, so $\varepsilon_N < 2/(N+2)$ there. The last claim of the proposition about $N \ge 8$ is Corollary 3.6. □

As an independent check, the same script runs (3) in exact integer arithmetic for 20 values of N between 8 and 1686. In each case $C_N < E_N$ exactly, and the double-precision ratio differs from

the exact one by a relative error below 10^{-13}. At $N = 8$ the exact ratio is $\frac{67903}{71680}$. By Lemma 3.4, P_N is always a mixture of two unimodal laws (namely f_N and its mirror image), as is the limit law in [10, eq. (1)].

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Let CP_N be the law of $\sum_{d=1}^{N-1} d\,\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. On average $x/(d(d+1))$ flipped vertices have a family of size d. If no flipped step has a flipped descendant, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum.

The law CP_N is a known one. Without the truncation at $d = N-1$ its generating function is $(1-s)^{x(1-s)/s}$, which is the Lea–Coulson form of the Luria–Delbrück distribution with parameter x [15, Eq. (14)]. Its limit after centering at $x \log x$ and dividing by x is known as well (see [15, Eq. (19)] and [16]), and the same stable law appears for the root cluster of a random recursive tree [14]. So the limit law in the two theorems below is not new. What they add is the comparison with the elephant walk in sup norm, with an explicit error in Theorem 3.9 and an absolute constant in Theorem 3.10. Numerical values of f_L and of its mode are standard (see Kölbig and Schorr [36]). What Lemma 3.11 adds is a certified enclosure.

Theorem 3.9 (compound Poisson bulk)**.** *For $N \geq 2$ and $0 < \varepsilon \leq \frac{1}{4}$,*

$$\sup_{m \geq 0} \bigl| f_N(m) - \mathrm{CP}_N(m) \bigr| \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}.$$

In words, bin by bin the law of the number of disagreeing steps is within δ_B of the compound Poisson law, and $\delta_B = O(x^2 \log^2 N/N)$ when $x^2 \leq N$.

Idea of the proof. The proof is in Appendix C. Let Z_+ be the total size of the flipped families. Then $M_N = Z_+$ unless some flipped step has a flipped descendant, and this has probability at most $x^2 \log N/N$. Given the tree, Z_+ is a sum of independent terms, so its characteristic function is a product, and the product is close to $e^{\varepsilon F}$ with $F = \sum_v (e^{i\theta |\mathcal{T}_v|} - 1)$. The mean of εF is exactly the logarithm of the characteristic function of CP_N. So it remains to show that F is close to its mean. For this we use the Efron–Stein inequality [25]. Changing the parent of one vertex changes the family sizes only along two paths to the root, and these paths have length of order $\log N$.

The families of size up to about x add up to nearly $x \log x$, and the few families of size of order x give fluctuations of order x. So we center M_N at $x \log x$ and divide by x. Put $\lambda_m = (m - x \log x)/x$, and let

$$f_L(\lambda) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\lambda t} e^{B(t)}\, dt, \qquad B(t) = -it \log|t| - \frac{\pi}{2}|t|.$$

This is the Landau density, a stable density with index one. We show in Appendix C that e^B is the characteristic function of a probability law of class L, so that f_L is a unimodal probability density by [24].

Theorem 3.10 (Landau limit)**.** *There is an absolute constant C such that the following hold under the standing assumptions of this section.*

(a) For every integer $m \geq 0$, $\bigl| x\,\mathrm{CP}_N(m) - f_L(\lambda_m) \bigr| \leq C\bigl(\frac{1+\log x}{x} + \frac{x}{N}\bigr)$.

(b) *For every integer* $m \geq 0$, $\left|x f_N(m) - f_L(\lambda_m)\right| \leq C\left(\frac{1+\log x}{x} + \frac{x}{N}\right) + x\delta_B$.

(c) *If* $x \to \infty$ *and* $x^3 \log^2 N/N \to 0$, *then* $(M_N - x\log x)/x$ *converges in law to the law with density* f_L. *Both conditions hold at the crossing.*

In words, near the crossing the number of disagreeing steps is about $x \log x$, with fluctuations of order x that follow the Landau law, bin by bin.

Idea of the proof. The proof is in Appendix C. If we allow every jump size $d \geq 1$, the characteristic function of the compound Poisson law is $e^{x\chi(e^{i\theta})}$ with $\chi(s) = (1-s)\log(1-s)/s$, and the jumps $d \geq N$ change it little. For small θ we have $\chi(e^{i\theta}) = -i\theta\log(-i\theta) + O(\theta^2|\log|\theta||)$. With $\theta = t/x$ the exponent becomes $B(t) + it\log x$ plus an error of order $t^2 \log x/x$. The term $it\log x$ is the centering at $x\log x$. Both characteristic functions decay like $e^{-c|t|}$, so Fourier inversion turns this into a bound for each bin. Part (b) then follows from Theorem 3.9.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 3.11 (the mode of the Landau density)**.** *(a) The density* f_L *has exactly one mode* λ_0, *and* $-0.22278299 < \lambda_0 < -0.22278297$.

(b) $f_L'(\lambda_0) = 0$, *and* $-0.0792 \leq f_L''(\lambda) \leq -0.0786$ *for* $-0.2229 \leq \lambda \leq -0.2227$.

(c) *There is a constant* $\kappa_L > 0$ *with* $f_L(\lambda_0) - f_L(\lambda) \geq \kappa_L \min\{1, (\lambda - \lambda_0)^2\}$ *for every real* λ.

The proof is in Appendix C. It is computer-assisted. We turn the Fourier integral into the real integral $f_L(\lambda) = \frac{1}{\pi}\int_0^\infty e^{-s\log s - \lambda s}\sin(\pi s)\,ds$. Then we enclose f_L' at two points and f_L'' on the interval between them in ball arithmetic, with the rigorous integration algorithm of [28] on $[2^{-20}, 30]$ and explicit bounds for the rest (`certify_elephant.py`). Part (c) follows from (b) and the unimodality of f_L.

Corollary 3.12 (where the modes are)**.** *Let* $N \to \infty$ *over even* N. *At the crossing, with* $x = x_N$, *every mode* $a^* \geq N/2$ *of* P_N *has*

$$m^* = x\log x + \lambda_0 x + O\big(\sqrt{x\log x}\big),$$

and so has the largest mode m_h *of* f_N.

In words, each hump of the law at the crossing sits at distance about $x\log x - 0.2228\,x$ from its end. The proof is in Appendix C. It compares f_N at a mode with f_N at the integer nearest to $x\log x + \lambda_0 x$, and uses Theorem 3.10(b) and Lemma 3.11(c).

Remark 3.13 (how good the bulk approximation is). The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\mathrm{CP}_N(m) - 1| \leq 3.6 \cdot 10^{-4}$ for $1 \leq m \leq 10^4$. At every tested crossing m^* equals the mode of CP_N. The error term in Corollary 3.12 is not effective. Its proof needs the bound of Theorem 3.10(b) to be below $\kappa_L/2$, where κ_L is the constant of Lemma 3.11(c), and numerically the best κ_L is about 0.026. These numbers are computed in double precision and are not used in any proof.

3.4 The tail

Theorem 3.14 (the tail)**.** *Let N be even, and let $x = N\varepsilon \geq 1$ depend on N with $x \log^2 N/N \to 0$. Let $\omega_N \to \infty$ with $\omega_N x \log^2(x+1) = o(N)$, and let $\delta'_N \to 0$. Then*

$$f_N(m) = \frac{x}{m(m+1)}\big(1 + o(1)\big) \qquad \text{and} \qquad P_N(N-m) = \frac{x}{2m^2}\big(1 + o(1)\big),$$

uniformly for $\omega_N x \log^2(x+1) \leq m \leq \delta'_N N$.

In words, the chance of landing at distance m from an end is the chance that a flipped family has m steps, halved because the first step must point to that end.

Idea of the proof. The proof is in Appendix B. For the lower bound we use the first flipped family, which has size m with probability about $x/(m(m+1))$ (Lemma B.4), while the other flips change the count only by a relative amount of order $x \log m/m$. For the upper bound we show $\mathbb{P}(M_N \geq m) \leq \frac{x}{m}(1 + O(x \log^2 m/m))$, and the unimodality of f_N turns this into a local bound. The mirror term $f_N(N-m)$ is at most $C_N = O(x/N^2)$, which gives the constant $\frac{1}{2}$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the exact ratio $P_N(N-m)/(x/m^2)$ is 0.51889. At moderate m, CP_N suggests $f_N(m)\, m(m+1)/x \approx 1 + x(2\log m + 2\gamma - 3)/m$, and this factor is still about 2 near $m = 70$ when $N = 1600$ (Figure 2(c)).

3.5 The center

Theorem 3.15 (shape at the crossing)**.** *There are absolute constants C_{w} and N_0 with the following property. Let $N \geq N_0$ be even with $x_N \in [2, N^{1/4}]$. Then at the crossing $P_N(a) > C_N$ for every $1 \leq a \leq N-1$ with $|a - h| \geq C_{\mathrm{w}}(N x_N \log N)^{1/2}$.*

By Theorem 2.7, $x_N \in [2, N^{1/4}]$ for large N, so at the crossing the window has width of order $N^{1/2} \log N$. In words, the middle bin is lower than every interior bin except possibly those within about $N^{1/2} \log N$ of it.

Idea of the proof. Up to the mode of f_N the law is above $f_N(1)/2$, which beats $C_N = E_N$ by Theorem 3.5. Farther in, it is close to $x/(2m^2)$, far above the center $4x/N^2$. Near the middle, the main term $x\Pi(m)$ below exceeds its value at h by a factor of about $1 + 12t^2/N^2$ at distance t from h, while we know C_N and the law only up to relative errors of order $x \log N/N$. The two meet at t of order $(Nx \log N)^{1/2}$.

Proof. Write $x = x_N$ and $\varepsilon = \varepsilon_N$, so that $C_N = E_N$. By symmetry we take $a = N - m$ with $1 \leq m \leq h$, and put $t = h - m$. We compare $P_N(N-m)$ with C_N in three ranges of m. In the first two a crude lower bound is enough. In the third, near the center, we need the comparison with $x\Pi(m)$ and a matching upper bound for C_N. Since $2 \leq x \leq N^{1/4}$, every error term below is bounded by a function of N alone that tends to 0, and we take N large.

We start with the upper bound for C_N. Put $\Pi(m) = \frac{1}{2}\big[\frac{1}{m(m+1)} + \frac{1}{(N-m)(N-m+1)}\big]$, so that $x\Pi(h) = \frac{4\varepsilon}{N+2}$. As $\varepsilon(1+\log N) \to 0$, Theorem 2.6(b) gives $C_N \leq x\Pi(h)(1+\eta)$ with $\eta = N^{2\varepsilon} e^{5200\varepsilon} - 1$. Put $y = \varepsilon(2\log N + 5200)$, so that $\eta = e^y - 1$. Then $y \leq \frac{1}{2}$ for large N, and $e^y - 1 \leq 2y$ for $0 \leq y \leq \frac{1}{2}$, so $\eta \leq 2y \leq 10^4 \varepsilon \log N$.

Step 1: $m \leq m_h$. Then $f_N(m) \geq f_N(1)$ by Lemma 3.4. The proof of Theorem 3.5 gives $f_N(1) \geq \frac{x}{2(1-\varepsilon)} f_N(0)$, and $f_N(0) = 2E_N = 2C_N$. So $P_N(N-m) \geq \frac{1}{2} f_N(m) \geq \frac{x}{2(1-\varepsilon)} C_N > C_N$, since $x \geq 2$.

Step 2: $m_h < m \le N/8$. Then $f_N(m) \ge f_N(k)$ with $k = \lfloor N/8 \rfloor$, by Lemma 3.4. Proposition B.5(a) at k gives $f_N(k) \ge \frac{x}{k(k+1)}(1 - O(\frac{x \log N}{N}))$. So $P_N(N-m) \ge \frac{32x}{N(N+8)}(1 - o(1))$, which exceeds $\frac{4x}{N(N+2)}(1+\eta) \ge C_N$.

Step 3: $N/8 < m \le h$. Then m and $N-m$ are at least $N/8$, and Proposition B.5 at m and at $N-m$ gives $P_N(N-m) \ge x\Pi(m)(1-\bar{e})$ with $\bar{e} = O(x \log N/N)$. It remains to compare $\Pi(m)$ with $\Pi(h)$. We measure the distance from the center by $s = 2t/\bar{N}$ with $\bar{N} = N+1$, so that $m + \frac{1}{2} = \frac{\bar{N}}{2}(1-s)$ and $N - m + \frac{1}{2} = \frac{\bar{N}}{2}(1+s)$. Since $\frac{1}{n(n+1)} \ge (n+\frac{1}{2})^{-2}$ for every $n \ge 1$, and $\Pi(h) = \frac{1}{h(h+1)} = (\frac{\bar{N}}{2})^{-2}(1-\bar{N}^{-2})^{-1}$, we get

$$\frac{\Pi(m)}{\Pi(h)} \ge \tfrac{1}{2}\big[(1-s)^{-2} + (1+s)^{-2}\big](1-\bar{N}^{-2}) \ge (1+3s^2)(1-\bar{N}^{-2}).$$

The last step holds because $\frac{1}{2}[(1-s)^{-2} + (1+s)^{-2}] = \sum_{i \ge 0}(2i+1)s^{2i}$, and we keep the first two terms of this series. Put $b = \bar{e} + \bar{N}^{-2}$, so that $b \le \frac{1}{2}$ and $(1-\bar{e})(1-\bar{N}^{-2}) \ge 1 - b$. Then $P_N(N-m) \ge x\Pi(h)(1+3s^2)(1-b) \ge x\Pi(h)(1+\frac{3}{2}s^2 - b)$.

Step 4: the window. The bound of Step 3 exceeds $x\Pi(h)(1+\eta) \ge C_N$ as soon as $\frac{3}{2}s^2 > \eta + b$. Now $\eta + b \le A_2 x \log N/N$ for an absolute constant A_2, since $x \ge 2$, and $s \ge t/N$. So the condition holds when $t \ge C_{\mathrm{w}}(Nx \log N)^{1/2}$ with $C_{\mathrm{w}}^2 > 2A_2/3$. □

We did not compute C_{w} or N_0. Inside the window the proof gives nothing, and closing it needs second differences of the law. We study them in Section 3.6, and Corollary 3.25 closes the window for large even N.

Conjecture 3.16 (convexity between the humps)**.** *For every $N \ge 10$, $x = x_N$ and every mode $a^* \ge N/2$ of P_N with $m^* = N - a^*$, the second difference $P_N(a-1) - 2P_N(a) + P_N(a+1)$ is positive for $2m^* + 1 \le a \le N - 2m^* - 1$.*

With the symmetry of the law, the conjecture makes P_N strictly convex on $[2m^*, N-2m^*]$, so h is its unique minimizer there (the two central bins for odd N). The next corollary shows that $m^* = o(N)$ at the crossing, so together with Theorem 3.15 the conjecture would make the center the unique interior minimum for large even N. Section 3.6 proves the conjecture for all large N, even (Theorem 3.24) or odd (Proposition 3.27), and, with a certified computation, for $10 \le N \le 2000$ (Proposition 3.26). For every N it reduces the conjecture to one inequality (Corollary 3.23).

Corollary 3.17 (the modes)**.** *For large even N the modes a^* of P_N at the crossing satisfy* $\min(a^*, N-a^*) = O(x_N \log^2 x_N)$.

Proof. We show that $f_N(m^*)$ is not much below the maximum of f_N, and then that f_N cannot be that large far to the right of its mode. Write $x = x_N$ and $\varepsilon = \varepsilon_N$. By Theorem 2.7, $x \sim 2\log N$, so for large N the hypotheses of Theorem 2.6(b) and Lemma B.7 hold. Let $\ell = \lceil Ax \log(x+2) \rceil$ be as in the proof of Lemma B.7, where it is shown that $f_N(\ell) \ge \frac{x}{2\ell(\ell+1)}$.

Step 1: $f_N(m^*)$ is large. The law is symmetric about h, so $m^* = \min(a^*, N-a^*)$ maximizes $g(m) = f_N(m) + f_N(N-m)$ over $0 \le m \le h$. Lemma B.7 gives $m_h \le h$ for large N. So for $m \le h$ we have $N - m \ge h \ge m_h$, and Lemma 3.4 and Theorem 2.6(b) give $f_N(N-m) \le f_N(h) = C_N = O(x/N^2)$. Hence

$$f_N(m^*) \ge g(m^*) - C_N \ge g(m_h) - C_N \ge f_N(m_h) - C_N \ge f_N(\ell) - C_N \ge \frac{x}{4\ell(\ell+1)}$$

for large N, because $\ell = O(\log^2 N)$ and so C_N is far below x/ℓ^2.

Step 2: m^* is not far beyond ℓ. Put $m' = \max(m_h, \ell)$ and suppose that $m^* > m'$. To the right of m_h the function f_N is nonincreasing, so Lemma B.1 (with $a = m'$ and $b = m^*$) and Lemma B.2 give

$$(m^* - m' + 1)\, f_N(m^*) \le \mathbb{P}(M_N \ge \ell) \le \frac{xH_\ell}{\ell}.$$

With Step 1 this gives $m^* - m' + 1 \le 4(\ell+1)H_\ell$. Hence $m^* \le \max(m_h, \ell) + 4(\ell+1)H_\ell$ in every case. Finally Lemma B.7 gives $m_h = O(x \log^2 x)$, and $\ell = O(x \log x)$ and $H_\ell = O(\log x)$. □

The conjecture is certified for every $10 \le N \le 2000$ (Proposition 3.26), and in double precision it holds for $N = 3200, 6400, 12800, 25600$. It leaves out the two endpoints $a = 2m^*$ and $a = N - 2m^*$, because there the second difference is not positive for some $N \le 51$. Heuristically, a concave stretch of f_N far from its mode is added to the tail $x/(2r^2)$ of the other branch, with $r = N - a$, and the second difference $3x/r^4$ of this tail beats the concave curvature, which is of order $x^2/(N^2 r^3)$. We leave open the tail between $x \log x$ and $x \log^2 x$, effective constants in Theorems 3.14 and 3.15, and the conjecture for the N between 2000 and the constants of Theorem 3.24 and Proposition 3.27.

3.6 Convexity between the humps

Throughout this subsection $0 < \varepsilon < \frac{1}{2}$ and $c = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + cb/t$. For $t \ge 1$ and every integer a we write $P_t(a) = \mathbb{P}(A_t = a)$, which is 0 unless $0 \le a \le t$, and

$$D_t(a) = P_t(a+1) - 2P_t(a) + P_t(a-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \ge 2$.

Lemma 3.18 (recurrence for the second difference)**.** *For every $t \ge 1$ and every integer a,*

$$P_{t+1}(a) = \big(1 - r_t(a)\big) P_t(a) + r_t(a-1)\, P_t(a-1), \tag{4}$$

$$D_{t+1}(a) = \big(1 - r_t(a+1)\big) D_t(a) + r_t(a-2)\, D_t(a-1). \tag{5}$$

Also $P_t(a) = P_t(t-a)$ and $D_t(a) = D_t(t-a)$. The same two recurrences hold with $f_t(m)$ in place of $P_t(a)$.

Proof. Given $A_t = a$, the copied step is $+1$ with probability a/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{a}{t} + \varepsilon(1 - \frac{a}{t}) = r_t(a)$. This gives (4) for $0 \le a \le t+1$, where at $a = 0$ and at $a = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1 - r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other a every term vanishes. The symmetry of P_t is Lemma 2.2, and it gives the symmetry of D_t. The chain M of Lemma 2.2 has the same transition probabilities, so (4) holds for f_t as well.

For (5) put $\varsigma = c/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (4) at $a+1$, a and $a-1$ into the definition of $D_{t+1}(a)$ and collect the coefficients of $P_t(a+1)$, $P_t(a)$, $P_t(a-1)$ and $P_t(a-2)$. They are

$$1 - r_t(a+1), \qquad 3r_t(a) - 2, \qquad 1 - 3r_t(a-1), \qquad r_t(a-2).$$

The right side of (5) has the coefficients $1 - r_t(a+1)$, $-2 + 2r_t(a+1) + r_t(a-2)$, $1 - r_t(a+1) - 2r_t(a-2)$ and $r_t(a-2)$. The second ones agree because $2r_t(a+1) + r_t(a-2) = 3r_t(a)$, and the third ones agree because $r_t(a+1) + 2r_t(a-2) = 3r_t(a-1)$. □

The recurrence (5), and its analogue for differences of any order, is used by Trinas and Valle [11, (4.10)] to bound the law and its differences at a fixed memory parameter. We use it for signs. The coefficients in (5) are positive when $2 \le a \le t-1$. So the second difference moves like a mass that is partly killed, and its sign can change only through what enters at the two edges. The next lemma is the elementary form of the variation diminishing property that we need.

For a finite list z of real numbers let $\mathrm{sp}(z)$ be its sign pattern. We delete the zeros of z, replace each entry by its sign, and merge equal neighbors. So $\mathrm{sp}(z)$ is a list of alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 3.19 (one step)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \le i \le n-1$ with $A_i \ge 0$ and $B_i \ge 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

Proof. We insert the numbers η_i into ζ one at a time. If $\eta_i = 0$, the sign pattern does not change. If $\eta_i > 0$, then $\zeta_i > 0$ or $\zeta_{i+1} > 0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The case $\eta_i < 0$ is the same. This proves the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

Lemma 3.20 (the sequence next to the end)**.** *For every $T \ge 2$ we have $\mathrm{sp}(s_2, s_3, \dots, s_T) \preceq \Pi_3$. Moreover s_3 and s_4 do not both vanish.*

Proof. Put $\rho_t = P_t(1)/P_t(0)$. By (4), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{c}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\rho_{t+1} - \rho_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{c\rho_t}{t}\Big), \qquad \rho_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\rho_t < \varepsilon t/c$ for $t \ge 2$. For $t = 2$ this holds because $c < 1-\varepsilon$. If it holds for t, then $0 < \rho_{t+1} - \rho_t \le \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{c}$, and it holds for $t+1$. So ρ_t is strictly increasing for $t \ge 2$, and $\rho_t < 1$ whenever $t \le c/\varepsilon$.

Now (5) with $a = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \ge 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{c}{t}\Big)(\rho_t - 2)\,P_t(0).$$

Here $1 - r_t(2) \ge 1 - \varepsilon - c = \varepsilon > 0$. If $t < c/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > c/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ρ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \ge \frac{1}{3}$, then $c/\varepsilon \le 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim we compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$, $P_3(0) = P_3(1) = \frac{1}{4}$ and $s_4 = \sigma_3 = (\varepsilon - \frac{c}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \ne \frac{1}{5}$. □

Theorem 3.21 (at most five blocks)**.** *Let $N \geq 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\operatorname{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$. The same holds for the second difference of f_N.

In words, for every $N \geq 3$ and every $0 < \varepsilon < \frac{1}{2}$ the law has at most two concave stretches, one for each hump, and what lies between them is convex.

Proof. For $3 \leq t \leq N$ let Z_t be the list

$$Z_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N - t - 1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list Z_3 is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and Z_N is the list in the statement.

We check that Z_{t+1} is obtained from Z_t as in Lemma 3.19, for $3 \leq t \leq N-1$. The first $N - t$ entries of Z_{t+1} are $s_N, \dots, s_{t+1}$, and they are copies of the first $N - t$ entries of Z_t. So $A_i = 1$ and $B_i = 0$ there. The next $t - 2$ entries are $D_{t+1}(b)$ for $2 \leq b \leq t-1$. By (5),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big) D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of Z_t, at the right places. The coefficients satisfy $r_t(b-2) \geq \varepsilon$ and $1 - r_t(b+1) \geq 1 - \varepsilon - c = \varepsilon$, since $2 \leq b \leq t-1$. The last $N - t$ entries of Z_{t+1} are $s_{t+1}, \dots, s_N$, and each is a copy of the entry of Z_t one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 3.19, $\operatorname{sp}(Z_N) \preceq \operatorname{sp}(Z_{N-1}) \preceq \cdots \preceq \operatorname{sp}(Z_3)$. By Lemma 3.20 the sign pattern p of $(s_3, \dots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of Z_3 is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D_t^f(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$Z_t^f = \big(s_N^f, \dots, s_{t+1}^f,\ s_t^f, D_t^f(2), \dots, D_t^f(t-1),\ d_t, d_{t+1}, \dots, d_{N-1}\big), \qquad 2 \leq t \leq N,$$

where $s_t^f = D_t^f(1)$ and $d_u = (1 - r_u(u+1))D_u^f(u) = (\varepsilon - \frac{c}{u}) f_u(u-1)$. The entry $D_{t+1}^f(t)$ of Z_{t+1}^f equals $r_t(t-2)D_t^f(t-1) + d_t$ by (5), and the other entries are as before. The proof of Lemma 3.20 applies to s_t^f without change, with $\rho_t = f_t(1)/f_t(0)$, $\rho_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s_2^f = 1 - 3\varepsilon$. The numbers d_u are negative for $u < c/\varepsilon$ and positive for $u > c/\varepsilon$. So $\operatorname{sp}(Z_2^f)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $Z_N^f = (D_N^f(1), \dots, D_N^f(N-1))$. □

Theorem 3.22 (no holes)**.** *Let $N \geq 5$ and $0 < \varepsilon < \frac{1}{2}$. Let $1 \leq a_0 < a_1 \leq N/2$ be integers with $D_N(a_0) < 0 < D_N(a_1)$. Then $D_N(a) > 0$ for every $a_1 \leq a \leq N - a_1$.*

In words, once the law has turned convex to the right of its left hump, it stays strictly convex up to the mirror image of that point.

Proof. By symmetry $D_N(N-a_1) > 0 > D_N(N-a_0)$. Suppose that $D_N(a) \le 0$ for some a with $a_1 < a < N - a_1$.

The case $D_N(a) < 0$. Then the entries of Z_N at $a_0 < a_1 < a < N-a_1 < N-a_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \mathrm{sp}(Z_N) \preceq \Pi_5$ by Theorem 3.21. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(a) = 0$. We use the lists Z_t of the proof of Theorem 3.21 and number the entries of Z_t from 1 to $2N-t-1$. So $D_t(b)$ stands at place $N-t+b$ for $1 \le b \le t-1$. For $3 \le t \le N-1$ let W_t be the list w of Lemma 3.19 for $\zeta = Z_t$ and $\eta = Z_{t+1}$. Then $\mathrm{sp}(W_t) = \mathrm{sp}(Z_t) \preceq \Pi_5$. The entry of Z_{t+1} at place k is a combination of the entries of Z_t at the places k and $k+1$, with nonnegative coefficients. We call these two entries its parents. For $2 \le b \le t-1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \le t \le N$ let B_t be the set of integers b with $\max(2, a-N+t) \le b \le \min(a, t-2)$. It is not empty, because $2 \le a \le N-2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in Z_t such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N-t+b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{a\}$, and we take the places a_0, a_1, $N-a_1$ and $N-a_0$.

Let the claim hold for $t+1$, where $4 \le t \le N-1$, with the places $p'_1 < p'_2 < p'_4 < p'_5$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one parent is negative. In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of Z_{t+1} at the places p'_2 and p'_4, and these stand between the negative entries at p'_1 and p'_5. So $(-,+,-,+,-) \preceq \mathrm{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b-1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of Z_{t+1} at p'_1 is negative, so one of its parents is negative. Let $p_1 \in \{p'_1, p'_1+1\}$ be its place in Z_t, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \le p'_1+1 \le p'_2 \le p_2$, and $p_1 \ne p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N-t-1+b$ with $b \in B_{t+1}$. The places $N-t+b$ with $b \in B_t$ lie between q^- and $q^+ + 1$. Since $p_2 \le p'_2 + 1 \le q^-$ and $p_4 \ge p'_4 \ge q^+ + 1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The same argument for the step from Z_5 to Z_4 shows that both parents of $D_5(b)$ vanish for $b \in B_5 \subseteq \{2,3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = 0$. The step from Z_4 to Z_3 shows that both parents of $D_4(2)$ vanish, and both are equal to s_3. So $s_3 = s_4 = 0$, which contradicts Lemma 3.20. □

Corollary 3.23 (one inequality is enough)**.** *Let $N \ge 8$ be even and $x = x_N$, or let $N \ge 5$ and $0 < \varepsilon < \frac{1}{2}$ be such that $P_N(0) < P_N(1)$. Let $a^* \ge N/2$ be a mode of P_N and $m^* = N - a^*$. If $D_N(a_1) > 0$ for some integer a_1 with $m^* < a_1 \le 2m^*+1$ and $a_1 \le N/2$, then $D_N(a) > 0$ for every $2m^*+1 \le a \le N-2m^*-1$. In particular the statement of Conjecture 3.16 holds for this N, this ε and this m^* as soon as $D_N(2m^*+1) > 0$. Note that $2m^*+1 \le N/2$ whenever the range $2m^*+1 \le a \le N-2m^*-1$ is not empty.*

Proof. At the crossing with $N \ge 8$ even, $P_N(0) = E_N < P_N(N-1) = P_N(1)$ by Proposition 3.8. Let a_0 be the smallest maximizer of P_N. Then $1 \le a_0 \le m^*$, because $P_N(0) < P_N(1)$ and $P_N(m^*) = P_N(a^*)$ is the maximum. Also $P_N(a_0-1) < P_N(a_0) \ge P_N(a_0+1)$, so $D_N(a_0) < 0$. Now Theorem 3.22 applies to $a_0 < a_1$. □

Theorem 3.24 (convexity between the humps for large N)**.** *There is N_0 such that Conjecture 3.16 holds for every even $N \ge N_0$. That is, for even $N \ge N_0$, $x = x_N$ and every mode $a^* \ge N/2$ of P_N*

with $m^* = N - a^*$,

$$P_N(a-1) - 2P_N(a) + P_N(a+1) > 0 \qquad \textit{for } 2m^* + 1 \le a \le N - 2m^* - 1.$$

Idea of the proof. By Corollary 3.23 it is enough to find one point of strict convexity between m^* and $2m^* + 1$. If there were none, P_N would be concave on $[m^*, 2m^* + 2]$, and then its value halfway would be at least half of its maximum. The Landau limit shows that the value halfway is much smaller.

Proof. Let N be even and large, and let $x = x_N$ and $\varepsilon = \varepsilon_N$. By Theorem 2.7, $x \to \infty$ and $x = O(\log N)$. So the standing assumptions of Section 3 hold, and $(1 + \log x)/x$, x/N and $x\delta_B = O(x^3 \log^2 N/N)$ tend to 0. So by Theorem 3.10(b) there are numbers $\eta_N \to 0$ with

$$\bigl|xf_N(m) - f_L(\lambda_m)\bigr| \le \eta_N \qquad \text{for every integer } m \ge 0. \tag{6}$$

The density f_L is continuous, because it is the Fourier transform of an integrable function. It is unimodal with mode λ_0 (Lemma 3.11), so it is nonincreasing on $[\lambda_0, \infty)$, and it is integrable. So $f_L(\lambda) \to 0$ as $\lambda \to \infty$. We write $o(1)$ for a quantity that is bounded in absolute value by a function of N alone that tends to 0.

Step 1, the mirror term. Let $0 \le m \le N/2$. Then $\lambda_{N-m} \ge \frac{N}{2x} - \log x \to \infty$. So (6) gives $xf_N(N-m) \le f_L(\lambda_{N-m}) + \eta_N = o(1)$. Since $P_N(m) = P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$, we get

$$2xP_N(m) = f_L(\lambda_m) + o(1) \qquad \text{uniformly for } 0 \le m \le N/2.$$

Step 2, the mode. Let m_0 be the integer part of $x \log x + \lambda_0 x$. Then $0 \le m_0 \le N/2$ and $|\lambda_{m_0} - \lambda_0| \le 1/x$, so Step 1 and the continuity of f_L give $2xP_N(m_0) = f_L(\lambda_0) + o(1)$. Now $P_N(m^*) \ge P_N(m_0)$ and $m^* \le N/2$, so Step 1 at m^* gives $f_L(\lambda_{m^*}) \ge f_L(\lambda_0) - o(1)$. By Lemma 3.11(c) this forces $|\lambda_{m^*} - \lambda_0| \le 1$ for large N. Hence

$$x \log x + (\lambda_0 - 1)x \le m^* \le x \log x + (\lambda_0 + 1)x, \qquad 2xP_N(m^*) \ge f_L(\lambda_0) - o(1).$$

In particular $m^* \to \infty$ and $2m^* + 2 \le N/2$ for large N.

Step 3, halfway. Let $v = m^* + \lfloor m^*/2 \rfloor$. Then $v \ge \frac{3}{2}m^* - 1$, so by Step 2

$$\lambda_v \ge \tfrac{1}{2} \log x + \tfrac{3}{2}(\lambda_0 - 1) - \tfrac{1}{x} \to \infty.$$

Also $v \le N/2$. So Step 1 gives $2xP_N(v) = f_L(\lambda_v) + o(1) = o(1)$. Since $f_L(\lambda_0) > 0$, Step 2 shows that $P_N(v) < \frac{1}{2}P_N(m^*)$ for large N.

Step 4, a point of convexity. Suppose that $D_N(a) \le 0$ for every $m^* < a \le 2m^* + 1$. Then P_N is concave on the integers of $[m^*, 2m^* + 2]$. With $u = m^*$ and $w = 2m^* + 2$ this gives

$$P_N(v) \ge \frac{w-v}{w-u}P_N(u) + \frac{v-u}{w-u}P_N(w) \ge \frac{w-v}{w-u}P_N(m^*) \ge \tfrac{1}{2}P_N(m^*),$$

because $w - v \ge \frac{1}{2}(m^* + 2) = \frac{1}{2}(w - u)$. This contradicts Step 3. So $D_N(a_1) > 0$ for some $m^* < a_1 \le 2m^* + 1$, and $a_1 \le N/2$ by Step 2. Corollary 3.23 gives the claim. □

Corollary 3.25 (the center is the unique interior minimum)**.** *For every large even N, at the crossing, $P_N(a) > C_N$ for every $1 \le a \le N - 1$ with $a \ne h$.*

Proof. Let N be large and even, and let m^* be as in Theorem 3.24. By that theorem and the symmetry of P_N, the first differences $P_N(a+1) - P_N(a)$ are strictly increasing for $2m^* \le a \le N - 2m^* - 1$, and $P_N(h+1) - P_N(h) = \frac{1}{2} D_N(h) > 0$. So P_N is strictly increasing on $[h, N - 2m^*]$ and strictly decreasing on $[2m^*, h]$. By Step 2 of the last proof, $m^* = O(\log N \log\log N)$. So for large N every a with $1 \le a \le N-1$ and $|a - h| < C_{\mathrm{w}}(N x_N \log N)^{1/2}$ lies in $[2m^*, N - 2m^*]$, and the other a are covered by Theorem 3.15. □

Proposition 3.26 (small N, computer-assisted)**.** *Conjecture 3.16 holds for every* $10 \le N \le N_{\mathrm{c}} = 2000$*, even or odd.*

Proof. The script `certify_convex.py` uses ball arithmetic with 160 bits (the library Arb). For each N it encloses ε_N in an interval of relative width at most $2 \cdot 10^{-8}$. It does so by evaluating the center and the end at the two ends of the interval, and Proposition 2.3 then places the crossing between them. It then runs (4) with ε equal to this interval. Each step only multiplies and adds enclosures of numbers in $[0, 1]$, so the radii stay small. From the resulting enclosure of P_N it lists every index that can be a mode, and for each of them it checks that the enclosure of $D_N(a)$ is positive for every $2m^* + 1 \le a \le N - 2m^* - 1$. It also checks that $D_N(a_0) < 0$ for some $a_0 \le 2m^*$ and that $D_N(2m^* + 1) > 0$. By Theorem 3.22 these two inequalities by themselves imply the claim. □

Proposition 3.27 (odd N)**.** *Let $N \ge 3$ be odd, and write $\widetilde{C}_N = P_N(\frac{N-1}{2}) = P_N(\frac{N+1}{2})$ for the two central bins. The ratio $\widetilde{C}_N / E_N$ is strictly increasing in ε, and it equals 1 at exactly one $\varepsilon_N \in (0, \frac{1}{2})$. Put $x_N = N\varepsilon_N$.*

(a) For every $0 < \varepsilon < \frac{1}{2}$,

$$\widetilde{C}_N = \Big(1 - \frac{c}{N}\Big)^{-1} C_{N+1},$$

where C_{N+1} is the center of the walk with $N+1$ steps and the same ε.

(b) As $N \to \infty$ over odd N, $x_N + \log x_N = \Lambda_N + O(L^2/N)$. So both parts of Theorem 2.7 hold for odd N as well.

(c) For every odd $N \ge 1689$ we have $x_N > 2$, and at the crossing $P_N(0) < P_N(1)$. So Corollary 3.23 applies to these N with $x = x_N$.

(d) There is N_0' such that Conjecture 3.16 holds for every odd $N \ge N_0'$.

In words, the center of a walk with an even number of steps is reached from the two central bins one step earlier, and from each of them with the same probability. So the two central bins of an odd N are the center of $N+1$ up to the factor $1 - c/N$, and what we know about the crossing for even N carries over.

Proof. The crossing. By Proposition 2.3, $P_N(k)/E_N$ is strictly increasing in ε for $k = \frac{N+1}{2}$. It tends to 0 as $\varepsilon \to 0$. At $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so $\widetilde{C}_N = \binom{N}{k} 2^{-N} > 2^{-N} = E_N$. So the ratio equals 1 at exactly one ε_N, and $\varepsilon_N < \frac{1}{2}$.

(a) Put $k = \frac{N+1}{2}$. Since $\varepsilon + \frac{c}{2} = \frac{1}{2}$, we have $r_N(k) = \frac{1}{2} + \frac{c}{2N}$ and $r_N(k-1) = \frac{1}{2} - \frac{c}{2N}$. So (4) with $t = N$ and $a = k$ gives

$$C_{N+1} = P_{N+1}(k) = \Big(\frac{1}{2} - \frac{c}{2N}\Big)\big(P_N(k) + P_N(k-1)\big) = \Big(1 - \frac{c}{N}\Big)\widetilde{C}_N.$$

(b) We follow the proof of Theorem 2.7(a), with $\widetilde{F}(x) = \log \widetilde{C}_N - \log E_N$ at $\varepsilon = x/N$ in place of F. Let $1 \le x \le 2\Lambda_N$. For large N, Theorem 2.6 applies to the even number $N+1$ at this ε and gives $\log C_{N+1} = \log \frac{4\varepsilon}{N+3} + O(\varepsilon L)$. By (a), $\log \widetilde{C}_N = \log C_{N+1} + O(1/N)$. Here $\log \frac{4\varepsilon}{N+3} = \log x + \log 4 - 2L + O(1/N)$, and $1/N \le xL/N$. So

$$\log \widetilde{C}_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big),$$

which is the estimate for $\log C_N$ in Step 1 of that proof. The formula $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$ holds for every N (Lemma 2.2). So $(*)$ holds for $\widetilde{F}$, and Step 2 of that proof gives $|x_N - g_N| = O(L^2/N)$ and the claim. Part (b) of Theorem 2.7 follows from part (a) as before.

(c) We argue as in the proof of Proposition 3.8. Let $N \ge 1689$ be odd and take $\varepsilon = 2/N$. Then $\varepsilon(1 + \log(N+1)) \le \frac{1}{100}$, so Theorem 2.6(b) for $N+1$ and part (a) give

$$\widetilde{C}_N \le \frac{N}{N-1}\, C_{N+1} \le \frac{N}{N-1} \cdot \frac{8}{N(N+3)} (N+1)^{4/N} e^{10400/N}.$$

The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1689$. Also $E_N = \frac{1}{2}(1 - \frac{2}{N})^{N-1} > 0.06$, as in that proof. So $\widetilde{C}_N < E_N$ at $\varepsilon = 2/N$, and hence $\varepsilon_N > 2/N$ and $x_N > 2 \ge 2(1-\varepsilon_N)$. Theorem 3.5 then gives $P_N(N-1) > E_N$ at the crossing, which is $P_N(0) < P_N(1)$ by symmetry. This is the hypothesis of the second case of Corollary 3.23.

(d) The proof of Theorem 3.24 uses that N is even in two places only. The first is $x \to \infty$ and $x = O(\log N)$, which holds for odd N by (b). The second is Corollary 3.23 at the end of Step 4, which applies by (c). The rest of that proof uses Theorem 3.10, Lemma 3.11 and $P_N(m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$, and these hold for every N. □

Remark 3.28 (what is still open). The constant N_0 of Theorem 3.24 is not explicit, because the constant in Theorem 3.10 is not. Even with an explicit constant, Step 3 needs $f_L(\frac{3}{2}\lambda_0 + \frac{1}{2}\log x) < \frac{1}{2} f_L(\lambda_0)$ up to the error η_N, and at the sizes we can compute the ratio $P_N(v)/P_N(m^*)$ is still about 0.8. Numerically f_L falls to half of its maximum at $\lambda = 2.432$. So even without the error η_N, Step 3 needs $\frac{3}{2}\lambda_0 + \frac{1}{2}\log x > 2.432$, that is, $x > 253$, and by Theorem 2.7 this is N near 10^{55}. So the N between N_c and the constants N_0 and N_0' of Theorem 3.24 and Proposition 3.27 are open, and no computation can reach these constants as the proof stands. By Corollary 3.23 and Proposition 3.27(c) the whole range would be closed by an effective proof of the single inequality $D_N(a_1) > 0$ at one point $m^* < a_1 \le 2m^* + 1$. Numerically the convex stretch starts near $1.5\, m^*$. This agrees with the Landau limit, because the inflection point of f_L to the right of its mode is at $\lambda = 1.104$ numerically, and $x \log x + 1.104\, x$ is below twice the mode $x \log x + \lambda_0 x$ as soon as $x \ge 4.8$. A proof along these lines would need the second differences of P_N near $2m^*$ with an error below their size, which is of order $1/x^3$, and the error $x\delta_B$ in Theorem 3.10 is far larger than that for the N in question.

4 Heavy-tailed runs

In this section the walk remembers the time since its last switch. The steps form maximal runs whose directions alternate. The first run has length D, and the later runs have i.i.d. lengths $\xi_1, \xi_2, \dots$ distributed as ξ, where

$$\bar{F}(k) = \mathbb{P}(\xi \ge k) = k^{-\alpha} \quad (k \ge 1)$$

for some $\alpha > 0$. So $\mathbb{P}(\xi = k) = k^{-\alpha} - (k+1)^{-\alpha} > 0$ for every $k \geq 1$. The first direction is fair, and D, (ξ_i) and the first direction are independent. With a fresh start D has the law of ξ. For $\alpha > 1$ the stationary start is $\mathbb{P}(D = d) = \bar{F}(d)/\mu$, where $\mu = \mathbb{E}\xi = \zeta(\alpha)$. This law is stationary in the following sense. The number of steps left in the current run, counting the current one, is a Markov chain that moves from $k \geq 2$ to $k-1$ and from 1 to a fresh copy of ξ. Since $\bar{F}(k) = \bar{F}(k+1) + \mathbb{P}(\xi = k)\bar{F}(1)$, the law $\bar{F}(k)/\mu$ is invariant for this chain. So the stationary start sees the runs at a typical time. We define A_N, P_N, C_N and E_N as in Section 1.2, put $R_N = E_N/C_N$, and write $\mathbb{P}_+$ for the law given that the first run is $+$. In this section the constants implied by $O(\cdot)$ may depend on α.

Proposition 4.1 (the edge)**.** *For every law of D, $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$. With a fresh start $E_N = \frac{1}{2}N^{-\alpha}$. With the stationary start*

$$E_N = \frac{\zeta(\alpha, N)}{2\zeta(\alpha)} = \frac{N^{1-\alpha}}{2(\alpha-1)\zeta(\alpha)}\Big(1 + \frac{\theta_N(\alpha-1)}{N}\Big)$$

for some $\theta_N \in [0,1]$, where $\zeta(\alpha, N) = \sum_{k \geq N} k^{-\alpha}$ is the Hurwitz zeta function.

Proof. All N steps are $+1$ exactly when the first direction is $+$ and $D \geq N$. Hence $E_N = \frac{1}{2}\mathbb{P}(D \geq N)$. For the fresh start $\mathbb{P}(D \geq N) = \bar{F}(N) = N^{-\alpha}$. For the stationary start $\mathbb{P}(D \geq N) = \sum_{d \geq N} d^{-\alpha}/\mu = \zeta(\alpha, N)/\zeta(\alpha)$. To estimate $\zeta(\alpha, N)$ we compare the sum with an integral. Since $t^{-\alpha}$ is decreasing,

$$\int_N^\infty t^{-\alpha}\, dt \leq \zeta(\alpha, N) \leq N^{-\alpha} + \int_N^\infty t^{-\alpha}\, dt.$$

The integral equals $N^{1-\alpha}/(\alpha-1)$, and $N^{-\alpha}$ is $(\alpha-1)/N$ times the integral. This gives the second form. □

So the edge only sees the first run, while the center depends on all the runs. For the center we need the renewal quantities

$$\tau_m = \xi_1 + \cdots + \xi_m, \quad \tau_0 = 0, \qquad u_m(k) = \mathbb{P}(\tau_m = k),$$
$$q_m(k) = \mathbb{P}(\tau_{m-1} < k \leq \tau_m) = \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j)$$

for $m, k \geq 1$. So $u_m(k)$ is the chance that the m-th run ends at level k, and $q_m(k)$ the chance that the m-th run covers level k. For the delayed versions let D be independent of (ξ_i), and put $u_m^D(k) = \mathbb{P}(D + \tau_{m-1} = k)$, $q_1^D(k) = \mathbb{P}(D \geq k)$ and $q_m^D(k) = \mathbb{P}(D + \tau_{m-2} < k \leq D + \tau_{m-1})$ for $m \geq 2$. For the stationary start $u_m^D(k) = \sum_d \frac{\bar{F}(d)}{\mu} u_{m-1}(k-d) = q_m(k)/\mu$.

The $+$ runs and the $-$ runs form two independent renewal processes. The walk is in bin a at time N exactly when one of them sits at its level and the other is inside a run that covers its level.

Proposition 4.2 (two renewal processes)**.** *Let D have any law on $\{1, 2, \dots\}$. For $a + b = N$ with $a, b \geq 1$,*

$$\mathbb{P}_+(A_N = a) = \sum_{m \geq 1} \Big[u_m(b)\, q_{m+1}^D(a) + u_m^D(a)\, q_m(b) \Big],$$

and $\mathbb{P}_+(A_N = N) = \mathbb{P}(D \geq N)$. So for even N and $h = N/2$, $C_N = \sum_{m \geq 1}[u_m(h) q_{m+1}^D(h) + u_m^D(h) q_m(h)]$. For the fresh start this is $C_N = \sum_m u_m(h)[q_m(h) + q_{m+1}(h)]$.

Proof. We split according to the run that contains step N. Let the $+$ runs have lengths $D, \xi_2^+, \xi_3^+, \dots$ and the $-$ runs $\xi_1^-, \xi_2^-, \dots$, and let $s_j^\pm$ be the sum of the first j lengths of each sign, with $s_0^\pm = 0$. So s_j^+ has law u_j^D and s_j^- has law u_j, and the two sequences are independent. The runs alternate and the first one is a $+$ run. So the j-th $+$ run comes after $j-1$ runs of each sign and covers the steps in $(s_{j-1}^+ + s_{j-1}^-,\, s_j^+ + s_{j-1}^-]$, and the j-th $-$ run covers $(s_j^+ + s_{j-1}^-,\, s_j^+ + s_j^-]$. Let $N = a + b$. There are three cases.

Suppose step N is in the first run. Then $A_N = N$, and this happens exactly when $D \geq N$.

Suppose step N is in the j-th $+$ run with $j \geq 2$. Then exactly $j-1$ of the $-$ runs lie before step N, and they are complete, so $A_N = N - s_{j-1}^-$. So $A_N = a$ means $s_{j-1}^- = b$, and then step $N = a + b$ is in this run exactly when $s_{j-1}^+ < a \leq s_j^+$. By independence this has probability $u_{j-1}(b)\, q_j^D(a)$.

Suppose step N is in the j-th $-$ run. Then the first j of the $+$ runs are complete and no other has begun, so $A_N = s_j^+$. So $A_N = a$ means $s_j^+ = a$, and then step N is in this run exactly when $s_{j-1}^- < b \leq s_j^-$. This has probability $u_j^D(a)\, q_j(b)$.

The cases are disjoint. Summing them, with $m = j-1$ in the second and $m = j$ in the third, gives the formula. Now let N be even. The first direction is fair, and changing the sign of every step sends A_N to $N - A_N$. Hence $C_N = \frac{1}{2}[\mathbb{P}_+(A_N = h) + \mathbb{P}_+(A_N = N - h)] = \mathbb{P}_+(A_N = h)$. With a fresh start $u_m^D = u_m$ and $q_m^D = q_m$, which gives the last formula. □

This identity is the coefficient form of the double transform of Godrèche and Luck [12, Section 7] for the occupation time of an alternating renewal process. To evaluate it we need the stable limit of τ_m and its local form.

Lemma 4.3 (stable and local limits)**.** *Let $1 < \alpha < 2$ and $a_m = m^{1/\alpha}$. Put $K_\alpha = \Gamma(2-\alpha)/(\alpha-1) = -\Gamma(1-\alpha) > 0$ and $\sigma_\alpha = 2K_\alpha|\cos\frac{\pi\alpha}{2}|$.*

(i) $\mathbb{E}e^{i\theta\xi} = 1 + i\mu\theta + K_\alpha(-i\theta)^\alpha + O(\theta^2)$ *as $\theta \to 0$, with the principal branch.*

(ii) $(\tau_m - m\mu)/a_m$ *converges in law to a variable G with $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$ for $s \geq 0$ and $\mathbb{E}e^{i\theta G} = \exp(K_\alpha(-i\theta)^\alpha)$. The law of G is totally skewed to the right and has a bounded, uniformly continuous density g with $g(z) \to 0$ as $|z| \to \infty$.*

(iii) (Gnedenko) The constant $\bar{u} := \sup_{m\geq 1}\sup_k a_m u_m(k)$ is finite, and $a_m u_m(k) - g((k - m\mu)/a_m) \to 0$ uniformly in k as $m \to \infty$.

(iv) If G' is an independent copy of G, then $G - G'$ is symmetric α-stable with $\mathbb{E}e^{i\theta(G-G')} = e^{-\sigma_\alpha|\theta|^\alpha}$, and $I_\alpha := \int_{\mathbb{R}} g(z)^2\, dz = \Gamma(1+1/\alpha)/(\pi\sigma_\alpha^{1/\alpha})$.

Proof. All four parts come from the expansion of the transform of ξ near 0. We find it first.

(i) Summation by parts gives $1 - \mathbb{E}w^\xi = (1-w)\sum_{k\geq 1}\bar{F}(k)w^{k-1}$ for $|w| \leq 1$. Put $w = e^{-v}$ with $\operatorname{Re} v \geq 0$, and write $\operatorname{Li}_\alpha(w) = \sum_{k\geq 1} k^{-\alpha}w^k$. Since $\bar{F}(k) = k^{-\alpha}$, the identity reads $1 - \mathbb{E}e^{-v\xi} = (e^v - 1)\operatorname{Li}_\alpha(e^{-v})$. For $0 < |v| < 2\pi$, [29, 25.12.12] gives $\operatorname{Li}_\alpha(e^{-v}) = \Gamma(1-\alpha)v^{\alpha-1} + \zeta(\alpha) + O(|v|)$. We multiply by $e^v - 1 = v + O(|v|^2)$ and use $\zeta(\alpha) = \mu$ and $\Gamma(1-\alpha) = -K_\alpha$. Since $\alpha + 1 > 2$,

$$1 - \mathbb{E}e^{-v\xi} = \mu v - K_\alpha v^\alpha + O(|v|^2) \qquad (v \to 0,\ \operatorname{Re} v \geq 0).$$

With $v = -i\theta$ this is (i).

(ii) Put $Z_m = (\tau_m - m\mu)/a_m$. Fix θ and let $t = \theta/a_m$. By (i) and $e^{-i\mu t} = 1 - i\mu t + O(t^2)$, and since $a_m^\alpha = m$,

$$e^{-i\mu t}\,\mathbb{E}e^{it\xi} = 1 + K_\alpha(-it)^\alpha + O(t^2) = 1 + \frac{K_\alpha(-i\theta)^\alpha}{m} + O\big(\theta^2 m^{-2/\alpha}\big).$$

The left side is the characteristic function of $(\xi-\mu)/a_m$ at θ, and $\mathbb{E}e^{i\theta Z_m}$ is its m-th power. As $2/\alpha>1$, the error is $o(1/m)$, so $\mathbb{E}e^{i\theta Z_m}\to\exp(K_\alpha(-i\theta)^\alpha)$. The real part of $(-i\theta)^\alpha$ is $|\theta|^\alpha\cos\frac{\pi\alpha}{2}$, which is negative because $1<\alpha<2$. So the limit has modulus $\exp(-\frac{1}{2}\sigma_\alpha|\theta|^\alpha)$. It is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a variable G, and Z_m converges in law to G. It is integrable, so by Fourier inversion G has a bounded, uniformly continuous density g, and g vanishes at infinity by the Riemann–Lebesgue lemma (the domain of attraction also follows from [22, Theorem XVII.5.2]). A stable law with this characteristic function is the one called totally skewed to the right.

For the Laplace transform let $s>0$ be real and take $v=s/a_m$ in the expansion above. In the same way $\mathbb{E}e^{-sZ_m}\to e^{K_\alpha s^\alpha}$. This also holds with $2s$ in place of s, so the second moments of e^{-sZ_m} are bounded in m, and these variables are uniformly integrable. Hence $\mathbb{E}e^{-sZ_m}\to\mathbb{E}e^{-sG}$, and so $\mathbb{E}e^{-sG}=e^{K_\alpha s^\alpha}$.

(iii) Every $\mathbb{P}(\xi=k)$ is positive, so ξ has maximal span 1. So Gnedenko's local limit theorem [21, Chapter 4] applies to the convergence in (ii) and gives the uniform limit. For $\bar{u}$, the limit gives an m_0 with $a_m u_m(k)\le\|g\|_\infty+1$ for all $m\ge m_0$ and all k. For $m<m_0$ we use $u_m(k)\le 1$, so $a_m u_m(k)\le a_{m_0}$. Hence $\bar{u}<\infty$.

(iv) The characteristic function of $-G'$ is the complex conjugate of that of G. So $G-G'$ has the characteristic function $|\mathbb{E}e^{i\theta G}|^2=e^{-\sigma_\alpha|\theta|^\alpha}$, which is that of a symmetric α-stable law. Plancherel's theorem gives

$$I_\alpha=\int_{\mathbb{R}}g(z)^2\,dz=\frac{1}{2\pi}\int_{\mathbb{R}}e^{-\sigma_\alpha|\theta|^\alpha}\,d\theta=\frac{1}{\pi}\int_0^\infty e^{-\sigma_\alpha\theta^\alpha}\,d\theta=\frac{\Gamma(1+1/\alpha)}{\pi\sigma_\alpha^{1/\alpha}},$$

where the last step is the substitution $x=\sigma_\alpha\theta^\alpha$. (Note that I_α is the density of $G-G'$ at 0. This is how it enters Theorem 4.4, where two independent renewal processes have to meet.) □

Theorem 4.4 (the center for $1<\alpha<2$)**.** *Let $1<\alpha<2$ and let D be any a.s. finite first run, in particular either start. As $N\to\infty$ over even N,*

$$C_N\sim 2I_\alpha\Big(\frac{\mu}{h}\Big)^{1/\alpha}=\frac{2\Gamma(1+1/\alpha)}{\pi}\Big(\frac{\mu}{K_\alpha|\cos(\pi\alpha/2)|\,N}\Big)^{1/\alpha}.$$

In words, the center decays like $N^{-1/\alpha}$, and the first run does not matter.

Idea of the proof. By Proposition 4.2 the walk ends in the center when, for some m, one renewal process ends its m-th run exactly at level h and the other is inside a run that covers level h. Level h takes about h/μ runs. Put $b_h=(h/\mu)^{1/\alpha}$, which is the scale on which τ_m spreads around its mean for such m. So the m-th run ends at h with probability about $g(z)/b_h$, where $z=(h-m\mu)/b_h$ says how far m is from the typical number of runs. A level is covered by a run μ times as often as a run ends there, so the m-th run covers h with probability about $\mu g(z)/b_h$. The values of m that matter have $|m\mu-h|$ of order b_h, and there are b_h/μ of them for each unit of z. Proposition 4.2 has two terms of this kind for each m, so C_N is about $2\cdot\frac{1}{b_h}\cdot\frac{\mu}{b_h}\cdot\frac{b_h}{\mu}\int g^2=2I_\alpha/b_h$.

In the proof (Appendix D) we split the sum in Proposition 4.2 by the number m of runs. The terms with $|m\mu-h|\le Mb_h$ form a Riemann sum for $2b_h^{-1}\int_{-M}^{M}g^2$ by Lemma 4.3(iii). The terms with $m\le h/(4\mu)$ add up to $O(h^{-\alpha})=o(b_h^{-1})$, since a sum of few runs that is far above its mean is most likely made by one long run. The others add up to at most $O(b_h^{-1})\mathbb{P}(|G|\ge M-1)+o(b_h^{-1})$, and we let $M\to\infty$.

α	$e(\alpha)$	$C(\alpha)$	$C_{\mathrm{fr}}(\alpha)$	R_{1600}	R_{10^4}	R_{10^5}
1.3	+0.469231	0.422717	0.498631	17.7593 [1.3180]	38.7537 [1.2171]	106.149 [1.1317]
1.5	+0.166667	0.647971	0.846371	2.14409 [0.9675]	2.9528 [0.9818]	4.37642 [0.9914]
φ	0	0.776132	1.073675	0.661763 [0.8526]	0.697331 [0.8985]	0.727689 [0.9376]
5/3	$-1/15$	0.835101	1.182237	0.411815 [0.8064]	0.389307 [0.8614]	0.352822 [0.9102]
1.7	−0.111765	0.880046	1.265508	0.298369 [0.7733]	0.262005 [0.8334]	0.215839 [0.8881]
1.9	−0.373684	1.438986	2.266075	0.0449706 [0.4923]	0.0256505 [0.5569]	0.0121612 [0.6242]

Table 2: Constants of Corollary 4.5 and exact values of $R_N = E_N/C_N$ for the stationary start, with $R_N/(C(\alpha)N^{e(\alpha)})$ in brackets.

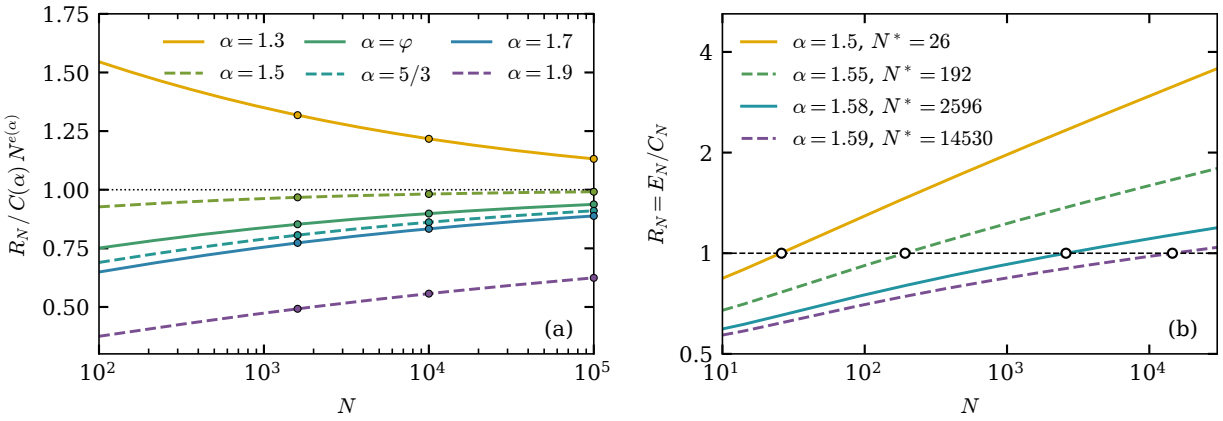


Figure 3: Alternating runs with a stationary start. (a) $R_N/(C(\alpha)N^{e(\alpha)})$ for even N from 100 to 10^5, with the entries of Table 2 as dots. (b) R_N for $\alpha = 1.5, 1.55, 1.58, 1.59$ below φ. Each curve crosses 1 once on $10 \le N \le 3 \cdot 10^4$, at $N^*(\alpha)$ (circles).

Corollary 4.5 (the golden-ratio exponent)**.** *Let $1 < \alpha < 2$ and $e(\alpha) = 1 - \alpha + 1/\alpha$. If $\mathbb{P}(D \ge N) \sim c_D N^{-\beta_D}$ with $c_D, \beta_D > 0$, then $R_N \sim c_D N^{1/\alpha - \beta_D}/(4I_\alpha(2\mu)^{1/\alpha})$. In particular $R_N \sim C(\alpha)N^{e(\alpha)}$ for the stationary start and $R_N \sim C_{\mathrm{fr}}(\alpha)N^{e(\alpha)-1}$ for the fresh start, where*

$$C(\alpha) = \frac{\pi\big(K_\alpha|\cos\tfrac{\pi\alpha}{2}|\big)^{1/\alpha}}{4(\alpha-1)\Gamma(1+\frac{1}{\alpha})\,\zeta(\alpha)^{1+1/\alpha}}, \qquad C_{\mathrm{fr}}(\alpha) = (\alpha-1)\zeta(\alpha)\,C(\alpha).$$

Since $e(\alpha) > 0$ exactly when $\alpha < \varphi$, with a stationary start the ends win for all large N if $\alpha < \varphi$, and the center wins if $\alpha > \varphi$. At $\alpha = \varphi$,

$$R_N \to C(\varphi) = \frac{\pi\varphi\big(\varphi\,\Gamma(\varphi^{-2})|\cos\tfrac{\pi\varphi}{2}|\big)^{1/\varphi}}{4\,\Gamma(\varphi)\,\zeta(\varphi)^{\varphi}} = 0.7761317206\ldots,$$

so the center still wins by the factor $1/C(\varphi) = 1.2884$. With a fresh start $e(\alpha) - 1 < 0$, and the center wins for every $1 < \alpha < 2$.

Proof. By Proposition 4.1 and Theorem 4.4, with $h = N/2$, we have $R_N = \mathbb{P}(D \ge N)/(2C_N)$, and this is asymptotic to $c_D N^{-\beta_D}/(4I_\alpha(2\mu/N)^{1/\alpha})$, which is the general formula. By Lemma 4.3(iv), $4I_\alpha(2\mu)^{1/\alpha} = 4\Gamma(1+\frac{1}{\alpha})\mu^{1/\alpha}/(\pi(K_\alpha|\cos\frac{\pi\alpha}{2}|)^{1/\alpha})$. The fresh start has $c_D = 1$ and $\beta_D = \alpha$, which gives $C_{\mathrm{fr}}(\alpha)$. By Proposition 4.1 the stationary start has $c_D = 1/((\alpha-1)\mu)$ and $\beta_D = \alpha - 1$, which gives $C(\alpha)$. Next, $e(\alpha) = -(\alpha^2 - \alpha - 1)/\alpha$, and φ is the positive root of $\alpha^2 - \alpha - 1$. At $\alpha = \varphi$ we

α	start	center C_N	ratio R_N	status
$\alpha > 2$	stationary	$\sim \sqrt{2\mu/(\pi v N)}$	$\sim C_2(\alpha) N^{3/2-\alpha} \to 0$	Theorem 4.7(a)
$\alpha = 2$	stationary	$\sim \sqrt{2\mu/(\pi N \log N)}$	$\sim \sqrt{\pi \log N/(8\zeta(2)^3 N)} \to 0$	Theorem 4.7(b)
$1 < \alpha < 2$	stationary	$\sim 2I_\alpha(\mu/h)^{1/\alpha}$	$\sim C(\alpha) N^{e(\alpha)}$	Corollary 4.5
$1 < \alpha < 2$	fresh	$\sim 2I_\alpha(\mu/h)^{1/\alpha}$	$\sim C_{\mathrm{fr}}(\alpha) N^{e(\alpha)-1} \to 0$	Corollary 4.5
$\alpha = 1$	fresh	$\approx 2/(\pi^2 \tilde{m}_h)$	$\approx \pi^2 \tilde{m}_h/(8h) \sim \pi^2/(8 \log N)$	heuristic, not proved (Remark 4.9)
$0 < \alpha < 1$	fresh	$\sim \frac{2}{\pi} \tan \frac{\pi\alpha}{2} \cdot \frac{1}{N}$	$\sim \frac{\pi}{4 \tan(\pi\alpha/2)} N^{1-\alpha} \to \infty$	Theorem 4.8

Table 3: The center and the ratio $R_N = E_N/C_N$ for each exponent. The center wins when $R_N \to 0$ and the ends win when $R_N \to \infty$. For $\alpha \geq 2$ the center formula holds for every a.s. finite first run, and a fresh start also gives $R_N \to 0$.

have $\alpha - 1 = 1/\varphi$, $1 + 1/\alpha = \varphi$ and $2 - \alpha = \varphi^{-2}$. So $K_\varphi = \varphi\,\Gamma(\varphi^{-2})$, and substituting these into $C(\alpha)$ gives the closed form. □

Corollary 4.5 is the lattice form, with its constant, of the heuristic of [20, Section 2.1]. The approach to $C(\varphi)$ is slow. At $\alpha = \varphi$ we have $R_{10^5}/C(\varphi) = 0.938$, and R_N increases on the computed grid (Table 2 and Figure 3).

The golden ratio also appears for the elephant walk of Section 2 with first step +1. There the law of A_N is log-concave for every N exactly when $1 - 2\varepsilon \leq 1/\varphi$ [10, Proposition 2.2]. We know of no connection. Log-concavity should be kept apart from unimodality, which is weaker. With first step +1, the limit law of the superdiffusive elephant walk (the case $\varepsilon < \frac{1}{4}$) is unimodal for every such ε [10, Theorem 1.1]. For log-concavity, the authors of [10] conjectured in their Remark 2.3 that for every $\varepsilon < \frac{1}{4}$ the law of A_N is log-concave for all large N. Trinas and Valle disproved this [11, Theorem 4.4]. They show that for $1 - 2\varepsilon > 0.80399$ the law of A_N is not log-concave for arbitrarily large N, and they prove a uniform local limit theorem for the superdiffusive walk [11, Theorem 3.1].

Remark 4.6 (crossover sizes and heat conduction). For $\alpha < \varphi$ and a stationary start the crossover is late when α is close to φ. For $\alpha = 1.5, 1.55, 1.58, 1.59$ and every even N in $[10, 3 \cdot 10^4]$, $R_N - 1$ changes sign once, at $N^*(\alpha) = 26, 192, 2596, 14530$. Heuristically, $C(\alpha) N^{e(\alpha)} = 1$ gives $\log N^*(\alpha) = 0.1834/(\varphi - \alpha) + O(1)$ as $\alpha \uparrow \varphi$. At the value $\alpha = 5/3$ of Lévy-walk models of heat conduction [19, 18], $R_N \sim C(5/3) N^{-1/15}$ with $C(5/3) = 0.8351$.

We turn to the other exponents. Table 3 collects them. For $\alpha \geq 2$ the sums τ_m have a Gaussian limit, and the center wins with either start.

Theorem 4.7 ($\alpha \geq 2$)**.** *Let D be any a.s. finite first run, and let $N \to \infty$ over even N.*

(a) Let $\alpha > 2$ and $v = \operatorname{Var} \xi = 2\zeta(\alpha - 1) - \zeta(\alpha) - \zeta(\alpha)^2$. Then $C_N \sim \sqrt{2\mu/(\pi v N)}$. With the stationary start $R_N \sim C_2(\alpha) N^{3/2-\alpha}$, where $C_2(\alpha) = \sqrt{\pi v/(2\mu)}/(2(\alpha - 1)\mu)$.

(b) Let $\alpha = 2$, so that $\mu = \zeta(2)$ and $\mathbb{E}\xi^2 = \infty$. Then $(\tau_m - m\mu)/\sqrt{m \log m}$ converges in law to the standard normal law, and $C_N \sim \sqrt{2\mu/(\pi N \log N)}$. With the stationary start $R_N \sim \sqrt{\pi \log N/(8\zeta(2)^3 N)}$.

In both cases $R_N \to 0$ with the fresh start too.

In words, for $\alpha \geq 2$ the center is of the Gaussian order $N^{-1/2}$, with an extra factor $(\log N)^{-1/2}$ at $\alpha = 2$, where the variance is infinite. With either start the end is at most of order $N^{1-\alpha}$, so the center wins. The exponent of R_N is continuous at $\alpha = 2$, since $e(2) = -\frac{1}{2} = \frac{3}{2} - 2$.

Idea of the proof. The proof of Theorem 4.4 uses the exponent α in three places only. These are the limit law of τ_m with its local form, the bound $u_m(k) \le C_B m k^{-\alpha-1}$ for a sum far above its mean, and the fact that the norming sequence varies regularly with index less than 1. All three hold for $\alpha \ge 2$ with the normal law in place of the stable law. For $\alpha > 2$ the norming is $a_m = \sqrt{vm}$. For $\alpha = 2$ the truncated second moment $\mathbb{E}[\xi^2; \xi \le x]$ grows like $2 \log x$, which gives the norming $a_m = \sqrt{m \log m}$. Since $\int g^2 = 1/(2\sqrt{\pi})$ for the standard normal density g, the center is $1/(\sqrt{\pi}\, b_h)$ to leading order, where b_h is the norming at $m = h/\mu$. We give the proof in Appendix D, where the proof of Theorem 4.4 is written so that it covers both theorems.

The convergence is slow at $\alpha = 2$. There R_N divided by the leading term in (b) is 0.885 at $N = 1600$ and 0.943 at $N = 10^5$. At $\alpha = 3$ the ratio of R_N to the leading term in (a) is 0.981 at $N = 16000$.

For $0 < \alpha < 1$ the runs have infinite mean, and only the fresh start makes sense. Then A_N/N has a limit law that is not concentrated at a point, namely Lamperti's occupation-time law [13, 12], with density

$$\ell_\alpha(y) = \frac{\sin \pi\alpha}{\pi} \frac{y^{\alpha-1}(1-y)^{\alpha-1}}{y^{2\alpha} + 2y^\alpha (1-y)^\alpha \cos \pi\alpha + (1-y)^{2\alpha}} \qquad (0 < y < 1).$$

So the center should be of order $1/N$. The next theorem is the local form of this at the center.

Theorem 4.8 $(0 < \alpha < 1)$**.** *Let $0 < \alpha < 1$, with a fresh start. As $N \to \infty$ over even N,*

$$NC_N \to \ell_\alpha(\tfrac{1}{2}) = \frac{2}{\pi} \tan \frac{\pi\alpha}{2}.$$

Hence $R_N \sim \frac{\pi}{4 \tan(\pi\alpha/2)} N^{1-\alpha} \to \infty$.

In words, for $\alpha < 1$ the center is an ordinary bin of size about $1/N$, while the end has the much larger probability $\frac{1}{2}N^{-\alpha}$ that the first run never stops. So the ends win for every $\alpha < 1$.

Idea of the proof. Now τ_m is of order $a_m = m^{1/\alpha}$ with no centering, and τ_m/a_m converges to a positive stable variable G with density g. The number m of runs of each sign is no longer concentrated. It ranges over all m with $m^{1/\alpha}$ of order h, and on this range the terms of Proposition 4.2 are $\frac{2}{\alpha h m} t^2 g(t)^2$ with $t = h/a_m$. Summing over m is integrating in $\log m$, which gives $NC_N \to 4 \int_0^\infty t g(t)^2 \, dt$. This integral is the density at $\frac{1}{2}$ of $G_1/(G_1 + G_2)$ for two independent copies of G, and a Mellin transform identifies the law of G_1/G_2 and hence this density. The proof is in Appendix D. It does not use Lamperti's theorem.

Here too the convergence is slow near $\alpha = 1$. At $N = 16000$ the ratio of NC_N to its limit is 1.042, 1.001, 0.928 and 0.554 for $\alpha = 0.3$, 0.5, 0.7 and 0.9. We do not prove the local limit at bins other than the center.

Remark 4.9 (the exponent $\alpha = 1$ and the fresh start). With a fresh start the threshold is $\alpha = 1$. The ends win for $\alpha < 1$ by Theorem 4.8, and the center wins for $\alpha > 1$ by Corollary 4.5 and Theorem 4.7. At $\alpha = 1$ we only have a heuristic. Suppose we repeat the proof of Theorem 4.4 with a skewed Cauchy limit, the centering $m(H_{m+1} - 1)$, and the density $1/\pi^2$ at 0 of the difference of two copies of the limit. This suggests the row of Table 3, namely $R_N \approx \pi^2 \tilde{m}_h/(8h) \sim \pi^2/(8 \log N)$, where $\tilde{m}_h$ solves $m(H_{m+1} - 1) = h$. We have not controlled the error terms, so the claim that the center wins by a logarithm at $\alpha = 1$ is not proved. The exact R_N divided by $\pi^2 \tilde{m}_h/(8h)$ is 0.796, 0.850 and 0.886 at $N = 1600$, 10^4 and 10^5.

We do not know a rate in Theorem 4.4, whether R_N is monotone with a single sign change of $R_N - 1$, how to make the case $\alpha = 1$ rigorous, or the full shape of P_N for heavy-tailed runs.

5 Aging walks

In this section the walk remembers its age. After its first step it switches direction at step $k \geq 2$ with probability c/k, independently of the past, where $0 < c < 2$. So the walk switches less and less often, and the expected number of switches in N steps is $c(H_N - 1)$, which is about $c \log N$. With a fair first step, A_N/N converges in law to Beta(c, c) [30, Example 1.1 and Theorem 1.4] (their chain is this walk for $c \leq 1$), [31, Theorem 1], and the limit is uniform exactly when $c = 1$ [31, Remark 2]. At finite N the first step matters. We first fix it, then pass to d states, and then return to the fair first step.

We write $\mathbb{P}_+$ for the law given $X_1 = +1$, as in Section 4, and X_N for the last step. A path is a sequence $x \in \{\pm 1\}^N$, and it switches at time $t \geq 2$ if $x_t \neq x_{t-1}$. Step $k \geq 2$ contributes the factor c/k to the probability of a path if the path switches there, and $1 - c/k$ otherwise. So relative to no switch, a switch at time t multiplies the probability of a path by the odds

$$\omega_t = \frac{c/t}{1 - c/t} = \frac{c}{t - c} \qquad (t \geq 2),$$

and

$$\mathbb{P}_+(X = x) = \prod_{k=2}^{N} \Big(1 - \frac{c}{k}\Big) \prod_{t:\, x_t \neq x_{t-1}} \omega_t \qquad (x_1 = +1). \tag{7}$$

Each ω_t is positive and continuous on $(0, 2)$, increases strictly in c, and tends to 0 as $c \downarrow 0$.

5.1 A fixed first step

Proposition 5.1 (two states, fixed first step)**.** *Let $N \geq 2$ and let the first step be $+1$.*

(a) At $c = 1$, for $1 \leq s \leq N$,

$$\mathbb{P}_+(A_N = s,\, X_N = +1) = \frac{s - 1}{N(N-1)}, \qquad \mathbb{P}_+(A_N = s,\, X_N = -1) = \frac{N - s}{N(N-1)}.$$

So A_N is uniform on $\{1, \ldots, N\}$.

(b) For $1 \leq s \leq N - 1$ the ratio $\mathbb{P}_+(A_N = s)/\mathbb{P}_+(A_N = N)$ increases strictly in $c \in (0, 2)$, and it equals 1 exactly at $c = 1$.

In words, with a fixed first step every bin from 1 to $N - 1$ ties with the end bin N at $c = 1$, for every N. For $c < 1$ the end bin is the most likely bin, and for $c > 1$ it is the least likely of the bins $1, \ldots, N$ (bin 0 has probability 0). So the fixed-start crossing is exactly $c = 1$, and it does not depend on N or on the bin.

Proof. (a) We use induction on N. Put $u_N^{\pm}(s) = \mathbb{P}_+(A_N = s,\, X_N = \pm 1)$, which is 0 unless $1 \leq s \leq N$. For $N = 2$ the second step is $+1$ or -1 with probability $\frac{1}{2}$ each, so $u_2^+(2) = u_2^-(1) = \frac{1}{2}$ and $u_2^+(1) = u_2^-(2) = 0$, as claimed. At $c = 1$ step $N + 1$ switches with probability $1/(N + 1)$. Conditioning on X_N gives, for every s,

$$(N+1)\, u_{N+1}^+(s+1) = N\, u_N^+(s) + u_N^-(s), \qquad (N+1)\, u_{N+1}^-(s) = u_N^+(s) + N\, u_N^-(s).$$

Suppose the claim holds for N, and let $1 \leq s \leq N$. Then $N(s-1) + (N - s) = s(N-1)$ gives $u_{N+1}^+(s+1) = \frac{s}{(N+1)N}$, and $(s-1) + N(N-s) = (N-1)(N+1-s)$ gives $u_{N+1}^-(s) = \frac{N+1-s}{(N+1)N}$.

The two remaining values are $u^+_{N+1}(1) = 0$ and $u^-_{N+1}(N+1) = 0$, since $u^\pm_N(0) = u^\pm_N(N+1) = 0$. This is the claim for $N+1$. Summing over X_N gives $\mathbb{P}_+(A_N = s) = \frac{(s-1)+(N-s)}{N(N-1)} = \frac{1}{N}$.

(b) We write the ratio as a sum over paths. Only the path with every step $+1$ has $A_N = N$, and it never switches. So $\mathbb{P}_+(A_N = N)$ is the first product in (7), and dividing by it leaves

$$\frac{\mathbb{P}_+(A_N = s)}{\mathbb{P}_+(A_N = N)} = \sum_x \prod_{t:\, x_t \neq x_{t-1}} \omega_t, \tag{8}$$

where the sum is over the paths x with $x_1 = +1$ and s steps equal to $+1$. For example, for $N = 3$ and $s = 2$ the paths are $(+,+,-)$ and $(+,-,+)$, and the ratio is $\omega_3 + \omega_2\omega_3$. For $1 \le s \le N-1$ there is such a path, and each of them switches at least once. So the right side is a sum of products of the ω_t with at least one factor each. Each ω_t is positive and increases strictly in c, so the sum does too. By (a) it equals 1 at $c = 1$. □

After a shift of one step, part (a) is the exercise in [37, footnote 1 on p. 6]. That footnote also says that the fact can be related to Pólya urns (the authors credit M. Balázs) and that a more general connection of their model with Pólya urns will appear in a later paper. So the uniform law for two states is not new. What we add for two states is the joint law with the last step and part (b). The argument for (b) is the one in Proposition 2.3. Both parts of Proposition 5.1 are also proved in Lean (Section 6).

5.2 d states

The same thing happens for a chain on d states that leaves its current state for each other state at the same aging rate. For such chains Dietz and Sethuraman [30, Theorem 1.4] proved that the fractions of time converge in law to the Dirichlet law with every parameter θ, which is uniform on the simplex exactly at $\theta = 1$ (see also [33]). Their chain moves to each other state with probability θ/n at time n once $n > \lceil (d-1)\theta \rceil$, and it does not move before that. Ours, defined below for $\theta < d/(d-1)$, moves to each other state with probability $\theta/(k+d-2)$ at every step $k \ge 2$. If $\theta \le 1$ then $\lceil (d-1)\theta \rceil \le d-1$, so our step k has the transition probabilities of their time $k+d-2$ for every $k \ge 2$, and the two chains differ by a shift of $d-2$ steps. If $\theta > 1$ the ceiling is d. Then their chain stays at time d, where ours takes its second step, and the later steps agree. One step does not change the limit. Part (a) of the theorem below is an exact form of their limit at finite N. Fix $d \ge 2$ and θ with $0 < \theta < d/(d-1)$. The chain $X_1, X_2, \ldots$ on $\{1, \ldots, d\}$ starts at $X_1 = 1$. At step $k \ge 2$ it moves to each of the other $d-1$ states with probability $\theta/(k+d-2)$, and it stays with probability $1 - (d-1)\theta/(k+d-2)$. The condition on θ says that the probability of staying is positive at every step $k \ge 2$. Let $L_N = (L_N(1), \ldots, L_N(d))$ be the vector of visit counts, namely $L_N(i) = \#\{k \le N : X_k = i\}$. Its values are the compositions $n = (n_1, \ldots, n_d)$ of N into d nonnegative parts with $n_1 \ge 1$. There are $\binom{N+d-2}{d-1}$ of them (subtract one from n_1). We call them bins. The corner Ne_1, where e_i is the i-th unit vector, is the bin of the chain that never moves.

Theorem 5.2 (d states)**.** *Let $d \ge 2$, $0 < \theta < d/(d-1)$ and $N \ge 2$.*

(a) At $\theta = 1$, for every bin n and every state j,

$$\mathbb{P}(L_N = n,\, X_N = j) = \frac{n_j - \mathbf{1}\{j = 1\}}{(N-1)\binom{N+d-2}{d-1}}. \tag{9}$$

So L_N is uniform on the $\binom{N+d-2}{d-1}$ bins, the corner included.

(b) For every bin $n \neq Ne_1$ the ratio $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ increases strictly in θ, and it equals 1 exactly at $\theta = 1$.

In words, at $\theta = 1$ all bins are equally likely, and given the counts the last state is distributed as the state at a uniformly chosen one of the steps $2, \dots, N$ (the forced first visit to state 1 is left out). Every other bin ties with the corner exactly at $\theta = 1$, for every $N \geq 2$ and every d. Below $\theta = 1$ the corner is the most likely bin, and above it the corner is the least likely.

Idea of the proof. For (a) we check (9) by induction on N. At $\theta = 1$ step N stays with weight $N - 1$ and moves to each other state with weight 1. We read the weight $N - 1$ as $(N - 2) + 1$. So the chain copies its last state with weight $N - 2$, and it picks each of the d states with weight 1. By the induction hypothesis the last state is like the state at a uniformly chosen one of the steps $2, \dots, N - 1$. So given the counts m after $N - 1$ steps, the new state is j with weight $(m_j - \mathbf{1}\{j = 1\}) + 1$. This is the rule of Pólya's urn, and it keeps the counts uniform. Part (b) is the path-weight argument of Proposition 5.1(b).

Proof. (a) We use induction on N. Write $P_N(n, j) = \mathbb{P}(L_N = n,\, X_N = j)$ and $\delta_j = \mathbf{1}\{j = 1\}$. Let $N = 2$. At $\theta = 1$ the second step goes to each of the d states with probability $1/d$ (the probability of staying is $1 - (d - 1)/d$). So $P_2(2e_1, 1) = \frac{1}{d}$ and $P_2(e_1 + e_j, j) = \frac{1}{d}$ for $j \neq 1$, and P_2 vanishes otherwise. Since $\binom{d}{d-1} = d$, the right side of (9) is $(n_j - \delta_j)/d$. This is $\frac{1}{d}$ in these d cases. In every other case $n_j = \delta_j$, and it is 0.

Now let $N \geq 3$ and suppose that (9) holds for $N - 1$. Fix a bin n and a state j. If $n_j = \delta_j$, then the right side of (9) is 0. The left side is 0 too. Indeed, if $j \neq 1$ then state j is never visited, and if $j = 1$ then state 1 is visited only at the first step, so $X_N \neq j$ in both cases. Otherwise $m = n - e_j$ is a composition of $N - 1$ with $m_1 \geq 1$. At $\theta = 1$ step N moves to each other state with probability $1/(N + d - 2)$ and stays with probability $(N - 1)/(N + d - 2)$. Conditioning on X_{N-1} gives

$$P_N(n, j) = \frac{(N - 1)\, P_{N-1}(m, j) + \sum_{i \neq j} P_{N-1}(m, i)}{N + d - 2} = \frac{(N - 2)\, P_{N-1}(m, j) + \sum_i P_{N-1}(m, i)}{N + d - 2}.$$

By the induction hypothesis and $\sum_i (m_i - \delta_i) = N - 2$, the numerator is

$$\frac{(N - 2)(m_j - \delta_j) + (N - 2)}{(N - 2)\binom{N+d-3}{d-1}} = \frac{n_j - \delta_j}{\binom{N+d-3}{d-1}}.$$

Since $\binom{N+d-2}{d-1} = \binom{N+d-3}{d-1} \frac{N+d-2}{N-1}$, this gives (9) for N. Summing (9) over j and using $\sum_j (n_j - \delta_j) = N - 1$ gives $\mathbb{P}(L_N = n) = 1/\binom{N+d-2}{d-1}$.

(b) A path $x \in \{1, \dots, d\}^N$ with $x_1 = 1$ has probability $\prod_{k=2}^{N} \big(1 - \frac{(d-1)\theta}{k+d-2}\big)$ times the product of the odds

$$\omega_t^{(d)} = \frac{\theta/(t + d - 2)}{1 - (d - 1)\theta/(t + d - 2)} = \frac{\theta}{t + d - 2 - (d - 1)\theta}$$

over the times $t \geq 2$ with $x_t \neq x_{t-1}$. The denominator is positive and decreases in θ, so each $\omega_t^{(d)}$ increases strictly in θ. Only the constant path has counts Ne_1. So $\mathbb{P}(L_N = n)/\mathbb{P}(L_N = Ne_1)$ is the sum of the products of the odds over the paths with counts n. For a bin $n \neq Ne_1$ there is such a path, and each of them moves at least once. Hence the ratio increases strictly in θ, and by (a) it equals 1 at $\theta = 1$. □

Remark 5.3. For $d = 2$ the chain switches at step k with probability θ/k, so Theorem 5.2 with $\theta = c$ is Proposition 5.1, and (9) reads $(s-1)/(N(N-1))$ for $j = 1$ and $n = (s, N-s)$.

Take Pólya's urn with d colors, started with one ball of each color. At each draw we take a ball uniformly at random and return it with one more ball of its color. A sequence of $N-1$ draws with counts m has probability $(d-1)!\prod_i m_i!/(N+d-2)!$. So the counts are uniform on the compositions of $N-1$ into d parts, and given the counts m the last draw has color j with probability $m_j/(N-1)$. Hence (9) says that, for each fixed N, the pair $(L_N - e_1, X_N)$ has the law of the counts and the last color after $N-1$ draws. The two processes are different, though. The chain is a Markov chain, and the draws are exchangeable. For $d = 2$ and $N = 4$ the path $1, 1, 2, 2$ has probability $\frac{1}{8}$ for the chain, and the draws $1, 2, 2$ have probability $\frac{1}{12}$. Also, at $\theta = 1$ the chain moves at step k with probability $(d-1)/(k+d-2)$, independently over k. This is the probability that customer k opens a new table in the Chinese restaurant process with parameter $d-1$. So the expected number of moves in N steps at the tie is $\sum_{k=2}^{N} \frac{d-1}{k+d-2} = (d-1)\log N + O(1)$, which is the first-order scale of the vertex crossing in Paper B [3].

We checked (9) and the strict ordering in (b) in exact rational arithmetic for $d = 2, 3, 4$ and $2 \le N \le 8$ (`verify_aging.py`).

The urn is not special to $\theta = 1$. It appears for every weight once the aging rate is adjusted a little. The chain below either redraws its state or repeats it. This is the mechanism of the conservative random walk of Engländer and Volkov [34], who redraw a direction in $\mathbb{Z}^d$ with probability p_n and study recurrence and scaling limits. We do not claim the urn law below as new. The footnote quoted above announces a general connection with Pólya urns, and two chapters of the book [35] concern urn models. We have not seen them. What we use it for is the explicit joint law on d states and the tie. Fix $d \ge 2$ and $\theta > 0$, and let $Y_1, Y_2, \dots$ be the following chain on $\{1, \dots, d\}$. At step $t \ge 1$, with probability

$$\varphi_t = \frac{d\theta}{d\theta + t - 1}$$

the state Y_t is drawn afresh, uniformly from the d states, and otherwise $Y_t = Y_{t-1}$ (so Y_1 is uniform, since $\varphi_1 = 1$). In other words, at step $t \ge 2$ the chain moves to each other state with probability $\theta/(d\theta + t - 1)$. Let M_T be the vector of visit counts of $Y_1, \dots, Y_T$. At $\theta = 1$ this is the chain of Theorem 5.2 without its first step, namely $Y_t = X_{t+1}$ and $M_T = L_{T+1} - e_1$. We write $x^{(m)} = x(x+1)\cdots(x+m-1)$.

Proposition 5.4 (fresh or repeat)**.** *For every $T \ge 1$, every composition m of T into d nonnegative parts and every state j,*

$$\mathbb{P}(M_T = m,\, Y_T = j) = \frac{m_j}{T}\, D_T(m), \qquad D_T(m) = \frac{T!}{(d\theta)^{(T)}} \prod_{i=1}^{d} \frac{\theta^{(m_i)}}{m_i!}. \tag{10}$$

So M_T has the law D_T of the counts in Pólya's urn with d colors and initial weight θ on each color. For $d = 2$ and even T,

$$\frac{\mathbb{P}(M_T = (T/2, T/2))}{\mathbb{P}(M_T = (T, 0))} = \binom{T}{T/2} \prod_{i=0}^{T/2-1} \frac{\theta + i}{\theta + T/2 + i}, \tag{11}$$

which increases strictly in θ and equals 1 exactly at $\theta = 1$.

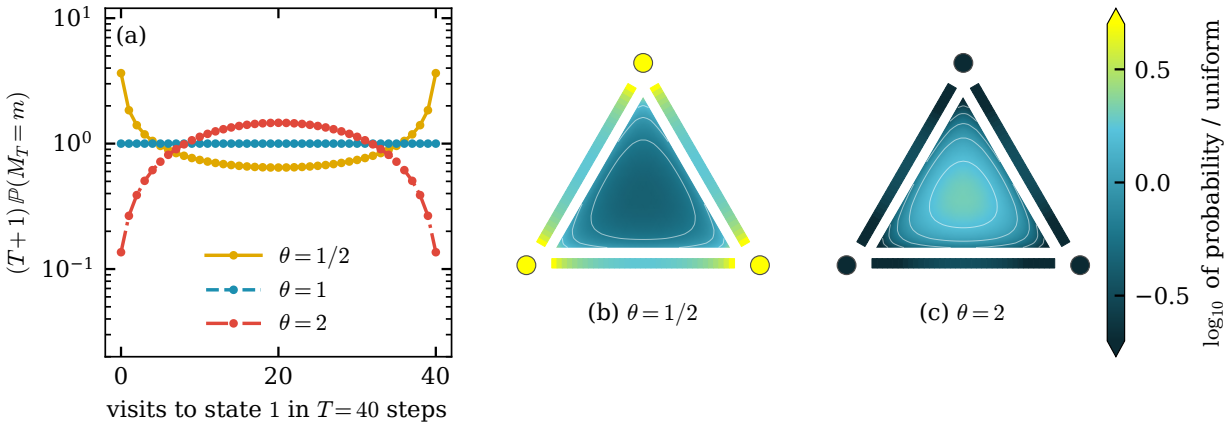


Figure 4: The exact law (10) of the counts of the chain that draws afresh or repeats. (a) Two states and $T = 40$, as a multiple of the uniform law. The law is flat at $\theta = 1$, the ends are the most likely bins for $\theta = \frac{1}{2}$, and the center is for $\theta = 2$. (b), (c) Three states and $T = 36$ on the simplex of visit fractions, colored by the probability of a bin relative to the uniform law. The triangle holds the bins that use all three states, the bands hold the bins that use two, and the disks are the bins that use one. The thin lines are level lines. For $\theta = \frac{1}{2}$ a corner is 9.6 times as likely as under the uniform law and the center 0.43 times. For $\theta = 2$ the numbers are 0.035 and 2.1.

In words, the chain repeats its last state where the urn repeats a uniformly chosen earlier draw, and at each fixed time this makes no difference to the counts or to the last state. For two states the center and the end tie at $\theta = 1$ for every T, with ends ahead below it and the center ahead above it, and now the whole law is explicit on both sides of the tie (Figure 4).

Proof. We use induction on T. For $T = 1$ both sides of (10) are $\frac{1}{d}$ when $m = e_j$ and 0 otherwise. Let $T \geq 2$ and suppose (10) holds for $T - 1$. If $m_j = 0$ both sides vanish. Otherwise put $m' = m - e_j$. Summing (10) over the last state gives $\mathbb{P}(M_{T-1} = m') = D_{T-1}(m')$. A fresh draw gives j with probability $\frac{1}{d}$ whatever the last state was, and a repeat gives j only when the last state was j (this is why the induction carries the last state along with the counts). So

$$\mathbb{P}(M_T = m,\, Y_T = j) = \frac{\varphi_T}{d}\, D_{T-1}(m') + (1 - \varphi_T)\, \frac{m_j - 1}{T - 1}\, D_{T-1}(m') = \frac{\theta + m_j - 1}{d\theta + T - 1}\, D_{T-1}(m').$$

On the other hand $D_T(m) = D_{T-1}(m') \cdot \frac{T}{m_j} \cdot \frac{\theta + m_j - 1}{d\theta + T - 1}$ by the definition of D_T. This gives (10). Summing over j gives $\mathbb{P}(M_T = m) = D_T(m)$, and the ratio (11) is $D_T(T/2, T/2)/D_T(T, 0)$ with $\theta^{(T)} = \theta^{(T/2)}(\theta + T/2)^{(T/2)}$. Each factor $(\theta + i)/(\theta + T/2 + i)$ increases strictly in θ, and at $\theta = 1$ the product is $(T/2)!^2/T!$. □

Remark 5.5. Here is another way to see the law of the counts in Proposition 5.4 (but not the joint law with the last state). The fresh draws cut $Y_1, \ldots, Y_T$ into runs, and each run gets a uniform state. The urn can be run in the same way, as a Chinese restaurant process in which each table gets a uniform color. There a table is opened with the same probabilities φ_t. The lengths of the runs are the spacings between the successes of independent trials with success probabilities φ_t, and it is a classical fact (the Feller coupling, see [32]) that the multiset of these spacings has the same law as the multiset of the table sizes (the two ordered lists do not). We do not know whether this is the relation to Pólya urns that the footnote of [37] has in mind. We checked (10) in exact arithmetic for four values of θ, with $T \leq 7$ for $d = 2, 3$ and $T \leq 5$ for $d = 4$ (`verify_aging_urn.py`).

5.3 A fair first step

We return to two states and let the first step be fair. Then both ends can be reached, and $E_N = \mathbb{P}(A_N = N) = \frac{1}{2}\prod_{k=2}^{N}(1 - c/k)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. So

$$\mathbb{P}(A_N = s) = \tfrac{1}{2}\big(\mathbb{P}_+(A_N = s) + \mathbb{P}_+(A_N = N - s)\big), \qquad E_N = \tfrac{1}{2}\mathbb{P}_+(A_N = N), \tag{12}$$

since $\mathbb{P}_+(A_N = 0) = 0$.

Proposition 5.6 (fair first step)**.** *Let $N \geq 2$ and $1 \leq s \leq N - 1$, with a fair first step.*

(a) At $c = 1$, $\mathbb{P}(A_N = s) = \frac{1}{N}$ and $E_N = \frac{1}{2N}$.

(b) The ratio $\mathbb{P}(A_N = s)/E_N$ is continuous and strictly increasing in $c \in (0, 2)$. It tends to 0 as $c \downarrow 0$ and equals 2 at $c = 1$. So there is exactly one c in $(0, 2)$ with $\mathbb{P}(A_N = s) = E_N$, and it is less than 1.

For even N we write c_N for the crossing in (b) with $s = h = N/2$, so that $C_N = E_N$ exactly at $c = c_N$. In words, at $c = 1$ every interior bin still has probability $\frac{1}{N}$, and the two ends share the probability $\frac{1}{N}$ that the fixed start gave to one end. So at $c = 1$ the center is twice as likely as an end, and the crossing moves below 1.

Proof. (a) This follows from (12) and Proposition 5.1(a). (b) By (12) the ratio is the sum of the right sides of (8) for s and for $N - s$. Both are finite sums of products of at least one ω_t. So the ratio is continuous and strictly increasing, and it tends to 0 as $c \downarrow 0$. By (a) it equals 2 at $c = 1$. Hence it equals 1 exactly once, at some $c < 1$. □

How far below 1 is c_N? The end is of order N^{-c} and the center is of order $1/N$, so they meet when N^{1-c} is about 2.

Proposition 5.7 (the fair-start crossing)**.** *For even $N \to \infty$,*

$$c_N = 1 - \frac{\log 2}{\log N + \gamma + 2 - 2\log 2} + O\big((\log N)^{-3}\big),$$

where γ is Euler's constant.

Idea of the proof. The end is explicit, namely $E_N = \Gamma(N+1-c)/(2\Gamma(2-c)\Gamma(N+1))$, which is $N^{-c}/(2\Gamma(2-c))$ with relative error $O(1/N)$. For the center we prove a local limit theorem with a rate, namely $NC_N = d_c + O(N^{-0.4})$ uniformly for c near 1, where $d_c = 4^{1-c}/\mathrm{B}(c, c)$ is the density of $\mathrm{Beta}(c, c)$ at $\frac{1}{2}$. So $\log(C_N/E_N) = (c-1)\log N + \Theta(c) + O(N^{-0.4})$ with $\Theta(c) = \log(2\Gamma(2-c)d_c)$. Here $\Theta(1) = \log 2$ and $\Theta'(1) = \gamma + 2 - 2\log 2$, and solving for the zero gives the expansion. The proof is in Appendix E. The local limit theorem comes from an averaging identity for C_N and a coupling of the walk with a sign process driven by a Poisson process of intensity $c\,dt/t$, whose occupation fraction is exactly $\mathrm{Beta}(c, c)$.

The convergence is slow. The exact crossing at $N = 200$ is $c_{200} = 0.8925$, against 0.8932 from the expansion without its error term. The constant in the error term is not computed.

6 Verification

All computations are in Python, and the main ones were done by two independent methods (for example, forward and backward recursions for the elephant center agree to about 10^{-14} at $N = 12800$). The proofs use computation in four places only. First, some constants are values of explicit sums and integrals, such as the 60-term sum in the proof of Theorem 2.6(b), which we evaluated in 30-digit arithmetic (`verify_elephant_crossing.py`). Second, Lemma 3.11 rests on enclosures of f_L' at two points and of f_L'' on the interval between them, computed in ball arithmetic with rigorous error bounds. Third, Proposition 3.8, which makes Corollary 3.6 hold for every even $N \geq 8$, rests for $N < 1688$ on a double-precision computation of C_N/E_N at $\varepsilon = 2/(N+2)$ whose rounding error is bounded in Lemma 3.7. The last two computations are in `certify_elephant.py`, which also repeats the third one in exact integer arithmetic for 20 values of N. Fourth, Proposition 3.26 rests on a computation in ball arithmetic (`certify_convex.py`, about an hour for the whole range, with its output in `data/certify_convex_output.txt`). The tables, the figures and the numbers quoted outside the proofs rest on computation alone, and so does the evidence for Conjecture 3.16 beyond what Section 3.6 proves. The script `verify_elephant_odd.py` checks the identity of Proposition 3.27(a) in exact rational arithmetic for six values of N, and it computes the numbers of Remark 3.28 for the Landau density in floating point.

Some of the exact and finite statements are also proved in Lean 4, among them Proposition 2.3, the value $C_N'(0) = 4/(N+2)$ of Proposition 2.5(b), Lemma 3.4, Theorem 3.5, Proposition 5.1 and the identities behind Corollary 4.5. The Lean files are in the folder `kagey-problems/problem131/lean/PaperD`. The file `MAP.md` in that folder lists each formalized statement with its Lean name. The asymptotic results and Theorem 5.2 were not formalized.

The scripts and the Lean files are in the repository named under "Data availability" at the end of the paper, which also says how to rerun the checks.

Acknowledgments. I thank Peter Kagey for posing the problem and Aidan Deshong for insightful conversations.

A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

A Proofs for Section 2

We first prove Theorem 2.6. Then Section A.1 proves Proposition 2.9, and Section A.2 proves Proposition 2.11. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1 - r_s(b))F(b) + r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1, and for $b \geq 1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h,\ 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the two colors, that is, with c extra balls of each color. For $b + c > 0$ and $s - b + c > 0$ let $\Phi_s^c(b) = \binom{N-s}{h-b}\mathrm{B}(h+c, h+c)/\mathrm{B}(b+c, s-b+c)$, where B is the beta function and the binomial coefficient is 0 unless $0 \leq h - b \leq N - s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \ge 0$, $1 \le s \le N$, $b + c > 0$ and $s - b + c > 0$.*

(a) $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N. So $\Phi_N^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.

(b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.

(c) If $0 < \varepsilon < \frac{1}{2}$ and $\beta = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\beta s}(b)$ for $0 \le b \le s$.

(d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \le c \le \beta s$, then $V_s(b) \ge \Phi_s^c(b)$ for $1 \le b \le s-1$. In particular $V_{t+1}(1) \ge \Phi_{t+1}^0(1) = \pi_t$ for $1 \le t \le h$.

Proof. (a) Let $x = b + c$, $y = s - b + c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 2.4, each order of the $N - s$ draws with $h - b$ draws of the first color has probability $(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = \mathrm{B}(h+c, h+c)/\mathrm{B}(x,y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $\mathrm{B}(x,y)/\mathrm{B}(x+1,y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1 + 2\beta = 1/(1-2\varepsilon)$, we have $\frac{b+\beta s}{s+2\beta s} = (1-2\varepsilon)(\frac{b}{s} + \beta) = r_s(b)$.

(d) We check the hypotheses of Lemma A.2(b) below for $F_u = \Phi_u^c$ on $\{1, \dots, u-1\}$, where $s \le u \le N$. The reason is that M has $\beta u \ge c$ extra balls at time u, so it is pulled toward $\frac{u}{2}$ at least as strongly as the Pólya(c) chain, and from a state closer to $\frac{u}{2}$ it is easier to end at h.

By (a), $F_N(b) = \mathbf{1}\{b = h\}$, and F_u vanishes off $\mathcal{S}_u$ because the binomial coefficient does. Let $s \le u < N$ and $b \in \mathcal{S}_u$. Subtracting the recursion in (a) from the definition of $\mathcal{L}_u$ gives

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = \big(r_u(b) - r_u^c(b)\big)\big(F_{u+1}(b+1) - F_{u+1}(b)\big).$$

We show that the two factors on the right never have opposite signs. For the first factor, $r_u^c(b) = \frac{1}{2} - (\frac{u}{2} - b)/(u + 2c)$, and by (c) the same formula with βu in place of c gives $r_u(b)$. So the first factor is $\frac{u}{2} - b$ times $\frac{1}{u+2c} - \frac{1}{u+2\beta u}$. The second of these is nonnegative because $c \le \beta s \le \beta u$. Hence the first factor has the sign of $\frac{u}{2} - b$ or vanishes. For the second factor there are three cases. If $b = h$ then $b \ge \frac{u}{2}$ and $F_{u+1}(b+1) = 0$, so the factor is ≤ 0. If $u - b = h$ then $b \le \frac{u}{2}$ and $F_{u+1}(b) = 0$, so the factor is ≥ 0. Otherwise both values are positive, and the formula for Φ gives $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$. With $b = \frac{u}{2} - d$ the numerator minus the denominator is $2d(h+c)$, so the ratio is ≥ 1 if $b \le \frac{u}{2}$ and ≤ 1 if $b \ge \frac{u}{2}$. In all three cases the second factor has the sign of $\frac{u}{2} - b$ or vanishes. Hence $\mathcal{L}_u F_{u+1} \ge F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) with $s_0 = s$ gives $V_s \ge F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. □

Lemma A.2. *Let $1 \le s_0 \le N$, and for $s_0 \le s \le N$ let $F_s \ge 0$ be a function on $\{1, \dots, s-1\}$.*

(a) If $F_N(h) \ge 1$ and $\mathcal{L}_s F_{s+1} \le K_s F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, with constants $K_s \ge 0$, then $V_s \le \big(\prod_{u=s}^{N-1} K_u\big) F_s$ on $\mathcal{S}_s$.

(b) If $F_N(b) \le \mathbf{1}\{b = h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, then $V_s \ge F_s$.

Proof. We use backward induction on s, from $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s = \mathcal{L}_s V_{s+1}$. Note that $\mathcal{L}_s$ is an average of two values, so it keeps inequalities between functions. In (a), suppose that $V_{s+1} \le \prod_{u>s} K_u\, F_{s+1}$ on $\mathcal{S}_{s+1}$. This also holds off $\mathcal{S}_{s+1}$, because there $V_{s+1} = 0$ and $F_{s+1} \ge 0$. So on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \le \prod_{u>s} K_u\, \mathcal{L}_s F_{s+1} \le \prod_{u \ge s} K_u\, F_s$. In (b), suppose that $V_{s+1} \ge F_{s+1}$ on $\{1, \dots, s\}$. Then $V_s = \mathcal{L}_s V_{s+1} \ge \mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$, and $F_s = 0 \le V_s$ off $\mathcal{S}_s$. □

Idea of the proof of the upper bound. We compare M with an urn whose number of extra balls, $\kappa(s+10)$ of each color at time s, grows a little faster than the βs of M. Its hitting probability $\Phi_s^{\kappa_s}$ is explicit. One step of M applied to it gives it back times a factor Δ_s, and Proposition A.4 shows that $\log \Delta_s \le \kappa/s$ up to a smaller term. Two effects enter Δ_s. Raising the extra balls from κ_s to κ_{s+1} changes Φ, and we bound this through the derivative of Φ_s^τ in τ (Lemma A.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\kappa H_N}$. Let ψ be the digamma function.

Lemma A.3. *Let $2 \le s \le N$, $b \in \mathcal{S}_s$ and $\tau \ge 0$. Put $X = b+\tau$, $Y = s-b+\tau$, $\Sigma = X+Y$ and $z = (s-2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1-z) + \frac{1+z}{(1-z)\Sigma} + \frac{2(1+z)}{3\Sigma^2(1-z)^2}.$$

Proof. We differentiate the beta functions and bound the digamma function by the first terms of its expansion. Only the two beta functions in $\Phi_s^\tau(b)$ depend on τ, and $\log \mathrm{B}(x,y) = \log\Gamma(x)+\log\Gamma(y)-\log\Gamma(x+y)$. So the derivative is $2\psi(\Sigma)-\psi(X)-\psi(Y)$ minus the difference $2\psi(N+2\tau)-2\psi(h+\tau)$. The duplication formula [29, 5.5.5] gives $2\psi(2y) = \psi(y)+\psi(y+\frac12)+2\log 2$, and ψ is increasing. With $y = h+\tau$ this shows that the difference we subtract is more than $2\log 2$. For $x>0$, [29, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2\int_0^\infty \frac{t\,dt}{(t^2+x^2)(e^{2\pi t}-1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t\,dt}{e^{2\pi t}-1} = \frac{1}{24}$. So $\psi(\Sigma) < \log\Sigma - \frac{1}{2\Sigma}$, and $\psi(x) > \log x - \frac{1}{2x} - \frac{1}{12x^2}$ for $x = X$ and $x = Y$. Hence the derivative is less than $\log\frac{\Sigma^2}{4XY} + (\frac{1}{2X}+\frac{1}{2Y}-\frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2}+\frac{1}{Y^2})$. Now $X-Y = 2b-s$, so $4XY = \Sigma^2-(X-Y)^2 = \Sigma^2(1-z)$ and $X^2+Y^2 = \Sigma^2(1+z)/2$. So the three terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. □

Proposition A.4. *Let $0<\varepsilon<\frac14$, $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$, $\kappa_s = \kappa(s+10)$ and $\Sigma_s = s+2\kappa_s$. For $1 \le s \le N-1$ and $b \in \mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s\Phi_{s+1}^{\kappa_{s+1}}(b)/\Phi_s^{\kappa_s}(b)$.*

(a) Let $c = \kappa_s$, $c' = \kappa_{s+1}$, $d = \frac{s}{2}-b$, $u = \frac{d}{s+2c'}$, $v = \frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\beta)}$. Then $|u| < \frac12$ and

$$\Delta_s(b) = \mathcal{J}\,\frac{\Phi_s^{c'}(b)}{\Phi_s^{c}(b)}, \qquad \mathcal{J} = 1-(\rho-1)\frac{u(u+v)}{\frac14-u^2}.$$

(b) $\log\Delta_s(b) \le \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) For $1 \le t \le h$, $V_{t+1}(1) \le \pi_t \exp\big(\kappa\big[H_{N-1}-H_t+\frac{2}{3t}\big] + \kappa_{t+1}\big[\log\frac{(t+1)^2}{4t}+\frac23\big]\big)$.

(d) $C_N \le U_N := \varepsilon c_1 e^{\kappa H_{N-1}} \sum_{t=1}^h w_t(1-\varepsilon)^{t-1}e^{\kappa\mathcal{G}(t)}$, where $\mathcal{G}(t) = (t+11)\big(\log\frac{(t+1)^2}{4t}+\frac23\big) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa-\beta)/(1+2\beta) = \kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c'.

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar r = \bar r_s(b)$ and $r = r_s(b)$. By Lemma A.1(b) with c', and the recursion in Lemma A.1(a), we have $r'\Phi_{s+1}^{c'}(b+1) = \bar r\,\Phi_s^{c'}(b)$ and $(1-r')\Phi_{s+1}^{c'}(b) = (1-\bar r)\Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar r)}{1-r'} + \frac{r\bar r}{r'}$. Here $r' = \frac12-u$, $\bar r = \frac12+v$ and $r = \frac12-\rho u$, since $\frac12 - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac14-u^2$ the numerator is $f(u,v)+f(-u,-v)$ with $f(u,v) = (\frac12+\rho u)(\frac12-v)(\frac12-u)$, and this equals $\frac14-\rho u^2-(\rho-1)uv$. Also $|d| \le \frac{s}{2}-1$, so $|u| < \frac12$.

(b) *Step 1: the factor $\mathcal{J}$.* Write $c = \kappa_s$, $c' = c + \kappa$ and $\Sigma = \Sigma_s$. Since $c' - \beta s = (\kappa - \beta)s + 11\kappa$, the choice of κ gives $\rho - 1 = \kappa + \nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\beta)}$. As u and v have the sign of d, $u(u+v) \geq u^2$. So $\mathcal{J} \leq 1 - (\kappa + \nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J} > 0$ by (a), since $0 < r < 1$ and $\bar{r}$, $1 - \bar{r}$ are not both 0. So $\log \mathcal{J} \leq -(\kappa + \nu_s)W'$.

Step 2: the ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau \in [c, c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ, while z decreases and Σ increases in τ. So we may take the bound at $\tau = c$, where $X + Y = \Sigma$, and multiply it by the length κ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1 + W$, $\frac{1+z}{1-z} = 1 + 2W$ and $-\log(1-z) \leq W$, so

$$\log \Delta_s(b) \leq \kappa\Big[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2}\Big] - (\kappa + \nu_s)W'. \tag{13}$$

Step 3: what is left to show. The terms of (13) without W and W' are $\frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2}$, which is the bound we want. Since $(1+2W)(1+W) = 1 + 3W + 2W^2$, the other terms are

$$\kappa(W - W') + \frac{2\kappa W}{\Sigma} + \frac{2\kappa W}{\Sigma^2} + \frac{4\kappa W^2}{3\Sigma^2} - \nu_s W',$$

and we must show that they add up to at most 0. If $W = 0$ then $d = 0$, so $W' = 0$ and there is nothing to do. So let $W > 0$. The first four terms are positive, and we show that each is a small multiple of $\frac{\kappa}{s}W$. The last term is negative (it comes from the shift 10 in κ_s), and it is a larger multiple of $\frac{\kappa}{s}W$. We need three facts, with $X = b + c$ and $Y = s - b + c$ as in Step 2.

(i) $1 + W \leq \frac{\Sigma}{2(1+c)}$. Indeed $X, Y \geq 1 + c$, since $b \geq 1$ and $s - b \geq 1$, and the larger of X and Y is at least $\frac{\Sigma}{2}$. So $XY \geq (1+c)\Sigma/2$, and $1 + W = \frac{1}{1-z} = \frac{\Sigma^2}{4XY}$.

(ii) $W - W' \leq \frac{2\kappa}{1+c}W$, and so $W' \geq \frac{9}{11}W$. Indeed z and z' have the same numerator $(s-2b)^2$, and their denominators are Σ^2 and $(s+2c')^2 = (\Sigma + 2\kappa)^2$. So $z' = \lambda z$ with $\lambda = (\frac{\Sigma}{\Sigma+2\kappa})^2 \geq 1 - \frac{4\kappa}{\Sigma}$. Then

$$W - W' = \frac{(1-\lambda)z}{(1-z)(1-\lambda z)} \leq (1-\lambda)W(1+W) \leq \frac{4\kappa}{\Sigma}\, W\, \frac{\Sigma}{2(1+c)} = \frac{2\kappa}{1+c}W,$$

where we used $1 - \lambda z \geq 1 - z$ and then (i). Since $c = \kappa(s+10) \geq 11\kappa$, we have $\frac{2\kappa}{1+c} \leq \frac{2}{11}$.

(iii) $\Sigma \geq s$, $1 + c > \kappa s$ and $\Sigma^2 \geq 3s$. The first two are clear. For the third, $W > 0$ means $s \neq 2b$, and with $b \geq 1$ and $s - b \geq 1$ this forces $s \geq 3$.

Step 4: the terms with W. We bound the four positive terms one by one.

$$\begin{aligned}
\kappa(W - W') &\leq \frac{2\kappa^2}{1+c}W \leq \frac{2\kappa}{s}W && \text{by (ii) and } 1 + c > \kappa s,\\
\frac{2\kappa W}{\Sigma} &\leq \frac{2\kappa}{s}W && \text{by } \Sigma \geq s,\\
\frac{2\kappa W}{\Sigma^2} &\leq \frac{2\kappa}{3s}W && \text{by } \Sigma^2 \geq 3s,\\
\frac{4\kappa W^2}{3\Sigma^2} &\leq \frac{4\kappa W(1+W)}{3\Sigma^2} \leq \frac{2\kappa W}{3\Sigma(1+c)} \leq \frac{2\kappa}{3s}W && \text{by (i) and } \Sigma \geq s.
\end{aligned}$$

The four bounds add up to $\frac{16\kappa}{3s}W$. For the negative term, (ii) gives $-\nu_s W' \leq -\frac{9}{11}\nu_s W = -\frac{18\kappa}{s(1+2\beta)}W$. So

$$\log \Delta_s(b) \leq \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W\Big[\frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\beta)}\Big],$$

and the bracket is negative because $1 + 2\beta < 2$ for $\varepsilon < \frac{1}{4}$, so $18/(1+2\beta) > 9 > 16/3$.

(c) We first compare $V_{t+1}(1)$ with $\Phi_{t+1}^{\kappa_{t+1}}(1)$, and then this with $\pi_t = \Phi_{t+1}^{0}(1)$. For the first step apply Lemma A.2(a) with $s_0 = t+1$, $F_s = \Phi_s^{\kappa_s}$ and $K_s = \exp(\frac{\kappa}{s} + \frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h) = 1$, $F_s > 0$ on $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} = \Delta_s F_s \le K_s F_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s \ge s$. The product of the K_s over $t < s < N$ is the exponential of $\kappa(H_{N-1} - H_t) + \frac{2\kappa}{3}\sum_{s>t} s^{-2}$, and $\sum_{s>t} s^{-2} \le \frac{1}{t}$. Since $1 \in \mathcal{S}_{t+1}$, we get $V_{t+1}(1) \le e^{\kappa[H_{N-1} - H_t + 2/(3t)]}\, \Phi_{t+1}^{\kappa_{t+1}}(1)$.

For the second step, $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau \in [0, \kappa_{t+1}]$ of the derivative in Lemma A.3 with $s = t+1$ and $b = 1$. As in Step 2 of (b), the bound of that lemma is largest at $\tau = 0$. There $X = 1$, $Y = t$ and $\frac{1}{1-z} = \frac{(t+1)^2}{4t}$, and the bound is $\log\frac{(t+1)^2}{4t} + \frac{t^2+1}{2t(t+1)} + \frac{t^2+1}{12t^2} \le \log\frac{(t+1)^2}{4t} + \frac{2}{3}$, because the two fractions are at most $\frac{1}{2}$ and $\frac{1}{6}$ for $t \ge 1$. Multiplying by the length κ_{t+1} of the interval gives (c).

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1} = \kappa(t+11)$ and $\pi_t = c_1 w_t$. □

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N \ge \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1}$. Now $(1-\varepsilon)^{t-1} \ge 1-(t-1)\varepsilon$ and $\sum_t w_t(t-1) = 2 - \frac{6}{h+2}$ by Proposition 2.5(c). So $C_N \ge \varepsilon c_1(1-2\varepsilon)$, and $c_1 = \frac{4}{N+2}$.

(b) *Step 1: the constants.* Let $N \ge 64$ and $\varepsilon(1+\log N) \le \frac{1}{100}$. Then $\varepsilon < 0.002$ and $\kappa \le 2\varepsilon(1+8\varepsilon) \le 2.1\varepsilon$. Since $\log\frac{(t+1)^2}{4t} \ge 0$ and $\frac{2}{3}(t+11) > 1 + \log t \ge H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \le \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \ge \frac{2}{3t}$, also $\mathcal{G}(t) \le (t+11)\log(t+1) \le 12t\log(t+1)$. So $\kappa\mathcal{G}(t) \le 25.2\,\varepsilon t \log(t+1)$, and this is at most $\frac{t}{2}\log 2$ for $t \le h$, since then $25.2\,\varepsilon\log(t+1) \le 25.2\,\varepsilon\log N \le 0.252 < \frac{1}{2}\log 2$.

Step 2: the sum over t. Since $e^y - 1 \le y e^y$ for $y \ge 0$, Step 1 gives $e^{\kappa\mathcal{G}(t)} - 1 \le \kappa\mathcal{G}(t) 2^{t/2}$. By Proposition A.4(d), $(1-\varepsilon)^{t-1} \le 1$, $\sum_t w_t = 1$ and $w_t \le 2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \le \varepsilon c_1 e^{\kappa H_{N-1}} \Big(1 + 50.4\,\varepsilon \sum_{t\ge 1} t^2 \log(t+1) 2^{-t/2}\Big),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1+\frac{1}{t})^2 \frac{\log(t+2)}{\log(t+1)} 2^{-1/2}$, which decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than four times the term at $t = 61$, and this is below 10^{-4}. The first 60 terms add up to 102.5864 to four decimals (computed in 30-digit arithmetic by `verify_elephant_crossing.py`), so the series is below 102.59.

Step 3: the factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \le 1 + \log N$, so $\kappa H_{N-1} \le 2\varepsilon\log N + 2\varepsilon + 16\varepsilon^2(1 + \log N) \le 2\varepsilon\log N + 3\varepsilon$. With $1 + y \le e^y$ this proves (b), since $3 + 50.4 \cdot 102.59 < 5200$. □

A.1 The bridge and the lower bound

Fix $1 \le t \le h$, and start the Pólya(c) chain at state 1 at time $t+1$. By Lemma A.1(b), given that it is at h at time N it moves up at time u from b with probability $\bar r_u(b) = \frac{h-b}{N-u}$, whatever c is. So its $h-1$ up steps are placed uniformly at random among the $N-t-1$ steps after time $t+1$. We call this law of the path $(M_u)_{t+1\le u\le N}$ the bridge, and we write $\mathbb{P}_t^{\mathrm{br}}$ and $\mathbb{E}_t^{\mathrm{br}}$ for it. For $1 \le b \le u-1$ let

$$\varpi_u^+(b) = 1 + \varepsilon\frac{u-2b}{b}, \qquad \varpi_u^-(b) = 1 - \varepsilon\frac{u-2b}{u-b},$$

$$\theta_u(b) = \frac{N(u-2b)^2}{2(N-u)\,b(u-b)}, \qquad Q_u(b) = (u-2b)^2\Big[\frac{\bar r_u(b)}{b^2} + \frac{1-\bar r_u(b)}{(u-b)^2}\Big],$$

and let $\varpi_u = \varpi_u^+(M_u)$ if $M_{u+1} = M_u + 1$ and $\varpi_u = \varpi_u^-(M_u)$ otherwise. Put

$$A_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\, \theta_u(M_u), \qquad B_t = \sum_{u=t+1}^{N-1} \mathbb{E}_t^{\mathrm{br}}\, Q_u(M_u).$$

Both are nonnegative, and neither depends on ε.

Proposition A.5 (bridge formula)**.** *Let $0 \le \varepsilon \le 1$ and $1 \le t \le h$. Then*

$$V_{t+1}(1) = \pi_t\, \mathbb{E}_t^{\mathrm{br}}\Big[\prod_{u=t+1}^{N-1} \varpi_u\Big].$$

If $0 < \varepsilon \le \frac{1}{2}$, then $\log V_{t+1}(1) \ge \log \pi_t + \varepsilon A_t - \varepsilon^2 B_t$.

Proof. We divide V by the hitting probability of Pólya's urn. The quotient satisfies a recursion whose weights are the step probabilities of the bridge, and each step leaves one factor ϖ_u. For $u \ge t+1$ and $b \in \mathcal{S}_u$ we have $\Phi_u^0(b) > 0$, and we put $R_u(b) = V_u(b)/\Phi_u^0(b)$. By Lemma A.1(a) and (b) with $c = 0$, $\frac{b}{u}\Phi_{u+1}^0(b+1) = \bar r_u(b)\Phi_u^0(b)$ and $(1 - \frac{b}{u})\Phi_{u+1}^0(b) = (1 - \bar r_u(b))\Phi_u^0(b)$. Dividing $V_u = \mathcal{L}_u V_{u+1}$ by $\Phi_u^0(b)$ gives

$$R_u(b) = (1 - \bar r_u(b))\, \frac{1 - r_u(b)}{1 - b/u}\, R_{u+1}(b) + \bar r_u(b)\, \frac{r_u(b)}{b/u}\, R_{u+1}(b+1),$$

where a term with an infeasible state is absent, and its bridge probability is 0 (if $b = h$ then $\bar r_u(b) = 0$, and if $u - b = h$ then $\bar r_u(b) = 1$). Also $\frac{r_u(b)}{b/u} = \varpi_u^+(b)$ and $\frac{1 - r_u(b)}{1 - b/u} = \varpi_u^-(b)$. So $R_u(b) = \mathbb{E}_t^{\mathrm{br}}[\varpi_u R_{u+1}(M_{u+1}) \mid M_u = b]$, and $R_N(h) = 1$. Iterating from $u = t+1$ and using $\Phi_{t+1}^0(1) = \pi_t$ gives the formula.

Now let $0 < \varepsilon \le \frac{1}{2}$. We take the logarithm of the formula and bound it from below in three steps. First, every factor is at least $\frac{1}{2}$. Indeed if $b \le u/2$ then $\varpi_u^+(b) \ge 1$ and $\varpi_u^-(b) - 1 \in [-\varepsilon, 0]$, and if $b \ge u/2$ the same holds with $+$ and $-$ exchanged. Second, by Jensen's inequality for the logarithm,

$$\log \mathbb{E}_t^{\mathrm{br}}\Big[\prod_u \varpi_u\Big] \ge \sum_u \mathbb{E}_t^{\mathrm{br}} \log \varpi_u \ge \sum_u \mathbb{E}_t^{\mathrm{br}}\big[(\varpi_u - 1) - (\varpi_u - 1)^2\big].$$

The second inequality is $\log(1+y) \ge y - y^2$ for $y \ge -\frac{1}{2}$, which holds since the difference vanishes at 0 and its derivative $y(1+2y)/(1+y)$ has the sign of y. Third, we compute the two means given $M_u = b$. Write $\bar r = \bar r_u(b)$. The bridge moves up with probability $\bar r$, and then $\varpi_u - 1 = \varepsilon\frac{u-2b}{b}$. Otherwise $\varpi_u - 1 = -\varepsilon\frac{u-2b}{u-b}$. Since $u\bar r - b = \frac{N(u-2b)}{2(N-u)}$, this gives

$$\mathbb{E}_t^{\mathrm{br}}[\varpi_u - 1 \mid M_u = b] = \varepsilon(u-2b)\frac{u\bar r - b}{b(u-b)} = \varepsilon\,\theta_u(b), \qquad \mathbb{E}_t^{\mathrm{br}}[(\varpi_u - 1)^2 \mid M_u = b] = \varepsilon^2 Q_u(b).$$

So the sum above is $\varepsilon A_t - \varepsilon^2 B_t$, which is the bound. □

Lemma A.6 (the weights w_t)**.** *For each fixed t, $w_t \to t2^{-t-1}$ as $N \to \infty$. For every even $N \ge 2$, $\sum_t w_t(t+1)^3 \le 124$.*

Proof. Each fraction in $\pi_t = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}$ tends to $\frac{1}{2}$, so $N\pi_t \to t2^{1-t}$, and $Nc_1 \to 4$. This is the first claim. For the second claim we write w_t and $(t+1)^3$ through binomial coefficients in t. By the proof of Lemma 2.4 (with $v = t+1$ and $d = h$), π_t is a constant times $\binom{t}{1}\binom{N-1-t}{h-1}$, and

the Chu–Vandermonde identity gives $\sum_t \binom{t}{j}\binom{N-1-t}{h-1} = \binom{N}{h+j}$. With $j=1$ this gives the constant, namely $w_t = \binom{t}{1}\binom{N-1-t}{h-1}/\binom{N}{h+1}$. Now $t(t+1)^3 = 8\binom{t}{1} + 38\binom{t}{2} + 54\binom{t}{3} + 24\binom{t}{4}$, because both sides are polynomials of degree 4 that agree at $t = 0, 1, 2, 3, 4$. So $\sum_t w_t(t+1)^3$ is the sum of 8, 38, 54 and 24 times $\binom{N}{h+j}/\binom{N}{h+1}$ for $j = 1, 2, 3, 4$. Since $\binom{N}{h+j} \le \binom{N}{h+1}$ for $j \ge 1$, it is at most $8 + 38 + 54 + 24 = 124$. □

Under $\mathbb{P}_t^{\mathrm{br}}$ let $K = M_u - 1$ and $K' = m - K$ be the numbers of up steps and of other steps among the $m = u - t - 1$ steps between times $t+1$ and u. So K is hypergeometric, with m draws from $N - t - 1$ steps of which $h - 1$ are up steps. Let $\Omega_u = \{u/8 \le M_u \le 7u/8\}$.

Lemma A.7. *Let $N \ge 16$, $1 \le t \le h$ and $t + 1 \le u \le N - 1$.*

(a) $\mathbb{E}_t^{\mathrm{br}}(u - 2M_u)^2 = \dfrac{(t-1)^2(N-u)^2}{(N-t-1)^2} + \dfrac{4(u-t-1)(h-1)(h-t)(N-u)}{(N-t-1)^2(N-t-2)} \le (t-1)^2 + u.$

(b) If $t \le \frac{N+2}{4}$, then $A_t \ge 2(H_{N-1} - H_t - 1 - \frac{1}{t}) \ge 2(\log N - \log t - 3)$.

(c) If $t \le \frac{N}{8}$ and $u \ge 8(t+1)$, then $\mathbb{P}_t^{\mathrm{br}}(\Omega_u^c) \le 2e^{-0.0943u}$.

(d) If $t \le \frac{N}{8}$, then $B_t \le 84(t + 1 + \log N) + 683(t+1)^3 + 19100$.

Proof. (a) Here $u - 2M_u = u - 2 - 2K$ and $u - 2 = m + t - 1$. Since $N - 2 = 2(h-1)$, the mean $\mathbb{E}K = \frac{m(h-1)}{N-t-1}$ gives

$$\mathbb{E}(u - 2M_u) = \frac{(m+t-1)(N-t-1) - m(N-2)}{N-t-1} = \frac{(t-1)(N-t-1-m)}{N-t-1} = \frac{(t-1)(N-u)}{N-t-1}.$$

The variance of a hypergeometric variable with m draws from $N - t - 1$ steps, of which $h - 1$ are up steps and $h - t$ are not, is $\operatorname{Var} K = \frac{m(h-1)(h-t)(N-t-1-m)}{(N-t-1)^2(N-t-2)}$, and $N - t - 1 - m = N - u$. So the second term is $4 \operatorname{Var} K$, and the identity is $\mathbb{E}(u - 2M_u)^2 = (\mathbb{E}(u - 2M_u))^2 + 4 \operatorname{Var} K$. For the bound, the first term is at most $(t-1)^2$ because $N - u \le N - t - 1$. In the second term $4(h-1)(h-t) = (N-2)(N-2t) = (N-t-1)^2 - (t-1)^2 \le (N-t-1)^2$, so it is at most $\frac{m(N-u)}{N-t-2}$. This is 0 for $u = t + 1$, and for $u \ge t + 2$ it is at most u, because $m \le u$ and $N - u \le N - t - 2$.

(b) We bound θ_u from below by a multiple of $(u - 2b)^2$ and use only the variance term of (a). Since $b(u-b) \le u^2/4$, $\theta_u(b) \ge \frac{2N(u-2b)^2}{(N-u)u^2}$. By (a), $\mathbb{E}_t^{\mathrm{br}}(u - 2M_u)^2 \ge 4 \operatorname{Var} K$, and in $4 \operatorname{Var} K$ the factor $N - u$ cancels. With $4(h-1)(h-t) = (N-2)(N-2t)$ this gives $A_t \ge P_t \sum_{u=t+1}^{N-1} \frac{u-t-1}{u^2}$ with $P_t = \frac{2N(N-2)(N-2t)}{(N-t-1)^2(N-t-2)}$. We show that $P_t \ge 2$. Here $P_1 = \frac{2N}{N-3} > 2$, and for real $t \in [1, \frac{N+2}{4}]$, $\frac{d}{dt}\log P_t = -\frac{2}{N-2t} + \frac{2}{N-t-1} + \frac{1}{N-t-2} \ge \frac{3}{N-t-1} - \frac{2}{N-2t} \ge 0$, since the last inequality is equivalent to $4t \le N + 2$. Next, the sum is $H_{N-1} - H_t - (t+1)\sum_{u=t+1}^{N-1} u^{-2}$, which is at least $H_{N-1} - H_t - \frac{t+1}{t}$. The sum is also nonnegative. So multiplying by $P_t \ge 2$ gives the first inequality, whether this lower bound is positive or not. For the second inequality use $H_{N-1} - H_t \ge \log \frac{N}{t+1} \ge \log N - \log t - \frac{1}{t}$.

(c) Here $m \ge \frac{7}{8}u$, so $\frac{u}{8} \le \frac{m}{7}$. Since $t \ge 1$, $\mathbb{E}K \ge \frac{m}{2}$, and since $t \le \frac{N}{8}$, $\mathbb{E}K' = \frac{m(h-t)}{N-t-1} \ge \frac{3m}{8}$. Hoeffding's inequality also holds for sampling without replacement [26, Section 6], so $\mathbb{P}(K - \mathbb{E}K \le -\sigma m) \le e^{-2\sigma^2 m}$ for $\sigma > 0$, and the same holds for K'. Now $\{M_u < \frac{u}{8}\} \subset \{K - \mathbb{E}K < -\frac{5m}{14}\}$, and since $u - M_u = t + K'$, also $\{u - M_u < \frac{u}{8}\} \subset \{K' - \mathbb{E}K' < -\frac{13m}{56}\}$. So $\mathbb{P}(\Omega_u^c) \le e^{-2(5/14)^2 m} + e^{-2(13/56)^2 m} \le 2e^{-0.10778m} \le 2e^{-0.0943u}$.

(d) We split each term $\mathbb{E}_t^{\mathrm{br}} Q_u(M_u)$ of B_t into the part on Ω_u and the part off Ω_u, and the three terms of the bound come from the parts on Ω_u, the parts off Ω_u with $u < 8(t+1)$, and the parts off Ω_u

with $u \ge 8(t+1)$. In all of them we use that $\bar r(u-b)^2+(1-\bar r)b^2 \le u^2$, so that $Q_u(b) \le \frac{(u-2b)^2u^2}{b^2(u-b)^2}$. On Ω_u we have $b(u-b) \ge \frac{7u^2}{64}$, so $Q_u \le \frac{4096}{49}\frac{(u-2b)^2}{u^2} \le 83.6\frac{(u-2b)^2}{u^2}$. By (a) these parts add up to at most $83.6\sum_{u>t}(\frac{(t-1)^2}{u^2}+\frac{1}{u}) \le 83.6(t+H_{N-1}) \le 84(t+1+\log N)$. Off Ω_u we only have $b(u-b) \ge u-1 \ge \frac{u}{2}$, which gives $Q_u \le 4(u-2b)^2 \le 4u^2$. For $u < 8(t+1)$ we bound the probability of Ω_u^c by 1, and these parts add up to at most $\sum_{u<8(t+1)} 4u^2 \le \frac{4}{3}(8(t+1))^3 \le 683(t+1)^3$. For $u \ge 8(t+1)$, (c) bounds the part by $8u^2e^{-0.0943u}$, and with $q = e^{-0.0943}$ we have $8\sum_{u\ge1} u^2q^u = 8q(1+q)/(1-q)^3 < 19100$. □

Proof of Proposition 2.9. The upper bound is Theorem 2.6(b). For the lower bound let $t_0 = \lfloor N/8 \rfloor \ge 8$. By Proposition 2.5(a), $C_N = \varepsilon c_1 \sum_t w_t(1-\varepsilon)^{t-1}V_{t+1}(1)/\pi_t$. This is εc_1 times a mean over the probability vector (w_t). By Jensen's inequality the logarithm of the mean is at least the mean of the logarithms. For $t \le t_0$ we bound $\log(V_{t+1}(1)/\pi_t)$ from below by Proposition A.5. For $t > t_0$ we use only that it is ≥ 0, which is Lemma A.1(d). This gives

$$\log C_N \ge \log(\varepsilon c_1) + \sum_t w_t(t-1)\log(1-\varepsilon) + \sum_{t\le t_0} w_t\big(\varepsilon A_t - \varepsilon^2 B_t\big).$$

The first sum. By $\log(1-\varepsilon) \ge -\varepsilon-\varepsilon^2$ and Proposition 2.5(c), it is at least $-(\varepsilon+\varepsilon^2)\sum_t w_t(t-1) \ge -2\varepsilon - 2\varepsilon^2$.

The terms with A_t. Since $t_0 \le \frac{N+2}{4}$, Lemma A.7(b) gives $A_t \ge 2(\log N - \log t - 3)$ for $t \le t_0$. Let $w_\le = \sum_{t\le t_0} w_t$. By Jensen's inequality $\sum_t w_t \log t \le \log \sum_t w_t t \le \log 3$, so

$$\sum_{t\le t_0} w_t\,\varepsilon A_t \ge 2\varepsilon(\log N - 3)w_\le - 2\varepsilon\log 3 \ge 2\varepsilon\log N - 2\varepsilon(1-w_\le)\log N - 6\varepsilon - 2\varepsilon\log 3.$$

By Proposition 2.5(c), $1 - w_\le \le 2\sum_{t>t_0} t2^{-t} = 2(t_0+2)2^{-t_0}$, and $N < 8(t_0+1)$. As $(t_0+2)2^{-t_0}\log(8t_0+8)$ decreases for $t_0 \ge 8$, we get $2\varepsilon(1-w_\le)\log N \le 4\cdot 10\cdot 2^{-8}\log 72\,\varepsilon < 0.67\,\varepsilon$.

The terms with B_t. Here $t \le t_0 \le \frac{N}{8}$, so Lemma A.7(d) applies. With $\sum_t w_t(t+1) \le 4$ (Proposition 2.5(c)) and Lemma A.6 it gives $\sum_{t\le t_0} w_tB_t \le 84(3+1+\log N) + 683\cdot 124 + 19100 = 84\log N + 104128$.

Collecting terms, and using $c_1 = \frac{4}{N+2}$,

$$\log C_N - \log\frac{4\varepsilon}{N+2} - 2\varepsilon\log N \ge -(2+0.67+6+2\log 3)\,\varepsilon - \varepsilon^2(84\log N + 104130),$$

which gives the lower bound since $2+0.67+6+2\log 3 < 11$. For the last claim, the hypothesis on ε gives $\varepsilon^2(84\log N + 1.05\cdot 10^5) \le \varepsilon\,\frac{84\log N + 1.05\cdot 10^5}{100(1+\log N)}$. The fraction decreases in N, and its value at $N = 64$ is below 205. Since $11 + 205 < 5200$, the two-sided bound follows. □

A.2 The coefficient of ε^2

We prove Proposition 2.11 in three parts. Lemma A.8(a) writes $k_2(N)$ through the sums A_t. Lemma A.8(b) finds the limit of $A_t - 2\log N$ for fixed t, and (c) bounds it uniformly in t. Then we sum over t.

Lemma A.8. *(a) For every even $N \ge 2$, $k_2(N) = \sum_{t=1}^{h} w_t\big[A_t - (t-1)\big]$.*

(b) For each fixed $t \ge 1$, as $N \to \infty$ over even N, $A_t - 2\log N \to A_t^\infty := 2(\gamma - 1 - \log 2) + (t-2)H_{t-1}$.

(c) For even $N \ge 64$ and $1 \le t \le h$, $-2\log(4t) - 6 \le A_t - 2\log N \le 2 + 2(t+11)\log(t+1)$.

Idea of the proof of (b). Fix t and let N be large. At a fixed time u the bridge looks like a walk with fair steps that starts at state 1 at time $t+1$. For such a walk $(u-2M_u)^2$ is about u and $M_u(u-M_u)$ is about $u^2/4$, so θ_u is about $2/u$, and the sum over $u<N$ is about $2\log N$. To find the constant we split θ_u into a part that is quadratic in $u-2M_u$, whose mean over the bridge we know exactly (Lemma A.7(a)), and a remainder whose means are summable over u, uniformly in N. So $A_t-2\log N$ has the same limit as the corresponding quantity for the fair walk. For the fair walk the mean of θ_u is an explicit integral over $[0,1]$, and we can sum these integrals over u.

Proof. (a) By Proposition 2.5(a) and (b), $c_2=\sum_t[-(t-1)\pi_t+\partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0}]$. In Proposition A.5 the bridge does not depend on ε, and each ϖ_u is affine in ε and equals 1 at $\varepsilon=0$. So by the conditional mean computed in its proof, $\partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0}=\pi_t\sum_u \mathbb{E}_t^{\mathrm{br}}[(\varpi_u-1)/\varepsilon]=\pi_t A_t$. Divide by c_1.

(c) Both sides of Proposition A.4(c) are differentiable in ε on $[0,\frac14)$ and equal π_t at $\varepsilon=0$, and $\kappa=2\varepsilon/(1-4\varepsilon)$ has derivative 2 at 0. Comparing the derivatives at 0 and using the proof of (a) gives $A_t\le 2H_{N-1}+2\mathcal{G}(t)$. By Step 1 in the proof of Theorem 2.6(b), $\mathcal{G}(t)\le(t+11)\log(t+1)$, and $H_{N-1}\le 1+\log N$. This is the upper bound. For $t\le\frac{N+2}{4}$ the lower bound follows from Lemma A.7(b). For $t>\frac{N+2}{4}$ we use $A_t\ge 0$ and $N<4t$.

(b) Fix t and let $N\ge 16(t+1)$. Recall that $m=u-t-1$ and $K=M_u-1$. Steps 1 to 5 show that the limit exists and is a limit for the fair walk. Steps 6 to 9 compute it.

Step 1: split θ_u. For $1\le b\le u-1$ we have $u^2=4b(u-b)+(u-2b)^2$, so $\frac{1}{2b(u-b)}=\frac{2}{u^2}+\frac{(u-2b)^2}{2b(u-b)u^2}$. Multiplying by $\frac{N(u-2b)^2}{N-u}$ gives

$$\theta_u(b)=\frac{N}{N-u}\Big[\frac{2(u-2b)^2}{u^2}+\vartheta_u(b)\Big],\qquad \vartheta_u(b)=\frac{(u-2b)^4}{2b(u-b)u^2}\ge 0.$$

The first term in the bracket is what we get when we replace $b(u-b)$ by its largest value $u^2/4$, and ϑ_u is the correction. So $A_t=A_t'+A_t''$, where

$$A_t'=\sum_{u=t+1}^{N-1}\frac{2N\,\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^2}{(N-u)u^2},\qquad A_t''=\sum_{u=t+1}^{N-1}\frac{N}{N-u}\,\mathbb{E}_t^{\mathrm{br}}\vartheta_u(M_u).$$

Step 2: the quadratic part A_t'. Write $\varsigma_t=\sum_{u>t}u^{-2}$. We show that $A_t'=2\log N+2\gamma-2H_t+2((t-1)^2-t-1)\varsigma_t+o(1)$. We insert the formula of Lemma A.7(a). In its second term the factor $N-u$ cancels, and we get

$$A_t'=\sum_{u=t+1}^{N-1}\frac{2N(t-1)^2(N-u)}{(N-t-1)^2u^2}+P_t\sum_{u=t+1}^{N-1}\frac{u-t-1}{u^2},$$

with P_t as in the proof of Lemma A.7(b). In the first sum the factor $\frac{2N(N-u)}{(N-t-1)^2}$ is at most 8 (because $N-t-1\ge\frac{N}{2}$) and tends to 2 for fixed u. So this sum tends to $2(t-1)^2\varsigma_t$ by dominated convergence. The second sum is $H_{N-1}-H_t-(t+1)\sum_{u=t+1}^{N-1}u^{-2}=\log N+\gamma-H_t-(t+1)\varsigma_t+o(1)$, and $P_t=2+O(1/N)$. Adding the two gives the claim.

Step 3: the terms of A_t'' converge. Fix u. The variable K counts the up steps among m steps drawn without replacement from $N-t-1$ steps, of which $h-1$ (about half) are up steps. So as $N\to\infty$ its law tends to the binomial law with m trials and success probability $\frac12$. Let ν_u have this binomial law and put $\bar M_u=1+\nu_u$. This is the walk with fair steps from state 1 at time $t+1$. Since $\frac{N}{N-u}\to 1$, the u-th term of A_t'' tends to $\mathbb{E}\vartheta_u(\bar M_u)$.

Step 4: the terms of A_t'' are dominated. To exchange the limit and the sum we need a bound for the u-th term of A_t'' that is summable over u and does not depend on N. We use four facts.

First, $\mathbb{E}_t^{\mathrm{br}}(u-2M_u)^4 \le 64\min(u,N-u)^2 + 8t^4$. To see this, note that $u-2M_u-\mathbb{E}(u-2M_u) = -2(K-\mathbb{E}K)$, where K is hypergeometric with $m \le u$ draws. The last $N-u$ steps of the bridge also form a uniform sample, so $h-M_u$ is hypergeometric with $N-u$ draws, and $u-2M_u-\mathbb{E}(u-2M_u) = 2(h-M_u-\mathbb{E}(h-M_u))$. By Hoeffding's comparison with sampling with replacement [26, Theorem 4], applied to the convex function $x \mapsto (x-x_0)^4$, the fourth central moment of a hypergeometric variable with n draws is at most that of the binomial variable with n trials and the same mean. The latter is $np(1-p)(1+3(n-2)p(1-p)) \le n^2/2$, where p is the success probability. We may take $n=m$ or $n=N-u$, so the fourth central moment of $u-2M_u$ is at most $8\min(u,N-u)^2$. Now use $|\mathbb{E}(u-2M_u)| \le t$ (by the proof of Lemma A.7(a)) and $(x_1+x_2)^4 \le 8(x_1^4+x_2^4)$.

Second, $\frac{N}{N-u} \le 2u$ and $\frac{N}{N-u}\min(u,N-u)^2 \le 2u^2$. The first holds because $u(N-u) \ge N-1 \ge \frac{N}{2}$. The second is clear for $u \le \frac{N}{2}$, and for $u > \frac{N}{2}$ it follows from $N(N-u) \le 2u^2$. Third, on Ω_u we have $2M_u(u-M_u)u^2 \ge \frac{7}{32}u^4$, so $\vartheta_u(M_u) \le \frac{32}{7u^4}(u-2M_u)^4$. By the first two facts the part of the u-th term on Ω_u is at most $\frac{32}{7u^4}(128u^2+16t^4u)$. Fourth, off Ω_u we only have $M_u(u-M_u) \ge \frac{u}{2}$, which gives $\vartheta_u \le u$. So the part of the u-th term off Ω_u is at most $2u^2\,\mathbb{P}_t^{\mathrm{br}}(\Omega_u^c)$. This is at most $4u^2e^{-0.0943u}$ for $u \ge 8(t+1)$ by Lemma A.7(c), and at most $2u^2$ for the finitely many smaller u. The sum of the two bounds is summable over u and does not depend on N.

Step 5: the limit as a sum for the fair walk. By Steps 2 to 4 and dominated convergence,

$$A_t - 2\log N \to \tilde{A}_t := 2\gamma - 2H_t + 2\big((t-1)^2-t-1\big)\varsigma_t + \sum_{u>t}\mathbb{E}\vartheta_u(\bar{M}_u).$$

We now undo the split of Step 1 for the fair walk. Let $\bar{\theta}_u = \mathbb{E}\frac{(u-2\bar{M}_u)^2}{2\bar{M}_u(u-\bar{M}_u)} = \mathbb{E}\big[\frac{2(u-2\bar{M}_u)^2}{u^2} + \vartheta_u(\bar{M}_u)\big]$, where the second form is the identity of Step 1. This is the mean of θ_u for the fair walk, without the factor $\frac{N}{N-u}$. The variable $u-2\bar{M}_u = (m+t-1)-2\nu_u$ has mean $t-1$ and variance m, so $\mathbb{E}(u-2\bar{M}_u)^2 = (t-1)^2+(u-t-1)$. So the computation of Step 2 gives $\sum_{u=t+1}^{u_1}\mathbb{E}\frac{2(u-2\bar{M}_u)^2}{u^2} = 2\log u_1 + 2\gamma - 2H_t + 2((t-1)^2-t-1)\varsigma_t + o(1)$ as $u_1 \to \infty$. We subtract $2\log u_1$ and add the convergent sum of the $\mathbb{E}\vartheta_u(\bar{M}_u)$. Hence

$$\tilde{A}_t = \lim_{u_1\to\infty}\Big(\sum_{u=t+1}^{u_1}\bar{\theta}_u - 2\log u_1\Big). \tag{14}$$

Step 6: $\bar{\theta}_u$ as an integral. It remains to compute the limit in (14). Write $\nu = \nu_u$, and for $0 \le q \le 1$ let $y = \frac{1+q}{2}$, so that $\mathbb{E}q^\nu = y^m$. Since $(u-2b)^2 = u^2 - 4b(u-b)$ and $\frac{1}{b(u-b)} = \frac{1}{u}(\frac{1}{b}+\frac{1}{u-b})$, we have $\frac{(u-2b)^2}{2b(u-b)} = \frac{u}{2}(\frac{1}{b}+\frac{1}{u-b}) - 2$. We take $b = \bar{M}_u = 1+\nu$. Then $u-b = t+(m-\nu)$, and $m-\nu$ has the law of ν. So $\bar{\theta}_u = \frac{u}{2}\mathbb{E}[\frac{1}{1+\nu}+\frac{1}{t+\nu}] - 2$. For an integer $a \ge 1$ we have $\frac{1}{a+\nu} = \int_0^1 q^{a-1+\nu}\,dq$, so $\mathbb{E}\frac{1}{a+\nu} = \int_0^1 q^{a-1}y^m\,dq$. Hence

$$\bar{\theta}_u = u\int_0^1 \frac{1+q^{t-1}}{2}\,y^m\,dq - 2.$$

Step 7: split off the part with q^{t-1}. Let $\sigma(q) = \sum_{j=0}^{t-2}q^j$, with $\sigma = 0$ for $t = 1$. Since $1-y = \frac{1-q}{2}$, we have $(1-y)\sigma(q) = \frac{1-q^{t-1}}{2}$, that is, $\frac{1+q^{t-1}}{2} = 1-(1-y)\sigma(q)$. Also $\int_0^1 y^m\,dq = \frac{2(1-2^{-m-1})}{m+1}$ and $u = m+1+t$. So

$$\bar{\theta}_u = \Big[\frac{2t}{m+1} - \Big(1+\frac{t}{m+1}\Big)2^{-m}\Big] - \mathcal{T}_m, \qquad \mathcal{T}_m = u\int_0^1 (1-y)\sigma(q)y^m\,dq.$$

To compute $\mathcal{T}_m$ we split $u = (m+1) + t$. For the part with $m+1$ we integrate by parts, using $\frac{d}{dq} y^{m+1} = \frac{m+1}{2} y^m$. For $j \ge 0$ this gives

$$(m+1) \int_0^1 q^j \, \frac{1-q}{2} \, y^m \, dq = -\delta_{j0} 2^{-m-1} + \int_0^1 y^{m+1} \big((j+1) q^j - j q^{j-1} \big) \, dq,$$

where the first term on the right is the boundary term at $q = 0$. We sum over $0 \le j \le t-2$. The integrands telescope to $(t-1) q^{t-2} y^{m+1}$. For the part with t we use $(1-y)\sigma(q) = \frac{1-q^{t-1}}{2}$ again. With $I(j,k) = \int_0^1 q^j y^k \, dq$ this gives

$$\mathcal{T}_m = -[t \ge 2] \, 2^{-m-1} + (t-1) I(t-2, m+1) + \frac{t}{2} \int_0^1 (1 - q^{t-1}) y^m \, dq,$$

where $[t \ge 2]$ is 1 if $t \ge 2$ and 0 if $t = 1$, and the term with I is absent for $t = 1$.

Step 8: sum over u. We sum the formulas of Step 7 over $0 \le m \le m_0$, so that $u_1 = m_0 + t + 1$. The terms $\frac{2t}{m+1}$ add up to $2t H_{m_0+1}$, and the terms 2^{-m-1} add up to $1 - 2^{-m_0-1}$. Under the integrals we use $\sum_{m \le m_0} y^m = \frac{2(1-y^{m_0+1})}{1-q}$ and $\frac{1-q^{t-1}}{1-q} = \sigma(q)$. So

$$\sum_{u=t+1}^{u_1} \bar\theta_u = 2t H_{m_0+1} - \sum_{m=0}^{m_0} \Big(1 + \frac{t}{m+1} \Big) 2^{-m} + [t \ge 2](1 - 2^{-m_0-1}) \\ - (t-1) \sum_{k=0}^{m_0} I(t-2, k+1) - t \int_0^1 \sigma(q) \big(1 - y^{m_0+1} \big) dq.$$

Step 9: let $m_0 \to \infty$. We go through the terms of Step 8. The first sum tends to $\sum_{m \ge 0} (1 + \frac{t}{m+1}) 2^{-m} = 2 + 2t \log 2$, since $\sum_{m \ge 0} \frac{2^{-m}}{m+1} = 2 \log 2$. The last integral tends to $\int_0^1 \sigma \, dq = H_{t-1}$. The sum of the $I(t-2, k+1)$ grows like $2 \log m_0$, and we need it up to $o(1)$. For $t \ge 2$ we sum the geometric series as before and write $q^{t-2} = 1 - (1 - q^{t-2})$. So the sum is twice the difference of $\int_0^1 y \frac{1-y^{m_0+1}}{1-q} dq$ and $\int_0^1 (1 - q^{t-2}) y \frac{1-y^{m_0+1}}{1-q} dq$. In the first integral we substitute $q = 2y - 1$. It becomes $\int_{1/2}^1 (y + \cdots + y^{m_0+1}) \, dy = \sum_{j=2}^{m_0+2} \frac{1-2^{-j}}{j} = H_{m_0+2} - \frac{1}{2} - \log 2 + o(1)$. In the second integral $\frac{1-q^{t-2}}{1-q} = \sum_{j=0}^{t-3} q^j$ is bounded, so by dominated convergence it tends to $\int_0^1 \sum_{j=0}^{t-3} q^j \frac{1+q}{2} dq = \frac{1}{2}(H_{t-2} + H_{t-1} - 1)$. So $\sum_{k \le m_0} I(t-2, k+1) = 2H_{m_0+2} - 2\log 2 - H_{t-2} - H_{t-1} + o(1)$. We collect the terms. Since $2t H_{m_0+1} - 2(t-1) H_{m_0+2} = 2H_{m_0+1} + o(1)$, we get

$$\sum_{u=t+1}^{u_1} \bar\theta_u = 2H_{m_0+1} - 2 - 2\log 2 + [t \ge 2] + (t-1) H_{t-2} - H_{t-1} + o(1).$$

Here $[t \ge 2] + (t-1) H_{t-2} - H_{t-1} = (t-2) H_{t-1}$ for all $t \ge 1$ (for $t \ge 2$ use $H_{t-2} = H_{t-1} - \frac{1}{t-1}$), and $H_{m_0+1} = \log u_1 + \gamma + o(1)$. So by (14), $\tilde A_t = 2(\gamma - 1 - \log 2) + (t-2) H_{t-1} = A_t^\infty$. □

Proof of Proposition 2.11. We pass to the limit under the sum over t and then sum the series. Since $\sum_t w_t = 1$, Lemma A.8(a) gives $k_2(N) - 2\log N = \sum_{t \le h} w_t [A_t - 2\log N - (t-1)]$. For fixed t the t-th term tends to $t 2^{-t-1} [A_t^\infty - (t-1)]$ by Lemma A.8(b) and Lemma A.6. By Lemma A.8(c) and $w_t \le 2t 2^{-t}$, for $N \ge 64$ the t-th term is at most $C_0 (1 + t \log(t+1)) \, t 2^{-t}$ in absolute value, with a constant C_0 that does not depend on N or t. This bound is summable over t. So by dominated convergence,

$$k_2(N) - 2\log N \to \sum_{t \ge 1} t 2^{-t-1} \big[A_t^\infty - (t-1) \big].$$

To sum the series, note that $\sum_t t2^{-t-1} = 1$ and $\sum_t t(t-1)2^{-t-1} = 2$. Let $\mathcal{H}(q) = \sum_{k\ge 0} H_k q^k = \frac{\ell}{1-q}$ with $\ell = -\log(1-q)$, and let $D = \frac{d}{dq}$. With $k = t-1$,

$$\sum_{t\ge 1} t(t-2)2^{-t-1}H_{t-1} = \frac{1}{4}\sum_{k\ge 0}(k^2-1)H_k 2^{-k} = \frac{1}{4}\big[(qD)^2\mathcal{H} - \mathcal{H}\big]_{q=1/2}.$$

Here $(qD)^2\mathcal{H} = q\big[\frac{1+\ell}{(1-q)^2} + \frac{q}{(1-q)^3} + \frac{2q(1+\ell)}{(1-q)^3}\big]$ equals $8 + 6\log 2$ at $q = \frac{1}{2}$, and $\mathcal{H}(\frac{1}{2}) = 2\log 2$. So the sum is $2 + \log 2$, and the limit is $2(\gamma - 1 - \log 2) + (2 + \log 2) - 2 = 2\gamma - 2 - \log 2$. □

B Proofs for Section 3

We keep the conventions of Section 3. We write C for an absolute constant whose value may change from line to line, and $\bar{c}$, A_0, A_1 for fixed absolute constants. Since M is independent of X_1, $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The family $\mathcal{T}_v$ has the law β_v of Lemma 2.4. Let $\bar{\beta}_v(s) = \mathbb{P}(|\mathcal{T}_v| \ge s)$, with $\bar{\beta}_v(s) = 1$ for $s \le 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \le s \le N-1$,

$$\bar{\beta}_v(s) = \binom{N-s}{v-1}\Big/\binom{N-1}{v-1} = \prod_{i=1}^{v-1}\Big(1 - \frac{s-1}{N-i}\Big) \le e^{-(v-1)(s-1)/(N-1)}, \qquad \beta_v(s) = \frac{v-1}{N-s}\,\bar{\beta}_v(s), \tag{15}$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \qquad \frac{\bar{\beta}_v(s-1)}{\bar{\beta}_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \tag{16}$$

Lemma B.1 (monotone sandwich)**.** *Let $0 \le a \le b \le N$. If $m_h \le a$, then $(b-a+1)f_N(b) \le \mathbb{P}(a \le M_N \le b)$. If $b \le m_h$, then $(b-a+1)f_N(a) \le \mathbb{P}(a \le M_N \le b)$.*

Proof. By Lemma 3.4, $f_N(\ell) \ge f_N(b)$ for $a \le \ell \le b$ in the first case, and $f_N(\ell) \ge f_N(a)$ in the second. Sum over ℓ. □

Lemma B.2 (tail)**.** *Let $0 < \varepsilon < 1$, $k_0 \ge 1$ and $t \ge k_0$. For every integer $z \ge 1$,*

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \frac{\varepsilon t H_z}{z}, \qquad \mathbb{E}M_t^{[k_0]} \le \varepsilon t \log\frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \ge z) \le xH_z/z$, and $\mathbb{P}(Z_s^{(j)} \ge z) \le xH_z/z$ for $j-1 \le s \le N$.

Proof. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \le \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \ge z$, then some marked family is large, or the small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \ge z) + \frac{\varepsilon}{z}\sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

By the first sum in Lemma 2.4, with t in place of N, on average $\frac{t}{d(d+1)}$ vertices have a family of size d. So the first sum over v is $\sum_{d\ge z}\frac{t}{d(d+1)} \le \frac{t}{z}$, and the second is $\sum_{d=1}^{z-1}\frac{t}{d+1} = t(H_z - 1)$.

Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \le \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \le \varepsilon t \log \frac{t}{k_0}$. □

Lemma B.3 (late start)**.** *Let $j \ge 2$, $t \ge j-1$ and $z \ge 0$. Then $\mathbb{P}(M_t^{[j-1]} \ge z) \le \exp\{-\frac{j-1}{2}(\frac{z}{t} - \varepsilon \log \frac{2t}{j-1})\}$.*

Proof. This is a Chernoff bound with an exponent that changes with time. A chain that starts at 0 at the late time $j-1$ is like an urn that already holds $j-1$ balls of one color, so the fraction $M_k^{[j-1]}/k$ moves slowly. For this reason we take an exponent θ_k of size about $(j-1)/(2k)$.

Write $G_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k \ge j-1$ let $\theta_k = \log \frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. We claim that for $k \ge j-1$

$$\mathbb{E}e^{\theta_{k+1} G_{k+1}} \le e^{\varepsilon\nu/(k+1-\nu)}\, \mathbb{E}e^{\theta_k G_k}.$$

To see this, condition on $G_k = c$. The chain moves up with probability $r_k(c) \le \varepsilon + c/k$, and $1 + y \le e^y$. So

$$\mathbb{E}\big[e^{\theta_{k+1}G_{k+1}} \mid G_k = c\big] = e^{\theta_{k+1}c}\big(1 + r_k(c)(e^{\theta_{k+1}} - 1)\big) \le \exp\Big\{c\Big(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}\Big) + \frac{\varepsilon\nu}{k+1-\nu}\Big\}.$$

It remains to check that $\theta_{k+1} + \frac{\nu}{k(k+1-\nu)} \le \theta_k$. Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \ge \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. This proves the claim.

We start from $G_{j-1} = 0$ and use the claim for $k = j-1, \dots, t-1$. Since $\sum_{k=j}^{t} \frac{1}{k-\nu} \le \log \frac{t-\nu}{\nu} \le \log \frac{2t}{j-1}$, we get $\mathbb{E}e^{\theta_t G_t} \le \exp\{\varepsilon\nu \log \frac{2t}{j-1}\}$. Finally $\theta_t = -\log(1 - \frac{\nu}{t}) \ge \frac{\nu}{t}$, so Markov's inequality gives $\mathbb{P}(G_t \ge z) \le e^{-\nu z/t}\, \mathbb{E}e^{\theta_t G_t}$, which is the claim. □

Lemma B.4 (the first flipped family)**.** *For $1 \le d \le N-1$,*

$$\frac{x}{d(d+1)}\Big(1 - \frac{2x(N-d-1)}{N(d+2)}\Big) \le \mathbb{P}(D = d) \le \frac{x}{d(d+1)}, \qquad \mathbb{P}(D \ge d) \le x\Big(\frac{1}{d} - \frac{1}{N}\Big).$$

Proof. By Lemma 3.3, $\mathbb{P}(D = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and $1 - (j-2)\varepsilon \le (1-\varepsilon)^{j-2} \le 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \ge d$. □

Proposition B.5 (local lower bound)**.** *There is an absolute constant C such that the following hold.*

- *(a) For $2 \le m \le N/4$, $f_N(m) \ge \frac{x}{m(m+1)}(1 - e_1(m))$ with $e_1(m) = C\big(\frac{x \log(m+1)}{m} + \varepsilon \log N + me^{-m/9}\big)$.*
- *(b) For $N/4 < m \le N - 16$ and $r = N - m$, $f_N(m) \ge \frac{x}{m(m+1)}(1 - e_2(m))$ with $e_2(m) = C\big(\frac{x \log N}{r} + \frac{x \log N}{m} + Ne^{-r/40}\big)$.*

Idea of the proof. We use the formula of Lemma 3.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z_{N-d}^{(j)}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. (If no step of the family flips back and the rest of the tree has no flip, then $Y = Z = 0$ and the quantity is d itself, which hits m at $d = m$.) The law β_j of d is nearly flat near m, and this gives almost the full weight $\beta_j(m)$. Then we sum over j. The errors e_1 and e_2 collect the chance that Y or Z is large, the small slope of β_j, and the first flips that come too late to be treated this way.

Proof. (a) We bound the inner sum of Lemma 3.3 from below for each early first flip, namely for $2 \le j \le J := \lfloor N/8 \rfloor$. The terms with $j > J$ are nonnegative and we drop them.

Step 1: one value of j. Fix $2 \le j \le J$ and put $\Gamma_d = d - Y_d + Z^{(j)}_{N-d}$ for $1 \le d \le N-j+1$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E}S_j$ with $S_j = \sum_d \beta_j(d)\mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z^{(j)}_{N-1} \le m-1,\ Y_{2m} \le m\}$. We claim that

$$S_j \ge \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N}\, Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z^{(j)}_{N-d}$ falls by 0 or 1. So every step of Γ lies in $\{-1,0,1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z^{(j)}_{N-1} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (16),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac{1}{4}$, and a product of factors $1 - q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z^{(j)}_{N-1} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z^{(j)}_{N-1} \ge m)\big] - \frac{2(j-2)}{N}\,\mathbb{E}Y_{2m}\,\beta_j(m).$$

Step 2: the chance that Y or Z is large. Lemma B.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z^{(j)}_{N-1} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$, so that $p \le C(\frac{x\log(m+1)}{m} + \varepsilon \log N)$. If $p > 1$, then $e_1(m) > 1$ (once C is large) and there is nothing to prove. So let $p \le 1$.

Step 3: sum over j. By Lemma 3.3, $f_N(m) \ge \sum_{j=2}^{J} \mathbb{P}(V = j)\, Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \ge (1-p)\sum_{j=2}^{J} \mathbb{P}(V=j)\,\beta_j(m) - \frac{4\varepsilon m \log N}{N}\sum_{j=2}^{J}\mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V = j) \le \varepsilon$ and the second sum in Lemma 2.4. So the second term is at most

$$\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)}\, 8\varepsilon \log N.$$

The first sum is $\mathbb{P}(D = m)$ without the terms $j > J$. So by Lemma B.4 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j>J} \beta_j(m)$.

Step 4: the late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (15), $\beta_j(m) \le \frac{j-1}{N-m} e^{-(j-1)a}$ with $a = \frac{m-1}{N-1} \le \frac{1}{4}$. We sum this over $j - 1 \ge J$, using $1 - e^{-a} \ge a/2$ and $N - m \ge \frac{3}{4}N$. Since $\lfloor N/8 \rfloor \ge \frac{N-1}{9}$ for $N \ge 64$, we have $aJ \ge \frac{m-1}{9}$, and so

$$\varepsilon \sum_{j>J} \beta_j(m) \le \frac{C\varepsilon}{N}\Big(\frac{J}{a} + \frac{1}{a^2}\Big)e^{-(m-1)/9} \le \frac{x}{m(m+1)}\, Cme^{-m/9}.$$

Step 5: collect. For $0 \le p \le 1$ and $q \ge 0$ we have $(1-p)(1-q) \ge 1-p-q$. So Steps 3 and 4 give

$$f_N(m) \ge \frac{x}{m(m+1)}\Big(1 - p - \frac{2x}{m+2} - Cme^{-m/9} - 8\varepsilon \log N\Big),$$

and by the bound for p in Step 2 this is (a).

(b) Now the first flipped family has more than $N/4$ vertices and leaves only $r = N - m$ vertices outside. So we run the same argument on the scale r instead of m. We take $J = w := \lfloor r/8 \rfloor$ (so that $w \ge 2$), the event $\mathcal{E} = \{Z^{(j)}_{N-1} \le m-1,\ Y_{m+w} \le w\}$ and $d^\circ = m + Y_{m+w} \le m + w$. The index d° is allowed, because $m + w + j \le m + 2w \le N$. As in Step 1, on $\mathcal{E}$ we have $\Gamma_1 \le m$ and $\Gamma_{d^\circ} \ge m$, so $\Gamma_d = m$ for some $d \le d^\circ$. Three estimates change.

The slope of β_j. For $i \le y \le w$ we have $N - m - i \ge r - w \ge \frac{7}{8}r$. So each fraction in the product of Step 1 is at most $\frac{8(j-2)}{7r} \le \frac{1}{7}$, and $\beta_j(m+y) \ge \beta_j(m)(1 - \frac{8y(j-2)}{7r})$.

The chance that Y or Z is large. Since $w + 1 \ge r/8$ and $m + w \le N$, Lemma B.2 gives $\mathbb{P}(Y_{m+w} > w) \le 8xH_N/r$ and $\mathbb{E}Y_{m+w} \le x \log N$, and as before $\mathbb{P}(Z^{(j)}_{N-1} \ge m) \le xH_m/m$. So p is replaced by $p' = 8xH_N/r + xH_m/m \le C(\frac{x \log N}{r} + \frac{x \log N}{m})$, and we may assume $p' \le 1$. With $\mathbb{P}(V = j) \le \varepsilon$ and the second sum in Lemma 2.4, the terms with $\mathbb{E}Y_{m+w}$ add up to at most

$$\varepsilon \cdot \frac{8x \log N}{7r} \cdot \frac{2N(r-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)} \cdot \frac{3x \log N}{m}.$$

The late first flips. Now $a = \frac{m-1}{N-1} \ge \frac{1}{5}$ and $N - m = r$, so (15) gives $\beta_j(m) \le \frac{j-1}{r} e^{-(j-1)/5}$. We sum over $j - 1 \ge J$ and use $J \ge \frac{r-7}{8}$ and $J + 1 \le r$. This gives

$$\varepsilon \sum_{j>J} \beta_j(m) \le \frac{C\varepsilon(J+1)}{r} e^{-J/5} \le C\varepsilon e^{-r/40} \le \frac{x}{m(m+1)} CNe^{-r/40},$$

since $\varepsilon = \frac{x}{m(m+1)} \cdot \frac{m(m+1)}{N}$ and $m + 1 \le N$. Collecting the terms as in Step 5, with $\frac{2x}{m+2} \le \frac{x \log N}{m}$, gives (b). □

Proposition B.6 (cumulative upper bound)**.** *There is an absolute constant C such that for $6 \le m \le N/2$ with $m \ge 16x \log m$ we have $\mathbb{P}(M_N \ge m) \le \frac{x}{m}\big(1 + C\frac{x \log^2 m}{m}\big)$.*

Idea of the proof. Given the first flip at time j, the count M_N is at most $D + Z_j$, the size of the first flipped family plus the count from the rest of the tree. The event $D \ge m$ gives the main term x/m. The other terms come from a shift of D by Z_j, from a large Z_j, and from a late first flip, and each is smaller by a factor of order $x \log^2 m/m$.

Proof. Step 1: split by the first flip. Let $J = \lfloor N/4 \rfloor$ and $Z_j = Z^{(j)}_{N-1}$. First let $2 \le j \le J$. By Lemma 3.3, given $V = j$ the count M_N has the law of $D - Y_D + Z^{(j)}_{N-D}$, where D has the law β_j and is independent of Y and $Z^{(j)}$. This is at most $D + Z_j$, because $Y \ge 0$ and $Z^{(j)}$ is nondecreasing in time. So $M_N \ge m$ needs $Z_j > m/8$, or $Z_j \le m/8$ and $D \ge m - Z_j$. Hence

$$\mathbb{P}(M_N \ge m \mid V = j) \le \mathbb{P}\Big(Z_j > \frac{m}{8}\Big) + \mathbb{E}\Big[\bar\beta_j(m - Z_j);\ Z_j \le \frac{m}{8}\Big].$$

We compare $\bar\beta_j(m-z)$ with $\bar\beta_j(m)$ for an integer $0 \le z \le m/8$. By (16), each step down from s to $s-1$ with $s \le m$ multiplies $\bar\beta_j$ by at most $\varrho_j := 1 + \frac{j-1}{N-m-j+2}$, and $\varrho_j \le 1 + \frac{4(j-1)}{N}$ because

$N-m-j \ge N/4$. Since $\varrho^z - 1 \le z(\varrho-1)\varrho^z$ for $\varrho \ge 1$, we get $\bar\beta_j(m-z) \le \bar\beta_j(m)\varrho_j^z \le \bar\beta_j(m)\big(1+z(\varrho_j-1)\varrho_j^{m/8}\big)$. So the mean above is at most $\bar\beta_j(m)\big(1+(\varrho_j-1)\varrho_j^{m/8}\,\mathbb{E}Z_j\big)$.

On $\{V>J\}$ there is no mark on $2,\dots,J$, so by Lemma 3.1 M_N is a copy of $M_N^{[J]}$ independent of $\{V>J\}$. Altogether

$$\mathbb{P}(M_N \ge m) \le \sum_{j=2}^{J} \mathbb{P}(V=j)\Big[\bar\beta_j(m)\big(1+(\varrho_j-1)\varrho_j^{m/8}\,\mathbb{E}Z_j\big)+\mathbb{P}\Big(Z_j > \frac{m}{8}\Big)\Big]+\mathbb{P}(V>J)\,\mathbb{P}\big(M_N^{[J]}\ge m\big).$$

This gives four terms, which we bound in turn. The main term is $\sum_j \mathbb{P}(V=j)\bar\beta_j(m) \le \mathbb{P}(D \ge m) \le \frac{x}{m}$ by Lemma B.4.

Step 2: the shift term. This is the sum over $2 \le j \le J$ of $\mathbb{P}(V=j)\,\bar\beta_j(m)\,(\varrho_j-1)\,\varrho_j^{m/8}\,\mathbb{E}Z_j$. We bound the five factors by

$$\mathbb{P}(V=j) \le \varepsilon, \quad \bar\beta_j(m) \le e^{-\frac{(j-1)(m-1)}{N-1}}, \quad \varrho_j - 1 \le \frac{4(j-1)}{N}, \quad \varrho_j^{m/8} \le e^{\frac{(j-1)m}{2N}}, \quad \mathbb{E}Z_j \le x\log\frac{N}{j-1}.$$

The second is (15), the fourth follows from the third and $1+y \le e^y$, and the last is Lemma B.2. The two exponentials combine. Since $m \ge 6$ we have $\frac{m-1}{N-1} - \frac{m}{2N} \ge \frac{m-2}{2N} \ge \frac{m}{3N}$. So with $a = \frac{m}{3N}$ and $i = j-1$ the shift term is at most

$$\frac{4\varepsilon x}{N} \sum_{1 \le i < N/4} i\, e^{-ai} \log\frac{N}{i}.$$

Here $i \le J-1 < N/4$, so every term is nonnegative. In this sum only the first flips up to a time of order $1/a = 3N/m$ matter, because a later first flip rarely has a family of size m. To bound the sum we write $\log\frac{N}{i} = \log\frac{m}{3} + \log\frac{1}{ai}$. The first part is nonnegative, and $\sum_{i\ge1} ie^{-ai} \le a^{-2}$. The second part is positive only for $i < 1/a$. There $i\log\frac{1}{ai} \le \frac{1}{ea}$, so these at most $1/a$ terms add up to at most $\frac{1}{ea^2}$. Hence the shift term is at most

$$\frac{4\varepsilon x}{N}\cdot\frac{\log m+1}{a^2} = \frac{36x^2(\log m+1)}{m^2} \le \frac{x}{m}\cdot\frac{Cx\log m}{m}.$$

Step 3: the big-Z term. This is $\sum_{j=2}^{J}\mathbb{P}(V=j)\mathbb{P}(Z_j > m/8)$, and again $\mathbb{P}(V=j) \le \varepsilon$. Recall that Z_j counts the disagreeing steps of a walk whose first $j-1$ steps agree. For small j we only have the general tail bound of Lemma B.2, which is of order $x\log m/m$. For large j the late start makes $Z_j \ge m/8$ exponentially unlikely (Lemma B.3). We split at $j_{\mathcal K} = \lfloor \mathcal K N/m\rfloor + 1$ with $\mathcal K = 32\log(m/x)$. Note that $m \ge 16x\log 6 > 28x$, so $\mathcal K \ge 2$.

Let $2 \le j \le j_{\mathcal K}$. Lemma B.2 at the integer $\lfloor m/8\rfloor + 1$ gives $\mathbb{P}(Z_j > m/8) \le 8xH_m/m$. There are at most $\mathcal K N/m$ such j, and $\mathcal K \le 32\log m$. So they contribute at most

$$\varepsilon\cdot\frac{\mathcal K N}{m}\cdot\frac{8xH_m}{m} = \frac{x}{m}\cdot\frac{8\mathcal K x H_m}{m} \le \frac{x}{m}\cdot\frac{Cx\log^2 m}{m}.$$

Let $j > j_{\mathcal K}$, so that $j-1 > \mathcal K N/m$. We use Lemma B.3 with $t = N-1$ and the level $m/8$. Its exponent contains the term $\varepsilon\log\frac{2(N-1)}{j-1}$. Here $\frac{2(N-1)}{j-1} \le \frac{2m}{\mathcal K} \le m$, so this term is at most $\frac{x\log m}{N} \le \frac{m}{16N}$, by the hypothesis $m \ge 16x\log m$. The other term in the exponent is $\frac{m}{8(N-1)} \ge \frac{m}{8N}$. So the lemma gives $\mathbb{P}(Z_j \ge m/8) \le e^{-(j-1)m/(32N)}$, which is a geometric sequence in j. Summing over $j-1 \ge j_{\mathcal K} > \mathcal K N/m$, these j contribute at most

$$\varepsilon\,\frac{e^{-\mathcal K/32}}{1-e^{-m/(32N)}} \le \varepsilon\cdot\frac{x}{m}\Big(\frac{32N}{m}+1\Big) \le \frac{x}{m}\cdot\frac{33x}{m}.$$

Here $e^{-\mathcal{K}/32} = x/m$ by the choice of $\mathcal{K}$, and we used $\frac{1}{1-e^{-y}} \le \frac{1}{y} + 1$ for $y > 0$ and $\varepsilon \le x/m$.

Step 4: no early flip. If there is no flip up to time J, the walk agrees with X_1 for about $N/4$ steps, and Lemma B.3 makes $M_N \ge m$ exponentially unlikely in m. Namely $\mathbb{P}(V > J) = (1-\varepsilon)^{J-1} \le e^{\varepsilon - 3x/16}$, since $J \ge 3N/16$, and Lemma B.3 with $j - 1 = J \in [\frac{3N}{16}, \frac{N}{4}]$ and $t = N$ gives $\mathbb{P}(M_N^{[J]} \ge m) \le \exp\{-\frac{3m}{32} + \frac{x}{8}\log\frac{32}{3}\}$. As $x \le m/28$ and $\varepsilon \le \frac{1}{4}$, the product of the two is at most $e^{1/4 - m/13}$, and this is at most $\frac{x}{m} \cdot \frac{Cx\log^2 m}{m}$ because $x \ge 1$. □

Lemma B.7 (mode bound)**.** *There are absolute constants $\bar{c} > 0$ and A_0 such that if $\varepsilon \log N \le \bar{c}$ and $A_0 x \log(x+2) \le N/4$, then $m_h \le A_0 x \log^2(x+1)$.*

Proof. We pick a point ℓ of order $x \log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Up to its mode f_N is nondecreasing. So if the mode were far to the right of ℓ, the bins from ℓ to the mode would carry more mass than the tail $\mathbb{P}(M_N \ge \ell)$ has.

Let $\ell = \lceil Ax\log(x+2)\rceil$. Since $\log(\ell+1) \le \log(A+2) + 2\log(x+2)$, we have $\frac{x\log(\ell+1)}{\ell} \le \frac{\log(A+2)+2}{A}$, and $\ell e^{-\ell/9}$ is small when A is large. So we can fix the absolute constants A and $\bar{c}$ such that $e_1(\ell) \le \frac{1}{2}$ in Proposition B.5(a) whenever $\varepsilon \log N \le \bar{c}$. We take $A_0 \ge A + 1$, so that $2 \le \ell \le N/4$. Then $f_N(\ell) \ge \frac{x}{2\ell(\ell+1)}$. If $m_h \ge \ell$, Lemmas B.1 (with $a = \ell$, $b = m_h$) and B.2 give $(m_h - \ell + 1) f_N(\ell) \le \mathbb{P}(M_N \ge \ell) \le x H_\ell/\ell$. So $m_h \le (\ell+1)(1 + 2H_\ell)$ in every case. As $\log \ell \le C\log(x+2)$ and $\log(x+2) \le 2\log(x+1)$, this is at most $A_0 x \log^2(x+1)$ once A_0 is large. □

Proposition B.8 (local upper bound)**.** *There are absolute constants A_1 and C such that if $\varepsilon \log N \le \bar{c}$ and $A_1 x \log^2(x+1) \le m \le N/8$, then*

$$f_N(m) \le \frac{x}{m(m+1)}\big(1 + C\Delta(m)^{1/2}\big), \qquad \Delta(m) = \frac{x\log^2 m}{m} + \varepsilon \log N + \frac{m}{N}.$$

Idea of the proof. To the right of its mode f_N is nonincreasing, so $(k+1)f_N(m)$ is at most the probability of $\{m - k \le M_N \le m\}$, which is a difference of two tails. We bound the first tail from above by Proposition B.6 and the second from below by summing Proposition B.5. Then we choose k to balance the errors.

Proof. We take $A_1 \ge 2A_0$ so large that Lemma B.7 applies and every $m' \ge m/2$ satisfies $m' \ge 6$ and $m' \ge 16x\log m'$ (as $t/\log t$ increases for $t \ge e$).

Step 1: a difference of tails. Let $1 \le k \le m/2$. Then $m - k \ge m/2 \ge m_h$, and Lemma B.1 with $a = m - k$ and $b = m$ gives $(k+1)f_N(m) \le \mathbb{P}(M_N \ge m - k) - \mathbb{P}(M_N \ge m + 1)$.

Step 2: bound each tail. Proposition B.6 at $m - k$ bounds the first term by $\frac{x}{m-k} + \frac{Cx^2\log^2 m}{m^2}$. For the second we sum Proposition B.5(a) over $m < \ell \le N/4$. Since $\sum_{m<\ell\le N/4} \frac{1}{\ell(\ell+1)} \ge \frac{1}{m+1} - \frac{4}{N}$, $\frac{\log(\ell+1)}{\ell}$ and $\ell e^{-\ell/9}$ decrease for $\ell \ge 9$, and $m e^{-m/9} \le C/m \le C\Delta(m)$, we get $\mathbb{P}(M_N \ge m+1) \ge \frac{x}{m+1}(1 - C\Delta(m))$. As $\frac{x}{m-k} - \frac{x}{m+1} = \frac{x(k+1)}{(m-k)(m+1)}$, this gives $f_N(m) \le \frac{x}{(m-k)(m+1)} + \frac{Cx\Delta(m)}{(k+1)m}$.

Step 3: choose k. If $\Delta(m) \le \frac{1}{16}$, take $k = \lceil m\Delta(m)^{1/2}\rceil \le \frac{m}{2}$. Then $\frac{m}{m-k} \le 1 + \frac{2k}{m} \le 1 + 2\Delta^{1/2} + \frac{2}{m}$ and $\frac{\Delta}{k+1} \le \frac{\Delta^{1/2}}{m}$, which gives the claim, as $\frac{1}{m} \le \Delta(m) \le \Delta(m)^{1/2}$. If $\Delta(m) > \frac{1}{16}$, take $k = \lfloor m/2\rfloor$. Then $f_N(m) \le \frac{x}{m(m+1)}(2 + C\Delta(m))$, and the claim follows after enlarging C, since $\Delta(m)$ is bounded by an absolute constant ($\frac{x\log^2 m}{m}$ decreases for $m \ge e^2$ and is bounded at $m = A_1 x\log^2(x+1)$). □

Proof of Theorem 3.14. We check that Propositions B.5(a) and B.8 apply on the whole range of m and that their error terms tend to 0 uniformly there. Then we show that the mirror term $f_N(N-m)$ is negligible.

Step 1: the hypotheses hold. Put $\ell_x = \log(x+1) \geq \log 2$ and let $\mathcal{R}_N$ be the range $\omega_N x\ell_x^2 \leq m \leq \delta'_N N$. Since $x\log^2 N/N \to 0$, for large N we have $\varepsilon(1+\log N) \leq \min(\bar{c}, \frac{1}{100})$, $\omega_N \geq \max(A_1, 16)$, $\delta'_N \leq \frac{1}{8}$ and $A_0 x\log(x+2) \leq N/4$. The standing assumptions $N \geq 64$ and $\varepsilon \leq \frac{1}{4}$ hold as well.

Step 2: the errors are small on $\mathcal{R}_N$. The function $\frac{x\log^2 m}{m}$ decreases for $m \geq e^2$. At $m = \omega_N x\ell_x^2$ we have $m \geq e^2$ and $\log m \leq \log\omega_N + 3\ell_x$, as $\log x$ and $\log\ell_x$ are at most ℓ_x. So on $\mathcal{R}_N$, uniformly in $x \geq 1$, $\frac{x\log^2 m}{m} \leq \frac{(\log\omega_N + 3\ell_x)^2}{\omega_N \ell_x^2} \leq \frac{1}{\omega_N}(\frac{\log\omega_N}{\log 2} + 3)^2 \to 0$. Also $m \geq \omega_N \log^2 2 \to \infty$ on $\mathcal{R}_N$. Since $\log(m+1) \leq 2\log^2 m$ for $m \geq 3$, Proposition B.5(a) has $e_1(m) \leq C(\frac{x\log^2 m}{m} + \varepsilon\log N + me^{-m/9}) \to 0$ uniformly on $\mathcal{R}_N$. Also $\Delta(m) \leq \frac{x\log^2 m}{m} + \varepsilon\log N + \delta'_N \to 0$ there, and Proposition B.8 applies. So $f_N(m) = \frac{x}{m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$.

Step 3: the mirror term. Lemma B.7 gives $m_h \leq A_0 x\ell_x^2 \leq h$ for large N, since $x\ell_x^2 \leq x\log^2(N+1) = o(N)$. As $N - m \geq h$, Lemma 3.4 gives $0 \leq f_N(N-m) \leq f_N(h) = C_N$. By Theorem 2.6(b) and $\varepsilon\log N \to 0$, $C_N \leq \frac{4x}{N(N+2)} N^{2\varepsilon} e^{5200\varepsilon} = \frac{4x}{N^2}(1+o(1))$. On $\mathcal{R}_N$ this is at most $\frac{x}{m(m+1)} \cdot 8\delta'^2_N (1+o(1))$, since $m(m+1) \leq 2\delta'^2_N N^2$. Hence $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)] = \frac{x}{2m(m+1)}(1+o(1))$ uniformly on $\mathcal{R}_N$, and $m(m+1) = m^2(1+o(1))$ because $m \to \infty$. □

C Proofs for the bulk

We prove Theorems 3.9 and 3.10, Lemma 3.11 and Corollary 3.12. We keep the notation of Section 3 and of Appendix B. So $\mathcal{T}$ is the tree of Lemma 3.1, $\mathcal{T}_v$ is the family of v, and μ_v is the mark of v. By Lemma 2.4 and the discussion after Lemma 3.1, $|\mathcal{T}_v|$ has the law β_v, with mean N/v, and $\sum_{v=2}^N \beta_v(d) = N/(d(d+1))$ for $1 \leq d \leq N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d} - 1}{d(d+1)}, \qquad \Psi_\infty(\theta) = \sum_{d \geq 1} \frac{e^{i\theta d} - 1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\mathrm{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

C.1 Proof of Theorem 3.9

Here we assume only $N \geq 2$ and $0 < \varepsilon \leq \frac{1}{4}$. Let $Z_+ = \sum_{v=2}^N \mu_v |\mathcal{T}_v|$ be the total size of the flipped families.

Step 1: $M_N = Z_+$ with high probability. Two families are nested or disjoint. Suppose that no marked vertex has a marked strict descendant. Then the marked families are disjoint, each of their vertices has exactly one mark on its path to the root, and the other vertices have none. So $M_N = Z_+$ by Lemma 3.1. The number of pairs of a marked vertex and a marked strict descendant has mean $\varepsilon^2 \mathbb{E}\sum_v (|\mathcal{T}_v| - 1)$, since marks and tree are independent. Hence

$$\mathbb{P}(M_N \neq Z_+) \leq \varepsilon^2 \sum_{v=2}^N \Big(\frac{N}{v} - 1\Big) \leq \varepsilon^2 N(H_N - 1) \leq \frac{x^2 \log N}{N}.$$

Step 2: given the tree, a product. Fix $\theta \in [-\pi, \pi]$. Given $\mathcal{T}$ the marks are independent, so $\mathbb{E}[e^{i\theta Z_+} \mid \mathcal{T}] = \prod_{v=2}^N (1 + a_v)$ with $a_v = \varepsilon(e^{i\theta|\mathcal{T}_v|} - 1)$. Here $|a_v| \leq 2\varepsilon \leq \frac{1}{2}$ and $\mathrm{Re}\, a_v \leq 0$.

Put $W = \sum_v a_v$, so that $\operatorname{Re} W \le 0$. For complex a with $|a| \le \frac12$ the series of the logarithm gives $|\log(1+a) - a| \le |a|^2$. So $\prod_v (1+a_v) = e^{W+\varrho}$ with $|\varrho| \le \sum_v |a_v|^2 \le 4\varepsilon^2(N-1)$. Since $|e^w - 1| \le |w| e^{|w|}$,

$$\Big|\prod_v (1+a_v) - e^W\Big| \le e^{\operatorname{Re} W}\, 4\varepsilon^2(N-1)\, e^{4\varepsilon^2(N-1)} \le \frac{4x^2}{N} e^{4x^2/N}.$$

Step 3: replace W by its mean. Write $W = \varepsilon F$ with $F = \sum_{v=2}^{N} (e^{i\theta|\mathcal{T}_v|} - 1)$. Then $\mathbb{E}W = \varepsilon \sum_d (e^{i\theta d} - 1) \sum_v \beta_v(d) = x\Psi_N(\theta)$, which is $\log \widehat{\mathrm{CP}}_N(\theta)$. Put $\delta = W - \mathbb{E}W$. Taylor's formula for $s \mapsto e^{\mathbb{E}W + s\delta}$ on $[0,1]$ gives

$$e^W - e^{\mathbb{E}W} - e^{\mathbb{E}W}\delta = \delta^2 \int_0^1 (1-s)\, e^{\mathbb{E}W + s\delta}\, ds.$$

The exponent $\mathbb{E}W + s\delta = (1-s)\mathbb{E}W + sW$ has real part at most 0, so the right side has modulus at most $\frac12|\delta|^2$. As $\mathbb{E}\delta = 0$, we get $|\mathbb{E}e^W - e^{\mathbb{E}W}| \le \frac12 \varepsilon^2 \operatorname{Var} F$, where $\operatorname{Var} F = \mathbb{E}|F - \mathbb{E}F|^2$.

Step 4: the variance of F. The variable F is a function of the independent parents $U_1, \dots, U_{N-1}$. Let $F^{(k)}$ be F with U_k replaced by an independent copy U_k'. The Efron–Stein inequality [25] says that $\operatorname{Var} F \le \frac12 \sum_k \mathbb{E}|F - F^{(k)}|^2$. We use it for a complex function of independent variables that are not identically distributed, so we recall the proof. Let $\mathbb{E}_k$ be the average over U_k alone, and let $\zeta_k = \mathbb{E}[F - \mathbb{E}_k F \mid U_1, \dots, U_k]$. Then $\zeta_k = \mathbb{E}[F \mid U_1, \dots, U_k] - \mathbb{E}[F \mid U_1, \dots, U_{k-1}]$, so $F - \mathbb{E}F = \sum_k \zeta_k$ and the ζ_k are orthogonal. Hence $\operatorname{Var} F = \sum_k \mathbb{E}|\zeta_k|^2 \le \sum_k \mathbb{E}|F - \mathbb{E}_k F|^2$, and $\mathbb{E}|F - \mathbb{E}_k F|^2 = \frac12 \mathbb{E}|F - F^{(k)}|^2$, because given the other parents F and $F^{(k)}$ are independent with the same law.

Replacing U_k by U_k' moves the family of $k+1$ from U_k to U_k'. So $|\mathcal{T}_v|$ changes only for vertices $v \ge 2$ on the path from U_k to the root or on the path from U_k' to the root. These paths do not depend on the choice at step k, since $U_k, U_k' \le k$. Let $\mathrm{dp}(u)$ be the depth of u, namely the number of edges between u and the root. There are at most $\mathrm{dp}(U_k) + \mathrm{dp}(U_k')$ such vertices, and each term of F moves by at most 2. Hence $\mathbb{E}|F - F^{(k)}|^2 \le 8\,\mathbb{E}[\mathrm{dp}(U_k)^2 + \mathrm{dp}(U_k')^2] = 16\,\mathbb{E}\,\mathrm{dp}(U_k)^2$.

Since $\mathrm{dp}(u+1) = 1 + \mathrm{dp}(U_u)$, the generating function $g_u(s) = \mathbb{E}s^{\mathrm{dp}(u)}$ satisfies $g_1 = 1$ and $u\, g_{u+1}(s) = s \sum_{j \le u} g_j(s)$. Subtracting the same identity for $u-1$ gives $u\, g_{u+1}(s) = (u - 1 + s)\, g_u(s)$ for $u \ge 2$, and $g_2(s) = s$. So $g_u(s) = \prod_{i=1}^{u-1} \frac{i-1+s}{i}$. Thus $\mathrm{dp}(u)$ is a sum of independent Bernoulli$(1/i)$ variables with $1 \le i < u$, and $\mathbb{E}\,\mathrm{dp}(u)^2 \le H_{u-1}^2 + H_{u-1}$. The parent U_k is independent of the depths of $1, \dots, k$, so $\mathbb{E}\,\mathrm{dp}(U_k)^2 \le H_N^2 + H_N$. We get $\operatorname{Var} F \le \frac12 (N-1) \cdot 16 (H_N^2 + H_N) \le 8N(H_N^2 + H_N)$, and the bound of Step 3 is at most $4x^2(H_N^2 + H_N)/N$.

Step 5: Fourier inversion. For every integer m,

$$\mathbb{P}(Z_+ = m) - \mathrm{CP}_N(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-im\theta} \big(\mathbb{E}e^{i\theta Z_+} - \widehat{\mathrm{CP}}_N(\theta)\big)\, d\theta,$$

so the left side is at most $\sup_\theta |\mathbb{E}e^{i\theta Z_+} - \widehat{\mathrm{CP}}_N(\theta)|$ in absolute value. By Steps 2 to 4 this is at most $\frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$. Finally $|f_N(m) - \mathbb{P}(Z_+ = m)| \le \mathbb{P}(M_N \ne Z_+)$, and Step 1 bounds it. This proves Theorem 3.9. □

C.2 Proof of Theorem 3.10

We write $B(t) = -it \log|t| - \frac{\pi}{2}|t|$ for real $t \ne 0$, and $B(0) = 0$. For $t > 0$ this is the value of $u \log u$ at $u = -it$, with the principal logarithm, and $B(-t) = \overline{B(t)}$. We need three facts about Ψ_N and Ψ_∞.

Fact 1. For real θ, $|\Psi_N(\theta) - \Psi_\infty(\theta)| \le \sum_{d\ge N} \frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \le \pi$ we have $\Psi_\infty(\theta) = -i\theta\log(-i\theta) + R(\theta)$ with $|R(\theta)| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big)$. The reason is that Ψ_∞ has a closed form in which θ enters only through $1 - e^{i\theta}$, and $1 - e^{i\theta}$ is close to $-i\theta$ for small θ. The term R is what this replacement changes. We prove the fact in four steps.

We start with the closed form. For $0 < |s| \le 1$, $s \ne 1$,

$$\sum_{d\ge1} \frac{s^d - 1}{d(d+1)} = \chi(s) := \frac{(1-s)\log(1-s)}{s},$$

because $\frac{1}{d(d+1)} = \frac{1}{d} - \frac{1}{d+1}$, $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s) - s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_\infty(\theta) = \chi(e^{i\theta})$.

Next we compare $1 - e^{i\theta}$ with $-i\theta$. Write $1 - e^{i\theta} = (-i\theta)\,\mathsf{u}$ with $\mathsf{u} = \operatorname{sinc}(\theta/2)\,e^{i\theta/2}$, where $\operatorname{sinc} y = \sin y/y$ is positive for $|y| \le \frac{\pi}{2}$. So u is close to 1 when θ is small. The argument of $-i\theta$ is $\mp\frac{\pi}{2}$ for $\pm\theta > 0$ and that of u is $\theta/2$. They add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1 - e^{i\theta}) = \log(-i\theta) + \log\mathsf{u}$ for principal logarithms.

Now we get a formula for R. By the two steps above, $\chi(e^{i\theta}) = (-i\theta)\,\mathsf{u}e^{-i\theta}\big(\log(-i\theta) + \log\mathsf{u}\big)$. Subtracting $-i\theta\log(-i\theta)$ gives

$$R(\theta) = -i\theta\log(-i\theta)\big(\mathsf{u}e^{-i\theta} - 1\big) - i\theta\,\mathsf{u}e^{-i\theta}\log\mathsf{u}.$$

It remains to bound the three quantities in this formula. We show that

$$|\mathsf{u}e^{-i\theta} - 1| \le 0.631\,|\theta|, \qquad |\log\mathsf{u}| \le 0.53\,|\theta|, \qquad |\log(-i\theta)| \le |\log|\theta|| + \frac{\pi}{2}.$$

For the first, put $y = \theta/2$. Then $|\mathsf{u}e^{-i\theta} - 1| = |\operatorname{sinc} y - e^{iy}| \le (1 - \operatorname{sinc} y) + |1 - e^{iy}| \le \frac{y^2}{6} + |y| \le (\frac{1}{2} + \frac{\pi}{24})|\theta|$, where the last step uses $|\theta| \le \pi$, and $\frac{1}{2} + \frac{\pi}{24} \le 0.631$. For the second, $|\log\mathsf{u}|^2 = \log^2\operatorname{sinc}(\theta/2) + \theta^2/4$, and $-\log\operatorname{sinc} y = \sum_{k\ge1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log\operatorname{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log\mathsf{u}| \le \big(\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2}\big)^{1/2}|\theta| \le 0.53\,|\theta|$. The third holds because $-i\theta$ has modulus $|\theta|$ and argument $\pm\frac{\pi}{2}$. Since $|\mathsf{u}e^{-i\theta}| \le 1$, the formula for R now gives

$$|R(\theta)| \le |\theta|\Big(|\log|\theta|| + \frac{\pi}{2}\Big)\cdot 0.631\,|\theta| + |\theta|\cdot 0.53\,|\theta| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big),$$

as $0.631\cdot\frac{\pi}{2} + 0.53 \le 1.53$.

Fact 3. For $|\theta| \le \pi$, $-\operatorname{Re}\Psi_\infty(\theta) = \sum_d \frac{1-\cos\theta d}{d(d+1)} \ge \frac{|\theta|}{2\pi}$. We may take $\theta \ne 0$. Keep only the terms with $d \le D := \lfloor\pi/|\theta|\rfloor$ and use $1 - \cos v \ge 2v^2/\pi^2$ for $|v| \le \pi$. This gives at least $\frac{2\theta^2}{\pi^2}\sum_{d\le D}\frac{d}{d+1} \ge \frac{\theta^2 D}{\pi^2} \ge \frac{|\theta|}{2\pi}$, since $D \ge \frac{\pi}{2|\theta|}$.

The function e^B is a characteristic function of class L. Let S_x have the law CP_∞ with parameter x, namely the law of $\sum_{d\ge1} d\,\mathcal{N}_d$. Its characteristic function is $e^{x\Psi_\infty(\theta)}$. By Fact 2, for $t \ne 0$ and $x \ge \max(1, |t|/\pi)$,

$$x\Psi_\infty(t/x) - it\log x = B(t) + xR(t/x), \qquad |xR(t/x)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big). \quad (17)$$

So the characteristic function of $(S_x - x\log x)/x$ tends to $e^{B(t)}$ as $x \to \infty$, for every real t. The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a

probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 3.10. For $0 < c < 1$ and real t we have $B(ct) = cB(t) - ict\log c$, and hence

$$e^{B(t)} = e^{B(ct)} \cdot e^{B((1-c)t)}\, e^{it[(1-c)\log(1-c)+c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [24].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ and the substitution $\theta = t/x$,

$$x\,\mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)}\,dt, \qquad A(t) = x\Psi_N(t/x) - it\log x.$$

Let $|t| \le \pi x$. By (17) and Fact 1,

$$|A(t) - B(t)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\mathrm{Re}\,A(t), \mathrm{Re}\,B(t)) \le -\frac{|t|}{2\pi} + \frac{2x}{N}$. Now $|e^A - e^B| \le |A-B|\, e^{\max(\mathrm{Re}\,A, \mathrm{Re}\,B)}$, and $\frac{2x}{N} \le \frac{1}{2}$. The integrals of $t^2 e^{-|t|/2\pi}$, of $t^2|\log|t||e^{-|t|/2\pi}$ and of $e^{-|t|/2\pi}$ over the real line are finite. So

$$\frac{1}{2\pi}\int_{-\pi x}^{\pi x} |e^{A(t)} - e^{B(t)}|\,dt \le C\Big(\frac{1+\log x}{x} + \frac{x}{N}\Big).$$

Beyond πx we use $|e^{B(t)}| = e^{-\pi|t|/2}$, which gives $\frac{1}{2\pi}\int_{|t|>\pi x}|e^{B(t)}|\,dt = \frac{2}{\pi^2}e^{-\pi^2 x/2} \le \frac{C}{x}$. Since $f_L(\lambda_m) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\lambda_m t} e^{B(t)}\,dt$, this proves (a).

Proof of (b). By Theorem 3.9, $|x f_N(m) - x\,\mathrm{CP}_N(m)| \le x\delta_B$. Add (a).

Proof of (c). Let $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then $x\delta_B \to 0$ and $x/N \to 0$, so the bound of (b) tends to 0. Let U be uniform on $[0,1)$ and independent of M_N. Then $(M_N + U - x\log x)/x$ has the density g_N with $g_N(\lambda) = x f_N(m)$ for $\lambda_m \le \lambda < \lambda_m + \frac{1}{x}$. On a compact set of λ the relevant m are nonnegative once x is large, and $|\lambda - \lambda_m| < \frac{1}{x}$. As f_L is uniformly continuous, $g_N \to f_L$ uniformly on compact sets. By Scheffé's lemma $g_N \to f_L$ in L^1, and since $U/x \to 0$ this gives the convergence in law. At the crossing $x_N \sim 2\log N$ by Theorem 2.7, so both conditions hold. □

C.3 Proof of Lemma 3.11

Step 1: a real integral. We claim that for every real λ and $k \ge 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi}\int_0^\infty (-s)^k\, e^{-s\log s - \lambda s}\sin(\pi s)\,ds. \tag{18}$$

Since $B(-t) = \overline{B(t)}$, the definition of f_L gives $f_L(\lambda) = \frac{1}{\pi}\,\mathrm{Re}\int_0^\infty \Phi(-it)\,dt$ with $\Phi(u) = \exp(u\log u + \lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u = \rho e^{i\varphi}$ with $-\pi < \varphi \le -\frac{\pi}{2}$. Then $\mathrm{Re}(u\log u) = \rho(\log\rho\cos\varphi - \varphi\sin\varphi)$, where $\cos\varphi \le 0$ and $\varphi\sin\varphi \ge 0$. For $\log\rho \ge |\lambda| + 1$ this gives $\mathrm{Re}(u\log u + \lambda u) \le -\rho(|\cos\varphi| + \varphi\sin\varphi) \le -\rho$, because $|\cos\varphi| + \varphi\sin\varphi \ge 1$ on $[-\pi, -\frac{\pi}{2}]$ (for $\psi = \varphi + \pi \in [0, \frac{\pi}{2}]$ it equals $\cos\psi + (\pi - \psi)\sin\psi \ge \cos\psi + \sin\psi \ge 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u = -it$ to the lower side of the negative real axis, where $u = -s$ and $\log u = \log s - i\pi$. There $u\log u = -s\log s + i\pi s$, and $dt = i\,du = -i\,ds$. Hence

$$f_L(\lambda) = \frac{1}{\pi}\,\mathrm{Re}\Big[-i\int_0^\infty e^{-s\log s - \lambda s}\, e^{i\pi s}\,ds\Big] = \frac{1}{\pi}\int_0^\infty e^{-s\log s - \lambda s}\sin(\pi s)\,ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (18). Also f_L'' is bounded on the real line, because $t^2 e^{B(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2: certified enclosures. Let $-0.223 \le \lambda \le 0$ and $k \in \{1, 2\}$, and split the integral in (18) at $\delta = 2^{-20}$ and at $T = 30$. On $[0, \delta]$ we have $-s \log s \le 1/e$, $-\lambda s \le 0.223\, s$ and $|\sin \pi s| \le \pi s$, so this part is at most $\pi e^{1/e + 0.223\delta} \delta^{k+2}/(k+2)$ in absolute value. On $[T, \infty)$ we have $s \log s + \lambda s \ge cs$ with $c = \log 30 - 0.223 > 3$, so this part is at most $\int_T^\infty s^2 e^{-cs}\, ds \le T^2 e^{-cT}$. Both are below 10^{-17}. On $[\delta, T]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [28], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229) \in [9.232 \cdot 10^{-6}, 9.234 \cdot 10^{-6}], \qquad f_L'(-0.2227) \in [-6.548 \cdot 10^{-6}, -6.546 \cdot 10^{-6}],$$

and $f_L''(\lambda) \in [-0.0792, -0.0786]$ for every $\lambda \in [-0.2229, -0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297.

Step 3: the mode. Write $[a, b] = [-0.2229, -0.2227]$. By Step 2, f_L' is strictly decreasing on $[a, b]$ with $f_L'(a) > 0 > f_L'(b)$. So f_L' has exactly one zero λ_0 in $[a, b]$, and f_L is strictly increasing on $[a, \lambda_0]$ and strictly decreasing on $[\lambda_0, b]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty, \lambda_*]$ and nonincreasing on $[\lambda_*, \infty)$. If $\lambda_* > \lambda_0$, then f_L would be nondecreasing just to the right of λ_0, which is false. In the same way $\lambda_* < \lambda_0$ is impossible. So $\lambda_* = \lambda_0$. By Step 2 again, $\lambda_0 \in (-0.22278299, -0.22278297)$. This proves (a) and (b).

Step 4: the constant κ_L. Let $\delta_0 = \min(\lambda_0 - a, b - \lambda_0) > 0$. For $|\lambda - \lambda_0| \le \delta_0$, Taylor's formula with $f_L'(\lambda_0) = 0$ and $f_L'' \le -0.078$ on $[a, b]$ gives $f_L(\lambda_0) - f_L(\lambda) \ge 0.039(\lambda - \lambda_0)^2$. For $|\lambda - \lambda_0| > \delta_0$ the monotonicity of Step 3 gives $f_L(\lambda) \le f_L(\lambda_0 \pm \delta_0) \le f_L(\lambda_0) - 0.039\, \delta_0^2$. Since $\delta_0 < 1$, (c) holds with $\kappa_L = 0.039\, \delta_0^2$. □

C.4 Proof of Corollary 3.12

The method is to compare f_N at a mode with f_N at the integer m' nearest to the predicted place $x \log x + \lambda_0 x$. By Theorem 3.10(b) both values are close to values of f_L, and f_L falls off quadratically away from λ_0. So a mode cannot be far from m'.

We first prove a statement at finite N. Assume the standing assumptions of Section 3, let κ_L be as in Lemma 3.11(c), and let E_1 be the bound of Theorem 3.10(b). Put

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

We claim that if $E' < \kappa_L/2$ and $m_h \le N/4$, then every mode m of f_N satisfies $|m - x \log x - \lambda_0 x| \le x\sqrt{2E'/\kappa_L}$, and so does m^* for every mode $a^* \ge N/2$ of P_N.

Let m' be the integer nearest to $x \log x + \lambda_0 x$. This number increases in $x \ge 1$ and is at least $\lambda_0 > -\frac{1}{2}$, so $m' \ge 0$. Also $|\lambda_{m'} - \lambda_0| \le \frac{1}{2x}$, and since $f_L'(\lambda_0) = 0$, $f_L(\lambda_{m'}) \ge f_L(\lambda_0) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde{m}$ be a mode of f_N or $\tilde{m} = m^*$. We claim that $f_N(\tilde{m}) \ge f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m^* we have $f_N(m^*) + f_N(N - m^*) = 2P_N(N - m^*) \ge 2P_N(N - m') \ge f_N(m')$. Here $N - m^* \ge \lceil N/2 \rceil \ge \lceil N/4 \rceil \ge m_h$. So f_N is nonincreasing on $\{\lceil N/4 \rceil, \dots, N\}$ by Lemma 3.4. There are at least $N/4$ integers from $\lceil N/4 \rceil$ to $\lceil N/2 \rceil$. So Lemma B.1 (with $a = \lceil N/4 \rceil \ge m_h$ and $b = \lceil N/2 \rceil$) gives the second inequality below, and Lemma B.2 gives the third,

$$f_N(N - m^*) \le f_N(\lceil N/2 \rceil) \le \frac{\mathbb{P}(M_N \ge \lceil N/4 \rceil)}{N/4} \le \frac{16xH_N}{N^2}.$$

Now Theorem 3.10(b), used at $\tilde{m}$ and at m', gives

$$f_L(\lambda_{\tilde{m}}) \geq x f_N(\tilde{m}) - E_1 \geq x f_N(m') - \frac{16x^2 H_N}{N^2} - E_1 \geq f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2 H_N}{N^2} \geq f_L(\lambda_0) - 2E'.$$

Since $2E' < \kappa_L$, Lemma 3.11(c) gives $(\lambda_{\tilde{m}} - \lambda_0)^2 \leq 2E'/\kappa_L$, which is the claim.

Now take $x = x_N$ at the crossing. By Theorem 2.7, $x \sim 2\log N$. So $x\delta_B$, x/N, $16x^2 H_N/N^2$ and $\|f_L''\|_\infty/(8x^2)$ are all $O(\log x/x)$, and $E' = O(\log x/x)$. For large N the standing assumptions hold, $E' < \kappa_L/2$, and Lemma B.7 gives $m_h \leq A_0 x \log^2(x+1) \leq N/4$. Hence $|m^* - x\log x - \lambda_0 x| \leq x\sqrt{2E'/\kappa_L} = O(\sqrt{x\log x})$, and the same holds for m_h. □

D Proofs for Section 4

We use the notation of Section 4, with $\tau_0 = 0$. For a first run D let $\tau_m^D = D + \tau'_{m-1}$, where (τ'_m) is an independent copy of (τ_m), and $\tau_0^D = 0$. Then $u_m^D(k) = \mathbb{P}(\tau_m^D = k)$ and $q_m^D(k) = \mathbb{P}(\tau_{m-1}^D < k \leq \tau_m^D)$. We find each of these by conditioning on where the run before ends. For example, the m-th run covers level k exactly when the first $m-1$ runs end at some level $k-j$ with $j \geq 1$ and the m-th run has length at least j. In this way, for $m, k \geq 1$,

$$\begin{gathered} q_m(k) = \sum_{j\geq 1} \bar{F}(j)\, u_{m-1}(k-j), \qquad q_{m+1}^D(k) = \sum_{j\geq 1} \bar{F}(j)\, u_m^D(k-j), \\ u_m^D(k) = \sum_{d\geq 1} \mathbb{P}(D=d)\, u_{m-1}(k-d). \end{gathered} \tag{19}$$

Here $u_0(k) = \mathbf{1}\{k=0\}$. The sums τ_m and τ_m^D increase strictly in m. So exactly one run covers a given level $k \geq 1$, and at most one run ends there. Hence $\sum_m q_m(k) = \sum_m q_m^D(k) = 1$ and $\sum_m u_m(k) \leq 1$, $\sum_m u_m^D(k) \leq 1$. Also, the first m' runs reach level k exactly when one of them covers it, so $\sum_{m\leq m'} q_m(k) = \mathbb{P}(\tau_{m'} \geq k)$, and in the same way $\sum_{m\leq m'} q_m^D(k) = \mathbb{P}(\tau_{m'}^D \geq k)$.

Sections D.1 to D.4 treat $\alpha > 1$ and prove Theorems 4.4 and 4.7 together. Section D.5 treats $0 < \alpha < 1$ and proves Theorem 4.8.

D.1 The setting for $\alpha > 1$

Let $\alpha > 1$, so that $\mu = \zeta(\alpha) < \infty$. The proof of the center asymptotics uses the exponent only through the following three facts about a norming sequence (a_m) of positive numbers.

(F1) For $m \geq 3$ we have $a_m = a(m)$, where $a(x) = c_a x^\kappa (\log x)^\beta$ on $[3,\infty)$ with constants $c_a > 0$, $0 < \kappa < 1$ and $\beta \geq 0$.

(F2) The variables $(\tau_m - m\mu)/a_m$ converge in law to a variable G that has a bounded, uniformly continuous density g with $g(z) \to 0$ as $|z| \to \infty$. With $z_{m,k} = (k - m\mu)/a_m$, the numbers $\delta_m = \sup_k |a_m u_m(k) - g(z_{m,k})|$ tend to 0, and $\bar{u} = \sup_{m\geq 1} \sup_k a_m u_m(k)$ is finite.

(F3) There is a constant C_B such that $u_m(k) \leq C_B\, m\, k^{-\alpha-1}$ for all $m \geq 1$ and $k \geq 2m\mu$.

The function a in (F1) is increasing, and $a(\lambda x)/a(x) \to \lambda^\kappa$ as $x \to \infty$, uniformly for λ in a compact subset of $(0,\infty)$. So $a_m \to \infty$ and $a_m/a_{m-1} \to 1$. For $1 < \alpha < 2$ we take $a_m = m^{1/\alpha}$. Then (F1) holds with $\kappa = 1/\alpha$ and $\beta = 0$, (F2) is Lemma 4.3(ii) and (iii), and (F3) is Lemma D.1 below. We write ω for the modulus of continuity of g and $\bar{g}(x) = \sup_{|y|\geq x} g(y)$. So $\omega(\eta) \to 0$ as $\eta \to 0$ and $\bar{g}(x) \to 0$ as $x \to \infty$.

D.2 Two lemmas on renewal sums

A sum of m runs that is much larger than its mean is most likely made by a single long run. The first lemma turns this into a local bound.

Lemma D.1. *Let $\alpha > 1$. There is a constant $C_B = C_B(\alpha)$ such that $u_m(k) \le C_B\, m\, k^{-\alpha-1}$ for all $m \ge 1$ and $k \ge 2m\mu$.*

Proof. We split the event $\{\tau_m = k\}$ according to whether some run is long, namely at least $y = k/r$, where $r > 2$ is a constant that depends only on α and is chosen below. The first part is small because one run takes a given value $\ell \ge y$ with probability at most $\alpha y^{-\alpha-1}$. If no run is long, then τ_m is a sum of variables below y whose mean is at most $k/2$, and a Chernoff bound shows that such a sum rarely reaches k. Here C depends only on α and may change from line to line. Since $\mu \ge 1$ we have $m \le k/2$.

One big jump. By the mean value theorem $\mathbb{P}(\xi = \ell) = \ell^{-\alpha} - (\ell+1)^{-\alpha} \le \alpha \ell^{-\alpha-1}$. Fix an index i. Then

$$\mathbb{P}(\tau_m = k,\, \xi_i \ge y) = \sum_{\ell \ge y} \mathbb{P}(\xi = \ell)\, \mathbb{P}(\tau_m - \xi_i = k - \ell) \le \alpha y^{-\alpha-1} = \alpha r^{\alpha+1} k^{-\alpha-1},$$

because the first factor is at most $\alpha y^{-\alpha-1}$ and the second factors add up to at most 1. The m indices together give at most $\alpha r^{\alpha+1} m k^{-\alpha-1}$.

No big jump. We may take $y \ge 1$, since otherwise $\max_i \xi_i < y$ is impossible. Let $x_0 > 0$ and $s \ge 0$. For $0 \le x \le s$ we have $e^x \le 1 + x + \frac{x^2}{2} e^s$. We use this with $x = s\xi/y$ on the event $\{\xi < y\}$ and then $1 + x \le e^x$. This gives $\mathbb{E}[e^{s\xi/y}; \xi < y] \le \exp\bigl(\frac{s}{y}\mathbb{E}[\xi; \xi < y] + \frac{s^2 e^s}{2y^2}\mathbb{E}[\xi^2; \xi < y]\bigr)$. By independence and Markov's inequality for $e^{s\tau_m/y}$,

$$\mathbb{P}\bigl(\tau_m \ge x_0,\ \max_{i \le m} \xi_i < y\bigr) \le \exp\Bigl(-\frac{s x_0}{y} + m\Bigl(\frac{s}{y}\,\mathbb{E}[\xi; \xi < y] + \frac{s^2 e^s}{2y^2}\,\mathbb{E}[\xi^2; \xi < y]\Bigr)\Bigr). \tag{20}$$

We take $x_0 = k$. Since $\mathbb{E}[\xi; \xi < y] \le \mu$ and $k - m\mu \ge k/2$, the first two terms of the exponent add up to at most $-\frac{s}{y} \cdot \frac{k}{2} = -sr/2$. So the exponent is at most $-sr/2 + \frac{1}{2} r^2 s^2 e^s\, m\, k^{-2}\, \mathbb{E}[\xi^2; \xi < y]$. It remains to choose s and r so that the bound is at most $Cmk^{-\alpha-1}$. The choice depends on how fast the truncated second moment grows, so there are three cases.

Let $1 < \alpha < 2$. Then $\mathbb{E}[\xi^2; \xi < y] \le \alpha \sum_{\ell < y} \ell^{1-\alpha} \le C y^{2-\alpha}$, so the exponent is at most $-sr/2 + Cr^\alpha s^2 e^s m/k^\alpha$. Take $s = \frac{1}{2}\log(k^\alpha/m)$, which is positive since $m \le k/2 < k^\alpha$. Then $e^s m/k^\alpha = e^{-s}$ and $s^2 e^{-s} \le 1$, so the second term is at most Cr^α. The first term is $-\frac{r}{4}\log(k^\alpha/m)$. So the probability is at most $C(m/k^\alpha)^{r/4}$. We take $r = 4 + 4/(\alpha - 1)$, so that $r/4 = 1 + 1/(\alpha - 1)$. Since $m \le k$, the bound is at most $C(m/k^\alpha)(k^{1-\alpha})^{1/(\alpha-1)} = Cmk^{-\alpha-1}$.

Let $\alpha > 2$. Then $\mathbb{E}[\xi^2; \xi < y] \le \mathbb{E}\xi^2 < \infty$, so the exponent is at most $-sr/2 + Cr^2 s^2 e^s m/k^2$. Take $s = \frac{1}{2}\log(k^2/m)$, which is positive since $m < k^2$. As before the probability is at most $C(m/k^2)^{r/4}$. With $r = 4\alpha$ and $m \le k$ this is $C\, m\, m^{\alpha-1} k^{-2\alpha} \le Cmk^{-\alpha-1}$.

Let $\alpha = 2$. Then $\mathbb{E}[\xi^2; \xi < y] \le 2\sum_{\ell < y} \ell^{-1} \le 2\log(ek)$, so the exponent is at most $-sr/2 + r^2 s^2 e^s m \log(ek)/k^2$. Take $s = \frac{1}{2}\log\frac{k^2}{m\log(ek)}$, which is positive since $m\log(ek) \le \frac{1}{2}k\log(ek) < k^2$. As before the probability is at most $C(m\log(ek)/k^2)^{r/4}$. With $r = 20$ and $m \le k$ this is at most $Cmk^{-6}\log^5(ek) \le Cmk^{-3}$.

In each case the two bounds together give the lemma. □

Lemma D.2. *Let $\alpha > 1$ and assume* (F1) *and* (F2). *Let D be any a.s. finite first run. Then*

$$\sup_k \Big| \frac{a_m}{\mu} q_m(k) - g(z_{m,k}) \Big|, \qquad \sup_k \big| a_m u_m^D(k) - g(z_{m,k}) \big|, \qquad \sup_k \Big| \frac{a_m}{\mu} q_{m+1}^D(k) - g(z_{m,k}) \Big|$$

tend to 0 as $m \to \infty$. For $m \geq 2$, $\sup_k q_m(k) \leq \mu \bar{u}/a_{m-1}$ and $\sup_k u_m^D(k) \leq \bar{u}/a_{m-1}$.

Proof. By (19), q_m and u_m^D are sums of shifts of u_{m-1} with total weights $\sum_j \bar{F}(j) = \mu$ and 1. Since $u_{m-1}(k) \leq \bar{u}/a_{m-1}$, this gives the two bounds for $m \geq 2$.

For the limits we use that a shift by a bounded amount j moves the point $z_{m,k}$ only by $O(1/a_m)$, and that g is uniformly continuous. So the local limit for u_{m-1} passes to these sums. We give the details for q_m. Fix $J \geq 1$ and $Z > 0$, put $\sigma_J = \sum_{j>J} \bar{F}(j)$ and write $z = z_{m,k}$. We compare term by term the two sums

$$a_{m-1} q_m(k) = \sum_{j \geq 1} \bar{F}(j)\, a_{m-1} u_{m-1}(k-j) \qquad \text{and} \qquad \mu g(z) = \sum_{j \geq 1} \bar{F}(j)\, g(z).$$

The terms with $j > J$ add up to at most $\bar{u}\sigma_J$ in the first sum and $\|g\|_\infty \sigma_J$ in the second.

Let $j \leq J$. By (F2), $|a_{m-1} u_{m-1}(k-j) - g(z_{m-1,k-j})| \leq \delta_{m-1}$. Next we compare $g(z_{m-1,k-j})$ with $g(z)$. The definition of $z_{m,k}$ gives $z_{m-1,k-j} = z\, a_m/a_{m-1} + (\mu - j)/a_{m-1}$. If $|z| \leq Z$, then $|z_{m-1,k-j} - z| \leq \eta_m$ with $\eta_m = Z|a_m/a_{m-1} - 1| + (J + \mu)/a_{m-1}$, so the two values of g differ by at most $\omega(\eta_m)$. If $|z| > Z$, then $|z_{m-1,k-j}| \geq Z/2$ once m is large (how large depends only on J and Z), so both values lie in $[0, \bar{g}(Z/2)]$. In both cases they differ by at most $\omega(\eta_m) + \bar{g}(Z/2)$. Hence, uniformly in k for large m,

$$\big| a_{m-1} q_m(k) - \mu g(z) \big| \leq \mu\big(\delta_{m-1} + \omega(\eta_m) + \bar{g}(Z/2)\big) + (\bar{u} + \|g\|_\infty)\sigma_J.$$

Here $\eta_m \to 0$ as $m \to \infty$. So we let $m \to \infty$ and then $J, Z \to \infty$, and the right side tends to 0. Since $a_m/a_{m-1} \to 1$ and $a_{m-1} q_m(k) \leq \mu \bar{u}$, we may replace a_{m-1} by a_m on the left. This gives the first limit.

The second limit follows in the same way from the sum for u_m^D, with the weights $\mathbb{P}(D = d)$ and with $\mathbb{P}(D > J)$ in place of σ_J, which tends to 0 as $J \to \infty$ because D is a.s. finite. The third follows from the sum for q_{m+1}^D, with $a_m u_m^D$ in place of $a_{m-1} u_{m-1}$, the second limit in place of $\delta_{m-1} \to 0$, and the shift $z_{m,k-j} - z_{m,k} = -j/a_m$. □

D.3 The center for $\alpha > 1$

Proposition D.3. *Let $\alpha > 1$ and assume* (F1), (F2) *and* (F3). *Let D be any a.s. finite first run and put $b_h = a(h/\mu)$. As $N = 2h \to \infty$,*

$$C_N \sim \frac{2}{b_h} \int_{\mathbb{R}} g(z)^2\, dz.$$

Idea of the proof. We split the sum in Proposition 4.2 by the number m of runs into four regions. In the main region $|m\mu - h| \leq M b_h$ both factors are given by the local limit theorem, and the terms form a Riemann sum for $2b_h^{-1} \int_{-M}^{M} g^2$. For $m \leq h/(4\mu)$ the level h is far above the mean of τ_m, and (F3) makes these terms $O(h^{-\alpha})$. In the other two regions (the m between these two, and the m above the main region) one factor of each term is $O(1/b_h)$, and the other factors add up to a tail probability of G, which is small when M is large.

Proof. By Proposition 4.2, $C_N = \sum_{m\ge1}\big[u_m(h)q^D_{m+1}(h) + u^D_m(h)q_m(h)\big]$. Fix $M>1$, put $m_\pm = (h \pm Mb_h)/\mu$, and let $\Sigma_1,\dots,\Sigma_4$ be the parts of this sum over the regions $\mathcal{R}_1=\{m\le h/(4\mu)\}$, $\mathcal{R}_2=\{h/(4\mu)<m<m_-\}$, $\mathcal{R}_3=\{m_-\le m\le m_+\}$ and $\mathcal{R}_4=\{m>m_+\}$. We show that $b_h\Sigma_3$ tends to $2\int_{-M}^{M} g^2$ and that the other three parts are small.

We need three consequences of (F1). First, $b_h = o(h)$ and $h^{-\alpha} = o(1/b_h)$, since $\kappa<1<\alpha$. Second, $a(h/(8\mu))\ge b_h/9$ and $a(h/(2\mu))\ge b_h/3$ for large h, since $8^{-\kappa}>\frac19$ and $2^{-\kappa}>\frac13$. Third, $a_m/b_h\to1$ uniformly in m with $|m\mu-h|\le 2Mb_h$.

Region $\mathcal{R}_1$ (far too few runs). Here $h\ge 4m\mu$, so level h is far above the mean of τ_m. We show that $u_m(h)$ and $q_m(h)$ are both $O(h^{-\alpha})$ for $m\in\mathcal{R}_1$. By (F3), $u_m(h)\le C_B m h^{-\alpha-1}\le C_B h^{-\alpha}/(4\mu)$. Also $q_1(h)=h^{-\alpha}$. For $m\ge2$ we split the sum for $q_m(h)$ in (19) at $j=h/2$. The terms with $j\ge h/2$ have $\bar F(j)\le (h/2)^{-\alpha}$, and the factors $u_{m-1}(h-j)$ add up to at most 1. So these terms add up to at most $(h/2)^{-\alpha}$. For $j<h/2$ we have $h-j>h/2\ge 2\mu(m-1)$, so (F3) applies to $u_{m-1}(h-j)$, and the weights $\bar F(j)$ add up to at most μ. So the other terms add up to at most $\mu C_B(m-1)(h/2)^{-\alpha-1}\le 2^{\alpha-1}C_B h^{-\alpha}$. In Σ_1 each of these factors is multiplied by $q^D_{m+1}(h)$ or by $u^D_m(h)$, and $\sum_m q^D_{m+1}(h)\le1$ and $\sum_m u^D_m(h)\le1$. Hence $\Sigma_1=O(h^{-\alpha})$, which is $o(1/b_h)$.

Region $\mathcal{R}_2$ (too few runs). In each term one factor is at most $9\bar u/b_h$, and the other factors add up to the probability that few runs reach level h. For large h we have $m-1\ge h/(8\mu)$ here, so $a_m\ge a_{m-1}\ge b_h/9$ and $u_m(h), u^D_m(h)\le 9\bar u/b_h$ by (F2) and Lemma D.2. Let $m'=\lfloor m_-\rfloor$. Every $m\in\mathcal{R}_2$ is at most m', so $\sum_{m\in\mathcal{R}_2} q_m(h)\le \mathbb{P}(\tau_{m'}\ge h)$ and $\sum_{m\in\mathcal{R}_2} q^D_{m+1}(h)\le \mathbb{P}(D+\tau'_{m'}\ge h)$. The last probability is at most $\mathbb{P}(D\ge b_h)+\mathbb{P}(\tau_{m'}\ge h-b_h)$. Since $m'\mu\le h-Mb_h$ and $a_{m'}\le b_h$, the event $\{\tau_{m'}\ge h-b_h\}$ implies $(\tau_{m'}-m'\mu)/a_{m'}\ge M-1$. Now $\mathbb{P}(D\ge b_h)\to0$ because D is a.s. finite, and $(\tau_{m'}-m'\mu)/a_{m'}$ converges in law to G, which has a continuous law. So $\limsup_h b_h\Sigma_2\le 18\bar u\,\mathbb{P}(G\ge M-1)$.

Region $\mathcal{R}_4$ (too many runs). This is like $\mathcal{R}_2$. For large h we have $m-1\ge h/(2\mu)$ here, so $u_m(h), u^D_m(h)\le 3\bar u/b_h$. Let $m''=\lfloor m_+\rfloor$. The runs after the first m'' cover level h exactly when the first m'' do not reach it. So $\sum_{m\in\mathcal{R}_4} q_m(h)=\mathbb{P}(\tau_{m''}<h)$ and, since $D\ge1$, $\sum_{m\in\mathcal{R}_4} q^D_{m+1}(h)=\mathbb{P}(D+\tau'_{m''}<h)\le\mathbb{P}(\tau_{m''}<h)$. Now $m''\mu>h+Mb_h-\mu$, so $(h-m''\mu)/a_{m''}\le(\mu-Mb_h)/a_{m''}\to -M$. Hence $\limsup_h b_h\Sigma_4\le 6\bar u\,\mathbb{P}(G\le -M)$.

Region $\mathcal{R}_3$ (the main part). Here $m\to\infty$ and $a_m=b_h(1+o(1))$ uniformly. By (F2) and Lemma D.2, the factors $u_m(h)$ and $u^D_m(h)$ are $(g(z_{m,h})+o(1))/a_m$, and the factors $q_m(h)$ and $q^D_{m+1}(h)$ are $\mu(g(z_{m,h})+o(1))/a_m$. So, uniformly in $m\in\mathcal{R}_3$,

$$u_m(h)q^D_{m+1}(h)+u^D_m(h)q_m(h)=\frac{2\mu}{a_m^2}\big(g(z_{m,h})^2+o(1)\big).$$

Put $y_m=(h-m\mu)/b_h$. Then $z_{m,h}=y_m b_h/a_m$ and $|y_m|\le M$, so $|z_{m,h}-y_m|\le M|b_h/a_m-1|\to0$ and $g(z_{m,h})^2=g(y_m)^2+o(1)$. The y_m with $m\in\mathcal{R}_3$ are the points of an arithmetic progression with step μ/b_h that lie in $[-M,M]$. There are at most $2Mb_h/\mu+1$ of them, so the $o(1)$ terms below add up to $o(1)$, and the rest is a Riemann sum. Hence

$$b_h\Sigma_3=2\big(1+o(1)\big)\frac{\mu}{b_h}\sum_{m\in\mathcal{R}_3}\big(g(y_m)^2+o(1)\big)\to 2\int_{-M}^{M} g^2.$$

Adding the four regions,

$$\limsup_{h\to\infty}\Big|b_hC_N-2\int g^2\Big|\le 2\int_{|z|>M} g^2+18\bar u\,\mathbb{P}(G\ge M-1)+6\bar u\,\mathbb{P}(G\le -M),$$

and letting $M\to\infty$ gives $C_N\sim 2b_h^{-1}\int g^2$. □

Proof of Theorem 4.4. Let $1 < \alpha < 2$ and $a_m = m^{1/\alpha}$. As noted in Section D.1, (F1) to (F3) hold, with $b_h = (h/\mu)^{1/\alpha}$. So Proposition D.3 gives $C_N \sim 2I_\alpha(\mu/h)^{1/\alpha}$, and the second form follows from Lemma 4.3(iv) and $h = N/2$. □

D.4 Proof of Theorem 4.7

We apply Proposition D.3 with the standard normal density in place of the stable density. So let $\alpha \geq 2$ and let g be the standard normal density. We have to find a norming sequence (a_m) with (F1), (F2) and (F3). Fact (F3) is Lemma D.1, which holds for every $\alpha > 1$. For (F2) it is enough that $(\tau_m - m\mu)/a_m$ converges in law to the standard normal law. Indeed, g is bounded and uniformly continuous and vanishes at $\pm\infty$. Every $\mathbb{P}(\xi = k)$ is positive, so ξ has maximal span 1, and Gnedenko's local limit theorem [21, Theorem 4.2.1] turns the convergence in law into $\delta_m \to 0$. Then $\bar{u} < \infty$ follows as in the proof of Lemma 4.3(iii). So in each of the two cases we choose (a_m), check (F1) and the normal limit, and read off $b_h = a(h/\mu)$. Since $\int g^2 = 1/(2\sqrt{\pi})$, Proposition D.3 then gives $C_N \sim 1/(\sqrt{\pi}\, b_h)$.

(a) Let $\alpha > 2$. Then ξ has a finite variance, and we use the central limit theorem. Summation by parts gives $\mathbb{E}\xi^2 = \sum_{k\geq 1}(2k-1)\bar{F}(k) = 2\zeta(\alpha-1) - \zeta(\alpha)$, so $v = \mathbb{E}\xi^2 - \mu^2$ is as stated and finite. It is positive because ξ is not constant. We take $a_m = \sqrt{vm}$. This has (F1) with $\kappa = \frac{1}{2}$ and $\beta = 0$, and the central limit theorem gives the normal limit. So $b_h = \sqrt{vh/\mu}$, and with $h = N/2$ we get $C_N \sim (\pi v h/\mu)^{-1/2} = \sqrt{2\mu/(\pi v N)}$. For the stationary start Proposition 4.1 gives $E_N = N^{1-\alpha}(1 + O(1/N))/(2(\alpha-1)\mu)$. Dividing E_N by C_N gives $R_N \sim C_2(\alpha) N^{3/2-\alpha}$.

(b) Let $\alpha = 2$, so that $\mu = \zeta(2)$ and $\mathbb{E}\xi^2 = \infty$. The central limit theorem with the norming $\sqrt{m}$ fails, but the variance is infinite only by a logarithm, and the limit is still normal with a slightly larger norming. To see this we compute the truncated second moment

$$U(x) = \mathbb{E}[\xi^2; \xi \leq x] = \sum_{k\leq x} k^2\big(k^{-2} - (k+1)^{-2}\big) = \sum_{k\leq x} \frac{2k+1}{(k+1)^2} = 2\log x + O(1).$$

This function varies slowly. So ξ is in the domain of attraction of the normal law, and $(\tau_m - m\mu)/a_m$ converges in law to the standard normal law for every sequence (a_m) with $mU(a_m)/a_m^2 \to 1$ [22, Theorem XVII.5.3 and eq. (XVII.5.23)]. We try $a(x) = \sqrt{x \log x}$. Then $U(a(m)) = \log m + \log\log m + O(1)$ and $a(m)^2/m = \log m$, so $mU(a(m))/a(m)^2 \to 1$. We take $a_m = a(m)$ for $m \geq 3$ and $a_1 = a_2 = 1$. This has (F1) with $\kappa = \beta = \frac{1}{2}$. So $b_h^2 = (h/\mu)\log(h/\mu) \sim N \log N/(2\mu)$ and $C_N \sim \sqrt{2\mu/(\pi N \log N)}$. For the stationary start $E_N = N^{-1}(1 + O(1/N))/(2\mu)$ by Proposition 4.1. Dividing gives $R_N \sim \sqrt{\pi \log N/(8\mu^3 N)}$.

Finally we take the fresh start. In both cases it has the edge $E_N = \frac{1}{2}N^{-\alpha}$ by Proposition 4.1, and the center asymptotics are the same because Proposition D.3 holds for every a.s. finite first run. So R_N is of order $N^{1/2-\alpha}$ for $\alpha > 2$ and of order $N^{-3/2}(\log N)^{1/2}$ for $\alpha = 2$, and it tends to 0. □

D.5 Proof of Theorem 4.8

Let $0 < \alpha < 1$ with a fresh start, $a_m = m^{1/\alpha}$ and $z_{m,k} = k/a_m$. There is no centering now. In this section μ is not used, and constants may depend on α.

Lemma D.4. *Let $0 < \alpha < 1$.*

(i) τ_m/a_m converges in law to a positive variable G with $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ for $s \geq 0$. The law of G has a bounded, uniformly continuous density g that vanishes on $(-\infty, 0]$ and at $+\infty$.

(ii) (Gnedenko) The constant $\bar{u} = \sup_{m\ge1} \sup_k a_m u_m(k)$ *is finite, and* $\delta_m = \sup_k |a_m u_m(k) - g(k/a_m)| \to 0$ *as* $m \to \infty$.

(iii) There is a constant C_B *such that* $u_m(k) \le C_B\, m\, k^{-\alpha-1}$ *for all* $m \ge 1$ *and* $k \ge a_m$.

(iv) $g(y) \le C_B y^{-1-\alpha}$ *for* $y > 1$. *So* $\int_0^\infty t\, g(t)^2\, dt < \infty$.

Proof. (i) Summation by parts gives $1-\mathbb{E}x^\xi = (1-x)\sum_{k\ge1} \bar{F}(k)x^{k-1}$ for $0 \le x < 1$. With $x = e^{-v}$ this is $1-\mathbb{E}e^{-v\xi} = (e^v-1)\mathrm{Li}_\alpha(e^{-v})$, and [29, 25.12.12] gives $\mathrm{Li}_\alpha(e^{-v}) = \Gamma(1-\alpha)v^{\alpha-1} + \zeta(\alpha) + O(v)$ as $v \downarrow 0$. So $1-\mathbb{E}e^{-v\xi} = \Gamma(1-\alpha)v^\alpha + O(v)$. As $1/\alpha > 1$, this gives $m(1-\mathbb{E}e^{-s\xi/a_m}) \to \Gamma(1-\alpha)s^\alpha$ for each $s \ge 0$. Hence $\mathbb{E}e^{-s\tau_m/a_m} \to e^{-\Gamma(1-\alpha)s^\alpha}$, and the continuity theorem for Laplace transforms [22, Section XIII.1] gives the limit $G \ge 0$. Both sides of $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ are continuous on $\mathrm{Re}\, s \ge 0$ and analytic on $\mathrm{Re}\, s > 0$, with the principal branch on the right. So the identity holds at $s = -i\theta$, and $|\mathbb{E}e^{i\theta G}| = \exp\bigl(-\Gamma(1-\alpha)\cos(\frac{\pi\alpha}{2})|\theta|^\alpha\bigr)$ is integrable. Fourier inversion and the Riemann–Lebesgue lemma give g. It vanishes on $(-\infty, 0)$ because $G \ge 0$, and at 0 by continuity.

(ii) As in Lemma 4.3(iii), ξ has maximal span 1, and Gnedenko's local limit theorem [21, Theorem 4.2.1] gives $\delta_m \to 0$. Then $a_m u_m(k) \le \|g\|_\infty + 1$ for large m and $a_m u_m(k) \le a_m$ for the others.

(iii) We argue as in Lemma D.1, with $r = 4 + 4/\alpha$, $y = k/r$ and $t = k^\alpha/m$, which is at least 1 because $k \ge a_m$. One thing is new. For k of order a_m the claim is $u_m(k) \le C/a_m$, which is a local bound, and a bound for the tail of τ_m cannot give it. So we cut the m runs into two halves. One half must be large, which we bound as before. The other half must then take one exact value, and this gives the factor $1/a_m$.

For $m = 1$ the claim is $\mathbb{P}(\xi = k) \le \alpha k^{-\alpha-1}$. Let $m \ge 2$. The bound $\alpha r^{\alpha+1} m k^{-\alpha-1}$ for the event that some run is at least y holds as before. For the other event we may take $y \ge 1$. Split the indices into $I_1 = \{1, \dots, \lfloor m/2 \rfloor\}$ and $I_2 = \{\lfloor m/2 \rfloor + 1, \dots, m\}$, and write $\tau_I = \sum_{i\in I} \xi_i$. On $\{\tau_m = k\}$ one of τ_{I_1}, τ_{I_2} is at least $k/2$, and given that half the other half has to equal k minus it. So by independence

$$\mathbb{P}\bigl(\tau_m = k,\ \max_i \xi_i < y\bigr) \le \sum_{i=1,2} \mathbb{P}\bigl(\tau_{I_i} \ge k/2,\ \max_{j\in I_i} \xi_j < y\bigr) \sup_\ell \mathbb{P}(\tau_{I_{3-i}} = \ell).$$

Each half has at least $\lfloor m/2 \rfloor \ge m/3$ indices, so by (ii) the supremum is at most $\bar{u}/a_{\lfloor m/2 \rfloor} \le 3^{1/\alpha}\bar{u}/a_m$. For the first factor we use (20) for the sum over I_i, with $x_0 = k/2$ and at most m terms. Now $\mathbb{E}[\xi; \xi < y] \le \sum_{\ell<y} \bar{F}(\ell) \le C y^{1-\alpha}$ and $\mathbb{E}[\xi^2; \xi < y] \le C y^{2-\alpha}$, and $m y^{-\alpha} = r^\alpha/t$. So the exponent in (20) is at most $-sr/2 + C(s + s^2 e^s)/t$. We take $s = \frac{1}{2}\log t$. Then $e^s = \sqrt{t}$, so the second term is bounded, and the first is $-\frac{r}{4}\log t$. Hence the first factor is at most $Ct^{-r/4} = Ct^{-1-1/\alpha}$. Since $a_m = m^{1/\alpha}$, we have $t^{-1-1/\alpha}/a_m = m k^{-\alpha-1}$, which is the claim.

(iv) Fix $y > 1$ and let $k_m = \lceil y a_m \rceil$. By (iii) and $m a_m^{-\alpha} = 1$, $a_m u_m(k_m) \le C_B m a_m (y a_m)^{-\alpha-1} = C_B y^{-1-\alpha}$. By (ii) and the continuity of g, the left side tends to $g(y)$ as $m \to \infty$. The integral converges because g is bounded and $t g(t)^2 \le C_B^2 t^{-1-2\alpha}$ for $t > 1$. □

Idea of the proof of Theorem 4.8. With a fresh start, Proposition 4.2 reads $C_N = \sum_{m\ge1} u_m(h)\bigl[q_m(h) + q_{m+1}(h)\bigr]$. Put $t_m = h/a_m$. Step 1 finds $q_m(h)$ on the scale where t_m is of order 1, namely $m q_m(h) \approx t_m g(t_m)/\alpha$. Step 2 sums over m. The terms with t_m very large or very small are negligible, and the others form a Riemann sum in the variable $\log m$. This gives $N C_N \to 4\int_0^\infty t g(t)^2\, dt$. Step 3 identifies this limit as the density at $\frac{1}{2}$ of $Y = G_1/(G_1 + G_2)$, where G_1, G_2 are independent copies of G. Step 4 finds the law of Y from the moments of G_1/G_2, and with it the value of the limit.

Proof of Theorem 4.8. Fix $M > 1$.

Step 1. Let $g_1(t) = \int_0^t w^{-\alpha} g(t-w)\,dw$. We claim that $mq_m(h) = g_1(t_m) + o(1)$ as $h \to \infty$, uniformly in m with $t_m \in [1/(2M), 2M]$, and that $g_1(t) = tg(t)/\alpha$. The reason for the first claim is that the sum for q_m in (19) is the convolution of the tail $j^{-\alpha}$ with u_{m-1}, and on the scale a_m it becomes the convolution g_1. The m in this range tend to infinity with h. Since $|k/a_{m-1} - k/a_m| \le 2M(a_m/a_{m-1} - 1) \to 0$ for $0 \le k \le h$, Lemma D.4(ii) for u_{m-1} and the uniform continuity of g give $a_m u_{m-1}(k) = g(k/a_m) + o(1)$ uniformly in $0 \le k \le h$. As $m\,j^{-\alpha} = (j/a_m)^{-\alpha}$ and $u_{m-1}(0) = 0$, the sum for q_m in (19) becomes

$$mq_m(h) = \frac{1}{a_m} \sum_{j=1}^{h-1} \Big(\frac{j}{a_m}\Big)^{-\alpha} \Big(g\big(t_m - \tfrac{j}{a_m}\big) + o(1)\Big).$$

For $x > 0$ we have $\frac{1}{a_m}\sum_{j \le x a_m} (j/a_m)^{-\alpha} \le \int_0^x w^{-\alpha} dw = x^{1-\alpha}/(1-\alpha)$. With $x = t_m \le 2M$ this shows that the $o(1)$ terms add up to $o(1)$. With $x = \eta$ it shows that the terms with $j \le \eta a_m$ are $O(\eta^{1-\alpha})$, and so is $\int_0^\eta w^{-\alpha} g(t_m - w)\,dw$. The other terms form a Riemann sum with mesh $1/a_m$ of the function $w^{-\alpha} g(t-w)$, which is uniformly continuous on $\{\eta \le w \le t \le 2M\}$. Letting $\eta \to 0$ proves the first claim.

For the second claim we compare Laplace transforms. The transform of the convolution g_1 is the product of those of $w^{-\alpha}$ and g, namely $\Gamma(1-\alpha)s^{\alpha-1}e^{-\Gamma(1-\alpha)s^\alpha}$. This is $\frac{1}{\alpha}$ times $-\frac{d}{ds}e^{-\Gamma(1-\alpha)s^\alpha}$, and minus the derivative of the transform of g is the transform of $tg(t)$. Both g_1 and $tg(t)$ are continuous on $(0,\infty)$, so $g_1(t) = tg(t)/\alpha$.

Step 2. We sum $u_m(h)[q_m(h) + q_{m+1}(h)]$ over m in three ranges. If $t_m > M$, then $h > Ma_m$, so $m < (h/M)^\alpha$, and Lemma D.4(iii) gives $u_m(h) \le C_B m h^{-\alpha-1} \le C_B M^{-\alpha}/h$. Since $\sum_m (q_m(h) + q_{m+1}(h)) \le 2$, these m contribute at most $2C_B M^{-\alpha}/h$ to C_N. If $t_m < 1/M$, then $u_m(h) \le \bar{u}/a_m < \bar{u}/(Mh)$, and these m contribute at most $2\bar{u}/(Mh)$. For the remaining m we have $t_{m+1} = t_m(1+o(1)) \in [1/(2M), 2M]$ and $(m+1)/m \to 1$, and g_1 is uniformly continuous on $[1/(2M), 2M]$. So Step 1 applies to $q_m(h)$ and to $q_{m+1}(h)$, and both are $(t_m g(t_m)/\alpha + o(1))/m$. With $1/a_m = t_m/h$ and $\delta_m \to 0$ we get, uniformly,

$$u_m(h)\big[q_m(h) + q_{m+1}(h)\big] = \frac{t_m}{h}\big(g(t_m) + o(1)\big)\,\frac{2}{m}\Big(\frac{t_m g(t_m)}{\alpha} + o(1)\Big) = \frac{2}{\alpha h m}\big(t_m^2 g(t_m)^2 + o(1)\big).$$

As m runs over these values, $\log m = \alpha \log(h/t_m)$ runs over an interval of length $2\alpha \log M$ with steps $\log(1 + 1/m) = (1 + o(1))/m$. So the sum of $1/m$ is $O(\log M)$, the $o(1)$ terms contribute $o(1/h)$, and the rest is a Riemann sum in the variable $\log m$. With $dm/m = -\alpha\,dt/t$ it is $\frac{2}{\alpha h}\big(\int_{1/M}^M t^2 g(t)^2 \,\frac{\alpha\,dt}{t} + o(1)\big)$. Hence

$$hC_N = 2\int_{1/M}^M tg(t)^2\,dt + O(M^{-\alpha} + M^{-1}) + o(1),$$

where the O term is uniform in h. By Lemma D.4(iv) we may let $M \to \infty$, and we get $NC_N = 2hC_N \to 4\int_0^\infty tg(t)^2\,dt$.

Step 3. Let G_1, G_2 be independent copies of G and $Y = G_1/(G_1 + G_2)$. The map $(G_1, G_2) \mapsto (Y, G_1 + G_2)$ has inverse $(y, r) \mapsto (yr, (1-y)r)$ with Jacobian r. So Y has the density $f(y) = \int_0^\infty r\,g(yr)\,g((1-y)r)\,dr$ on $(0,1)$. With $r = 2t$ we get $f(\frac12) = \int_0^\infty r\,g(r/2)^2\,dr = 4\int_0^\infty tg(t)^2\,dt$. Hence $NC_N \to f(\frac12)$ by Step 2. The function f is continuous on $[\frac14, \frac34]$. Indeed, for such y both yr and $(1-y)r$ are at least $r/4$, so by Lemma D.4(iv) the integrand is at most $r\min(\|g\|_\infty^2, Cr^{-2-2\alpha})$. This is integrable, and dominated convergence applies.

Step 4. It remains to show that $f(\frac12) = \ell_\alpha(\frac12)$. We show that Y has the density ℓ_α. To do this we compute the moments $\mathbb{E}W^p$ of $W = G_1/G_2$ for small $|p|$, and we check that a certain explicit density ρ_α has the same moments. The law of Y does not change if G is multiplied by a constant, so we may take $\mathbb{E}e^{-sG} = e^{-s^\alpha}$. For $x > 0$ and $0 < p < 1$,

$$x^p = \frac{p}{\Gamma(1-p)} \int_0^\infty s^{-p-1}(1 - e^{-sx})\, ds, \qquad x^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty s^{p-1} e^{-sx}\, ds.$$

We put $x = G$, take expectations and substitute $w = s^\alpha$. The first formula gives $\mathbb{E}G^p$ for $0 < p < \alpha$ and the second gives $\mathbb{E}G^{-p}$ for $0 < p < 1$. Together, $\mathbb{E}G^p = \Gamma(1 - p/\alpha)/\Gamma(1-p)$ for $|p| < \alpha$ (see also [23, Theorem 2.6.3]). Now $Y = W/(1+W)$ and $\mathbb{E}W^p = \mathbb{E}G^p\, \mathbb{E}G^{-p}$ by independence. With $\Gamma(1-z)\Gamma(1+z) = \pi z/\sin(\pi z)$ this is

$$\mathbb{E}W^p = \frac{\Gamma(1 - \frac{p}{\alpha})\Gamma(1 + \frac{p}{\alpha})}{\Gamma(1-p)\Gamma(1+p)} = \frac{\sin \pi p}{\alpha \sin(\pi p/\alpha)}, \qquad 0 < |p| < \alpha.$$

Let $\rho_\alpha(w) = \frac{\sin \pi\alpha}{\pi} \frac{w^{\alpha-1}}{w^{2\alpha} + 2w^\alpha \cos \pi\alpha + 1}$ for $w > 0$. We use the classical integral

$$\int_0^\infty \frac{x^\beta\, dx}{x^2 + 2x\cos\theta + 1} = \frac{\pi \sin \beta\theta}{\sin\theta\, \sin \pi\beta}, \qquad 0 < |\beta| < 1,\ 0 < \theta < \pi,$$

whose value at $\beta = 0$ is the limit $\theta/\sin\theta$. For $-1 < \beta < 0$ it follows from the partial fractions $\frac{1}{x^2+2x\cos\theta+1} = \frac{1}{2i\sin\theta}\big(\frac{1}{x+e^{-i\theta}} - \frac{1}{x+e^{i\theta}}\big)$ and $\int_0^\infty x^\beta (x + e^{\pm i\theta})^{-1} dx = -\pi e^{\pm i\beta\theta}/\sin(\pi\beta)$. The substitution $x \mapsto 1/x$ shows that the left side is even in β, and so is the right side. So the formula holds for $0 < \beta < 1$ too. We use it with $x = w^\alpha$, $\beta = p/\alpha$ and $\theta = \pi\alpha$. Then $w^{\alpha-1}dw = dx/\alpha$, and we get

$$\int_0^\infty w^p \rho_\alpha(w)\, dw = \frac{\sin \pi\alpha}{\pi\alpha} \cdot \frac{\pi \sin \pi p}{\sin \pi\alpha\, \sin(\pi p/\alpha)} = \mathbb{E}W^p \qquad (0 < |p| < \alpha).$$

At $p = 0$ the value $\theta/\sin\theta$ of the classical integral gives $\int_0^\infty \rho_\alpha = 1$, so ρ_α is a probability density. So $\log W$ and the logarithm of a variable with density ρ_α have the same moment generating function near 0. A moment generating function that is finite near 0 determines the law, so W has density ρ_α. Then $Y = W/(1+W)$ has density $\rho_\alpha(\frac{y}{1-y})(1-y)^{-2}$, and multiplying the numerator and the denominator by $(1-y)^{2\alpha}$ shows that this is $\ell_\alpha(y)$. Both f and ℓ_α are densities of Y that are continuous near $\frac12$, so

$$f(\tfrac12) = \ell_\alpha(\tfrac12) = \frac{2\sin\pi\alpha}{\pi(1+\cos\pi\alpha)} = \frac{2}{\pi}\tan\frac{\pi\alpha}{2}.$$

Dividing $E_N = \frac12 N^{-\alpha}$ by C_N gives the formula for R_N. □

E Proof of Proposition 5.7

We use the setting of Section 5 with two states and a fair first step. So $0 < c < 2$, X_1 is a fair sign, and the switch indicators $\chi_k = \mathbf{1}\{X_k \neq X_{k-1}\}$, $k \geq 2$, are independent Bernoulli(c/k) variables, independent of X_1. Put

$$\rho(a,b) = \prod_{k=a+1}^{b} \Big(1 - \frac{c}{k}\Big) = \frac{\Gamma(b+1-c)\,\Gamma(a+1)}{\Gamma(b+1)\,\Gamma(a+1-c)}, \qquad \lambda_{t,N} = \frac{c}{t}\,\rho(t,N)$$

for integers $1 \le a \le b$ and $2 \le t \le N$. So $\lambda_{t,N}$ is the probability of a switch at time t and none in $(t, N]$, and $E_N = \frac{1}{2}\rho(1, N)$. Let T_s^+ be the time of the s-th step equal to $+1$. Let B be a Beta(c, c) variable, $d_c = 4^{1-c}/\mathrm{B}(c, c)$ its density at $\frac{1}{2}$, and $\Theta(c) = \log\bigl(2\Gamma(2-c)\, d_c\bigr)$. For even N we write $h = N/2$ and $C_N = \mathbb{P}(A_N = h)$. Changing the sign of every step preserves the law of the walk and sends A_N to $N - A_N$. We call this flip symmetry.

Idea of the proof. A walk that ends in the center with last step -1 takes its h-th step $+1$ at some time r, switches at time $r+1$, and never switches again. So C_N is an average of the explicit numbers $\lambda_{r+1,N}$ over the law of T_h^+ (Lemma E.1). For $c \le 1$ these numbers are monotone in r, which already traps c_N between two explicit roots and gives $1 - c_N = O(1/\log N)$ (Proposition E.2). For the constant we need the law of T_h^+. We have $T_h^+ \le r$ exactly when $A_r \ge h$, and A_r/r is close to a Beta(c, c) variable with an error that is a power of r (Lemma E.4). This is proved by coupling the walk with a sign process in continuous time that switches at the points of a Poisson process of intensity $c\, dt/t$, whose occupation fraction is exactly Beta(c, c) at every time (Proposition E.3). Summation by parts then gives $NC_N = d_c + O(N^{-0.4})$ (Proposition E.5).

E.1 A gamma ratio

Gautschi's inequality [29, 5.6.4] says that $x^{1-s} < \Gamma(x+1)/\Gamma(x+s) < (x+1)^{1-s}$ for $x > 0$ and $0 < s < 1$. We claim that for $0 < c < 2$ and real $y \ge 2$,

$$\frac{\Gamma(y+1-c)}{\Gamma(y+1)} = y^{-c}(1+\vartheta) \qquad \text{with} \qquad -\frac{1}{y} \le \vartheta \le \frac{2}{y}. \tag{21}$$

If $c = 1$ the left side is $1/y$ and $\vartheta = 0$. If $c < 1$ we take $x = y$ and $s = 1 - c$. Then the left side lies between $(y+1)^{-c}$ and y^{-c}, and $(1+1/y)^{-c} \ge 1 - c/y \ge 1 - 1/y$. If $c > 1$ we take $x = y$ and $s = 2 - c$ and use $\Gamma(y+1-c) = \Gamma(y+2-c)/(y+1-c)$. Then the left side lies between $(y+1)^{1-c}/(y+1-c)$ and $y^{1-c}/(y+1-c)$. Here $y/(y+1-c) = 1 + (c-1)/(y+1-c) \le 1 + 1/(y-1) \le 1 + 2/y$, and $(1+1/y)^{1-c}\, y/(y+1-c) \ge (1+1/y)^{-1} \ge 1 - 1/y$.

E.2 Averaging at the center and a bracket

Lemma E.1 (averaging at the center)**.** *Let $N \ge 2$ be even and $T = T_h^+$. Then*

$$C_N = 2\sum_{r=h}^{N-1} \mathbb{P}(T = r)\, \lambda_{r+1,N} \qquad \textit{and} \qquad \sum_{r=h}^{N-1} \mathbb{P}(T = r) = \frac{1}{2}. \tag{22}$$

If $0 < c \le 1$, then $c/N \le C_N \le \lambda_{h+1,N}$.

Proof. On the event $\{A_N = h,\, X_N = -1\}$ the last step equal to $+1$ is the h-th one, at a time $T = r \le N - 1$. It is followed by a switch at $r+1$ and no switch in $(r+1, N]$. Conversely, these conditions give $A_N = h$ and $X_N = -1$. The event $\{T = r\}$ depends only on $X_1, \chi_2, \dots, \chi_r$. So $\mathbb{P}(A_N = h,\, X_N = -1) = \sum_{r=h}^{N-1} \mathbb{P}(T = r)\, \lambda_{r+1,N}$, and flip symmetry gives the same value for $X_N = +1$. Next, $T \le N - 1$ exactly when $A_{N-1} \ge h$. This has probability $\frac{1}{2}$, because $N - 1$ is odd and A_{N-1} is symmetric about $(N-1)/2$. Finally $\lambda_{t+1,N}/\lambda_{t,N} = t/(t+1-c)$, which is at most 1 for $c \le 1$. So $\lambda_{N,N} = c/N \le \lambda_{t,N} \le \lambda_{h+1,N}$ for $h+1 \le t \le N$, and by (22) C_N is a weighted average of these values. □

For an integer $M \geq 2$ put $\eta_M(c) = M\rho(1,M)/(2c)$. Since $\rho(1,M)$ decreases in c and equals $1/M$ at $c=1$, the function η_M decreases strictly on $(0,1]$, from ∞ at 0 to $\eta_M(1) = \frac{1}{2}$. Let $c^{(M)} \in (0,1)$ be the root of $\eta_M = 1$.

Proposition E.2 (bracket)**.** *The numbers $c^{(M)}$ increase with M, and $1 - c^{(M)} = \frac{\log 2}{\log M + 1 + \gamma} + O\big((\log M)^{-3}\big)$. For every even $N \geq 2$ we have $c^{(h+1)} \leq c_N \leq c^{(N)}$. So $(1-c_N)\log N \to \log 2$, and $c_N \geq c^{(3)} = (9 - \sqrt{57})/2 > 0.725$ for $N \geq 4$.*

Proof. The equation $\eta_M(c) = 1$ says that $\frac{1}{2}\rho(1,M) = c/M$, namely that the end for M steps equals c/M. We compare E_N with the two bounds for C_N in Lemma E.1, and each comparison is an inequality for some η_M.

Monotonicity. For $c < 1$ we have $\eta_{M+1}/\eta_M = (M+1-c)/M > 1$. So $\eta_{M+1}(c^{(M)}) > 1$, and since η_{M+1} decreases, $c^{(M+1)} > c^{(M)}$.

The bracket. By Proposition 5.6(b), $C_N < E_N$ for $c < c_N$ and $C_N > E_N$ for $c > c_N$, and $c_N < 1$. So it is enough to compare C_N with E_N for $0 < c < 1$. Let $c^{(N)} < c < 1$. Then $\eta_N(c) < 1$, that is $E_N = \frac{1}{2}\rho(1,N) < c/N$, and $c/N \leq C_N$ by Lemma E.1. So $C_N > E_N$, which means $c > c_N$. Hence $c_N \leq c^{(N)}$. Now let $c < c^{(h+1)}$. Then $\eta_{h+1}(c) > 1$, that is $\frac{1}{2}\rho(1,h+1) > c/(h+1)$. With $\rho(1,N) = \rho(1,h+1)\rho(h+1,N)$ this gives $E_N > \frac{c}{h+1}\rho(h+1,N) = \lambda_{h+1,N} \geq C_N$. So $c < c_N$. Hence $c_N \geq c^{(h+1)}$.

The expansion. The number $c^{(3)}$ solves $3\rho(1,3) = (2-c)(3-c)/2 = 2c$, that is $c^2 - 9c + 6 = 0$. For $N \geq 4$ we have $h+1 \geq 3$, so $c_N \geq c^{(h+1)} \geq c^{(3)}$. By (21) with $y = M$ we have $M\rho(1,M) = M^{1-c}(1 + O(1/M))/\Gamma(2-c)$. So the equation $\eta_M(c) = 1$ reads $(1-c)\log M = \Theta_0(c) + O(1/M)$ with $\Theta_0(c) = \log\big(2c\,\Gamma(2-c)\big)$, uniformly for $0 < c \leq 1$. For $M \geq 3$ we have $c^{(M)} \in [c^{(3)}, 1)$, where Θ_0 is bounded. So $e = 1 - c^{(M)}$ is $O(1/\log M)$, and we may expand at $c = 1$. We have $\Theta_0(1) = \log 2$ and $\Theta_0'(1) = 1 - \psi(1) = 1 + \gamma$, where $\psi = \Gamma'/\Gamma$. Taylor's theorem gives $e(\log M + 1 + \gamma) = \log 2 + O(e^2) + O(1/M)$, which is the expansion. The limit of $(1-c_N)\log N$ follows since $\log(h+1) \sim \log N$. □

E.3 The continuum occupation

Proposition E.3 (continuum occupation)**.** *Let Π be a Poisson process on $(0,\infty)$ with intensity $c\,dt/t$, and let $Y(1)$ be a fair sign independent of Π. Put $Y(t) = Y(1)(-1)^{\Pi((1,t])}$ for $t \geq 1$ and $Y(t) = Y(1)(-1)^{\Pi((t,1])}$ for $0 < t < 1$. Then for every $r > 0$ the fraction $Z_r = \frac{1}{r}\int_0^r \mathbf{1}\{Y(t) = +1\}\,dt$ has exactly the Beta(c,c) law.*

This agrees with [37, Remark 6.2], which obtains the Beta(c,c) one-dimensional marginals of the zigzag process of [37, Definition 6.1] from a scaling limit and mentions a direct proof by E. Crane that is not written there. We give a direct proof.

Proof. We pass to logarithmic time, where the sign process is stationary, and then we condition on the last switch before time 1. This gives an equation for the law of Z_1 that has only one solution, and we check that Beta laws solve it.

Step 1 (the law of Z_r does not depend on r). The points $\log t$, $t \in \Pi$, form a Poisson process $\tilde{\Pi}$ of rate c on $\mathbb{R}$, and $W(u) = Y(e^u)$ satisfies $W(v) = W(u)(-1)^{\tilde{\Pi}((u,v])}$ for $u < v$. Fix a real ℓ. Then $W(\ell)$ is $W(0)$ times a function of $\tilde{\Pi}$, and $W(0)$ is a fair sign independent of $\tilde{\Pi}$. So $W(\ell)$ is a fair sign independent of $\tilde{\Pi}$ too. As $\tilde{\Pi} - \ell$ has the law of $\tilde{\Pi}$, the pair $(W(\ell), \tilde{\Pi} - \ell)$ has the law of $(W(0), \tilde{\Pi})$, and so the process $W(\ell + \cdot)$ has the law of W. With $\ell = \log r$ this says that $Y(r\,\cdot)$ has the law of Y. Hence $Z_r = \int_0^1 \mathbf{1}\{Y(rt) = +1\}\,dt$ has the law of Z_1.

Step 2 (an equation for the law of Z_1). The substitution $t = e^{-s}$ gives $Z_1 = \int_0^\infty e^{-s}\mathbf{1}\{V_s = +1\}\,ds$, where $V_s = W(-s)$. The process V looks backward from time 1. It starts at a fair sign and switches at the points of a Poisson process of rate c. Let $\mu_\pm$ be the law of Z_1 given $V_0 = \pm 1$, let τ be the first switch of V, and let $U = e^{-\tau}$ (in the original time, U is the last switch of Y before time 1). Then $\mathbb{P}(U \le u) = \mathbb{P}(\tau \ge -\log u) = u^c$ for $0 \le u \le 1$, so U is Beta$(c, 1)$. After τ the process V starts afresh from $-V_0$, independently of τ. We split the integral at τ. Given $V_0 = +1$ the part before τ is $\int_0^\tau e^{-s}ds = 1 - U$, and the part after τ is U times a copy of Z_1 for a process started at -1. Given $V_0 = -1$ the part before τ is 0, and the part after is U times a copy of Z_1 for a process started at $+1$. So μ_+ is the law of $1 - U + UZ_-$ and μ_- is the law of UZ_+, where $Z_\pm$ has law $\mu_\pm$ and is independent of U. In other words, (μ_+, μ_-) is a fixed point of the map Ψ that sends a pair (ν_+, ν_-) of laws on $[0, 1]$ to the laws of $1 - U + UZ_-$ and UZ_+, where $Z_\pm \sim \nu_\pm$ are independent of U.

Step 3 (Ψ has only one fixed point). We measure the distance between two laws ν, ν' on $[0, 1]$ by the Wasserstein distance, namely the smallest value of $\mathbb{E}|Z - Z'|$ over all pairs with $Z \sim \nu$ and $Z' \sim \nu'$. Take two pairs (ν_+, ν_-) and (ν'_+, ν'_-), and take $Z_\pm \sim \nu_\pm$ and $Z'_\pm \sim \nu'_\pm$ that attain these distances, independently of U. Then $\mathbb{E}|UZ_\pm - UZ'_\pm| = \mathbb{E}U\,\mathbb{E}|Z_\pm - Z'_\pm|$. So Ψ multiplies the larger of the two distances by at most $\mathbb{E}U = c/(c+1) < 1$. Two fixed points are then at distance 0, so they are equal.

Step 4 (Beta laws solve the equation). We claim that the pair Beta$(c+1, c)$, Beta$(c, c+1)$ is a fixed point of Ψ. (This is the natural guess. If the sign is $+$ at time 1, then more of the time before is spent at $+$.) We need one fact, namely that UZ is Beta$(c, c+1)$ whenever Z is Beta$(c+1, c)$ and independent of U. To prove it let G_a, G_b, G_d be independent Gamma variables with shapes a, b, d and unit scale. A change of variables shows that $G_a/(G_a + G_b)$ is Beta(a, b) and independent of $G_a + G_b$. Hence it is independent of the pair $(G_a + G_b, G_d)$ and of $(G_a + G_b)/(G_a + G_b + G_d)$, which is Beta$(a+b, d)$. Their product $G_a/(G_a + G_b + G_d)$ is Beta$(a, b+d)$. With $(a, b, d) = (c, 1, c)$ this is the fact.

Now let Z_+ be Beta$(c+1, c)$ and Z_- be Beta$(c, c+1)$, independent of U. By the fact UZ_+ is Beta$(c, c+1)$. Also $1 - Z_-$ is Beta$(c+1, c)$, so $U(1 - Z_-)$ is Beta$(c, c+1)$ and $1 - U + UZ_- = 1 - U(1 - Z_-)$ is Beta$(c+1, c)$. This proves the claim. By Step 3, μ_+ is Beta$(c+1, c)$ and μ_- is Beta$(c, c+1)$. Since V_0 is a fair sign, Z_1 has the equal mixture of these two laws. As $\mathrm{B}(c+1, c) = \mathrm{B}(c, c+1) = \frac{1}{2}\mathrm{B}(c, c)$, the mixture has density $x^{c-1}(1-x)^{c-1}(x + 1 - x)/\mathrm{B}(c, c)$, which is the Beta$(c, c)$ density. □

Lemma E.4 (coupling)**.** *Let $J \subset (0, 2)$ be compact, $a_J = \min(\min J, 1)$ and $\kappa_J = a_J/(1 + a_J)$. There is C_J such that for $c \in J$ and every integer $r \ge 2$,*

$$\sup_{y \in \mathbb{R}} \big|\mathbb{P}(A_r \ge y) - \mathbb{P}(rB \ge y)\big| \le C_J\,(\log r)\,r^{-\kappa_J}.$$

Proof. We build the walk and the process Y of Proposition E.3 on one probability space so that $X_k = Y(k)$ for most $k \le r$. This works because Y changes sign between the times $k - 1$ and k with probability close to c/k. Then A_r is close to rZ_r, which has the law of rB. The first K steps are left uncoupled, and we choose K at the end.

Step 1 (the coupling). Take Π and Y as in Proposition E.3 and an integer $1 \le K \le r$. For $k > K$ let $\chi'_k = \mathbf{1}\{\Pi((k-1, k]) \text{ is odd}\}$, so that $Y(k) = Y(k-1)(-1)^{\chi'_k}$. These variables depend on Π in disjoint intervals, so they are independent of each other and of $Y(K)$. The number $\Pi((k-1, k])$ is Poisson with mean $\nu = c \log \frac{k}{k-1}$, and a Poisson variable with mean ν is odd with probability $\frac{1}{2}(1 - e^{-2\nu})$. So $\mathbb{P}(\chi'_k = 1) = p'_k = \frac{1}{2}\big(1 - (1 - 1/k)^{2c}\big)$. By Taylor's theorem for the

function $\frac{1}{2}(1-(1-x)^{2c})$ at $x=1/k$, $|p'_k - c/k| = \frac{1}{2}c|2c-1|(1-\vartheta)^{2c-2}k^{-2}$ for some $0<\vartheta<1/k$. This is at most $3k^{-2}$. Indeed, if $c \geq 1$ then $\frac{1}{2}c|2c-1| < 3$ and the power of $1-\vartheta$ is at most 1. If $c<1$ then $\frac{1}{2}c|2c-1| \leq \frac{1}{2}$ and the power is at most $(1-\frac{1}{2})^{-2}=4$, as $k \geq 2$.

For $K < k \leq r$ we couple $\chi_k \sim \text{Bernoulli}(c/k)$ with χ'_k so that $\mathbb{P}(\chi_k \neq \chi'_k) = |p'_k - c/k|$, using independent extra randomness. We put $X_K = Y(K)$ and $X_k = X_{k-1}(-1)^{\chi_k}$ for $K < k \leq r$, and we draw $X_1, \dots, X_{K-1}$ from the law of the walk given X_K, again with independent randomness. In the walk X_K is a fair sign and $(\chi_k)_{k>K}$ is independent of $(X_1, \dots, X_K)$. The same holds here, so $(X_1, \dots, X_r)$ has the law of the walk.

Step 2 (A_r is close to rZ_r). Let $\mathcal{A}$ be the event that $\chi_k = \chi'_k$ for $K < k \leq r$. Then $\mathbb{P}(\mathcal{A}^c) \leq \sum_{k>K} 3k^{-2} \leq 3/K$, and on $\mathcal{A}$ we have $X_k = Y(k)$ for $K \leq k \leq r$. Now A_r is the sum of $\mathbf{1}\{X_k = +1\}$ over the steps $k \leq r$, and rZ_r is the sum of $\int_{k-1}^{k} \mathbf{1}\{Y(t)=+1\}\,dt$ over the same k. Let $K < k \leq r$. If $\Pi((k-1,k]) = 0$, then $Y = Y(k)$ on $(k-1,k]$, so on $\mathcal{A}$ the two terms for this k are equal. The other $k \leq r$ number at most $K + \Pi((K,r])$, and for each of them the two terms differ by at most 1. Hence $|A_r - rZ_r| \leq K + \Pi((K,r])$ on $\mathcal{A}$. The variable $\Pi((K,r])$ is Poisson with mean $c\log(r/K) \leq 2\log r$, so by Markov's inequality it exceeds K with probability at most $2\log r/K$. So with $b_K = (3+2\log r)/K$ we have $|A_r - rZ_r| \leq 2K$ outside an event of probability at most b_K.

Step 3 (the distribution functions). By Step 2, for every real y,

$$\mathbb{P}(rZ_r \geq y + 2K) - b_K \leq \mathbb{P}(A_r \geq y) \leq \mathbb{P}(rZ_r \geq y - 2K) + b_K,$$

and rZ_r has the law of rB by Proposition E.3. So $\mathbb{P}(A_r \geq y)$ and $\mathbb{P}(rB \geq y)$ differ by at most b_K plus the probability that rB lies in an interval of length $2K$. Hence the left side of the lemma is at most $b_K + \sup_x \mathbb{P}(x \leq B \leq x + 2K/r)$.

Step 4 (short intervals and the choice of K). We bound the mass that B gives to an interval of length δ. For $c \leq 1$ the density of B is at most $2^{1-c}\big(x^{c-1} + (1-x)^{c-1}\big)/\mathrm{B}(c,c)$, and x^{c-1} has integral at most δ^c/c over such an interval. For $c \geq 1$ the density is at most d_c. The mass of an interval is also at most 1. So there is C'_J such that an interval of length δ has mass at most $C'_J \delta^{a_J}$ for all $\delta > 0$ and $c \in J$. A large K makes b_K small and the interval long, and we balance the two. Take $K = \lceil r^{\kappa_J} \rceil$. Then $b_K \leq (3+2\log r) r^{-\kappa_J}$ and $2K/r \leq 4r^{\kappa_J - 1}$, and $(r^{\kappa_J-1})^{a_J} = r^{-\kappa_J}$ because $(1-\kappa_J)a_J = \kappa_J$. So both terms are at most a constant times $(\log r)\, r^{-\kappa_J}$. □

E.4 A local limit at the center

Proposition E.5 (local limit at the center)**.** *For each compact $J \subset (0,2)$ there is C_J such that $\big|NC_N - d_c\big| \leq C_J\,(\log N)\,N^{-\kappa_J}$ for $c \in J$ and even $N \geq 4$.*

Proof. We start from the averaging identity of Lemma E.1 and replace the law of $T = T_h^+$ by its continuum version. Since $T \leq r$ exactly when $A_r \geq h$, and A_r is close to rB by Lemma E.4, the time T is close in law to h/B. The weights $\lambda_{r+1,N}$ are explicit, so the average becomes an explicit integral against the Beta(c,c) density.

Let $F(r) = \mathbb{P}(T \leq r) = \mathbb{P}(A_r \geq h)$, $\Phi(r) = \mathbb{P}(rB \geq h)$ and $\Lambda_r = N\lambda_{r+1,N}$. By Lemma E.1,

$$NC_N = 2\sum_{r=h}^{N-1} \big(F(r) - F(r-1)\big)\Lambda_r.$$

Step 1 (replacing F by Φ). Put $\Delta(r) = F(r) - \Phi(r)$. Then $\Delta(h-1) = 0$, because $A_{h-1} < h$ and $B < 1$. For $h \leq r \leq N-1$ we have $r \geq 2$ and $N/2 \leq r < N$, so Lemma E.4 gives

$|\Delta(r)| \le \delta_N := C_J(\log N)N^{-\kappa_J}$ after enlarging C_J. Summation by parts gives

$$\sum_{r=h}^{N-1}\big(\Delta(r)-\Delta(r-1)\big)\Lambda_r = \Delta(N-1)\Lambda_{N-1} - \sum_{r=h}^{N-2}\Delta(r)\big(\Lambda_{r+1}-\Lambda_r\big).$$

The ratio $\lambda_{t+1,N}/\lambda_{t,N} = t/(t+1-c)$ shows that Λ_r is monotone in r (it decreases for $c \le 1$ and increases for $c \ge 1$). So the differences $\Lambda_{r+1}-\Lambda_r$ have one sign, and their absolute values add up to $|\Lambda_{N-1}-\Lambda_h|$. Hence the right side is at most $\delta_N\big(\Lambda_{N-1}+|\Lambda_{N-1}-\Lambda_h|\big)$ in absolute value. Now $\Lambda_{N-1} = N\lambda_{N,N} = c$ and $0 < \Lambda_h < 2c$. For $c \le 1$ the latter holds because $\Lambda_h = \frac{Nc}{h+1}\rho(h+1,N)$ with $\rho \le 1$ and $N < 2(h+1)$, and for $c > 1$ because Λ_r increases to c. So $\Lambda_{N-1}+|\Lambda_{N-1}-\Lambda_h| \le 2c$, and replacing F by Φ changes NC_N by at most $4c\,\delta_N$.

Step 2 (the new sum is an expectation). After the replacement the sum is $2\sum_{r=h}^{N-1}(\Phi(r)-\Phi(r-1))\Lambda_r$. The term with $r=h$ vanishes, since $\Phi(h)=\Phi(h-1)=0$. For $h < r \le N-1$ we have $\Phi(r)-\Phi(r-1) = \mathbb{P}(h/r \le B < h/(r-1))$, and on this event $\lceil h/B\rceil = r$. These events are disjoint, and their union is $\{h/(N-1) \le B < 1\}$. So the new sum is $2\,\mathbb{E}\big[\Lambda_{\lceil h/B\rceil};\, B \ge h/(N-1)\big]$.

Step 3 (evaluating the expectation). By (21) with $y=N$ and $y=r+1 \ge 3$, $\Lambda_r = c\big((r+1)/N\big)^{c-1}(1+O(1/N))$ for $h \le r \le N-1$. On the event above with $r = \lceil h/B\rceil$, the number $(r+1)/N$ lies in $[\frac12, 2]$ and within $2/N$ of $h/(BN) = 1/(2B)$, which also lies in $[\frac12,2]$. The function x^{c-1} is 4-Lipschitz on $[\frac12,2]$. So $\Lambda_{\lceil h/B\rceil} = c(2B)^{1-c}+O(1/N)$. Also $\mathbb{P}\big(\frac12 \le B < h/(N-1)\big) = O_J(1/N)$, since the density of B is bounded near $\frac12$ uniformly on J. So we may replace the event $\{B \ge h/(N-1)\}$ by $\{B \ge \frac12\}$. In the remaining integral the factor b^{1-c} cancels the factor b^{c-1} of the density. Hence NC_N is within $4c\,\delta_N$ of

$$2c\,\mathbb{E}\big[(2B)^{1-c};\, B \ge \tfrac12\big] + O_J(1/N) = \frac{2c\,2^{1-c}}{\mathrm{B}(c,c)}\int_{1/2}^{1}(1-b)^{c-1}\,db + O_J(1/N) = d_c + O_J(1/N). \quad \square$$

E.5 The expansion of the crossing

Proof of Proposition 5.7. We write $\log C_N - \log E_N$ as an explicit function of c with a small error, and we solve for its zero near $c=1$. Let $J = [c^{(3)}, 1]$. Then $\kappa_J = c^{(3)}/(1+c^{(3)}) > 0.42$, and d_c is bounded below on J by a positive constant. By (21) with $y=N$, $\log E_N = \log\big(\frac12\rho(1,N)\big) = -\log 2 - \log\Gamma(2-c) - c\log N + O(1/N)$. By Proposition E.5, $\log C_N = \log d_c - \log N + O(N^{-0.4})$ for large N. Both hold uniformly for $c \in J$, so

$$\log C_N - \log E_N = (c-1)\log N + \Theta(c) + O(N^{-0.4}) \qquad (c \in J). \tag{23}$$

Here $\Theta(1) = \log 2$ because $d_1 = 1$. Next, $\Theta'(c) = -\psi(2-c) - \log 4 - 2\psi(c) + 2\psi(2c)$, and with $\psi(1) = -\gamma$ and $\psi(2) = 1-\gamma$ this gives $\Theta'(1) = \gamma + 2 - 2\log 2$. Put $M = \log N + \Theta'(1)$. By Proposition E.2, $c_N \in J$ for $N \ge 4$ and $e = 1 - c_N$ is $O(1/\log N)$. The left side of (23) vanishes at $c = c_N$, because $C_N = E_N$ there. As Θ is smooth on J, Taylor's theorem gives $\Theta(c_N) = \log 2 - e\,\Theta'(1) + O(e^2)$. So (23) at $c = c_N$ reads $0 = -e\log N + \log 2 - e\,\Theta'(1) + O(e^2) + O(N^{-0.4})$, that is $eM = \log 2 + O(e^2) + O(N^{-0.4}) = \log 2 + O(M^{-2})$. So $e = \log 2/M + O(M^{-3})$, which is the expansion. $\square$

With one more term in Taylor's theorem and $e^2 = (\log 2)^2/M^2 + O(M^{-4})$, the same argument adds the term $-\Theta''(1)(\log 2)^2/(2M^3)$ to the expansion of c_N, with error $O(M^{-4})$. Here $\Theta''(1) = \pi^2/2 - 4$, since $\Theta''(c) = \psi'(2-c) - 2\psi'(c) + 4\psi'(2c)$, $\psi'(1) = \zeta(2)$ and $\psi'(2) = \zeta(2) - 1$. At $N = 200$ the bracket of Proposition E.2 is $[0.88749, 0.89878]$, Proposition 5.7 gives 0.89319, and with the M^{-3} term we get 0.89236. The high-precision value is $c_{200} = 0.89253$. The constants C_J and the constants in the O-terms of this appendix are not computed.

Statements and declarations

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Data availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of `https://github.com/apemm/Kagey-Problems`, at the release `paperD-arxiv-v1`. To rerun the checks, run `python -B verify_all.py --paper d` in that folder, which takes several minutes. The scripts are in the subfolder `paperD` and need `numpy`, `scipy`, `mpmath` and `python-flint`. The file `SUPPLEMENT.md` says what each script checks and where to find the predictions that we wrote down before each computation. Section 6 describes the checks.

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